CHAPTER 27

WIND LOADS ON BUILDINGS: MAIN WIND FORCE RESISTING SYSTEM (DIRECTIONAL PROCEDURE)

27.1 SCOPE

27.1.1 Building Types. This chapter applies to the determination of main wind force resisting system (MWFRS) wind loads on enclosed, partially enclosed, and open buildings of all heights using the Directional Procedure.

Part 1 applies to buildings of all heights where it is necessary to separate applied wind loads onto the windward, leeward, and sidewalls of the building to properly assess the internal forces in the MWFRS members.

Part 2 applies to a special class of buildings designated as enclosed simple diaphragm buildings, as defined in Section 26.2, with $h \le 160$ ft ($h \le 48.8$ m).

27.1.2 Conditions. A building that has design wind loads determined in accordance with this chapter shall comply with all of the following conditions:

1. The building is a regular-shaped building as defined in Section 26.2, and
2. The building does not have response characteristics that make it subject to across-wind loading, vortex shedding, or instability caused by galloping or flutter; nor does it have a site location for which channeling effects or buffeting in the wake of upwind obstructions warrant special consideration.

27.1.3 Limitations. The provisions of this chapter take into consideration the load magnification effect caused by gusts in resonance with along-wind vibrations of flexible buildings. Buildings that do not meet the requirements of Section 27.1.2 or that have unusual shapes or response characteristics shall be designed using recognized literature documenting such wind load effects or shall use the Wind Tunnel Procedure specified in Chapter 31.

27.1.4 Shielding. There shall be no reductions in velocity pressure caused by apparent shielding afforded by buildings and other structures or terrain features.

27.1.5 Minimum Design Wind Loads. The wind load to be used in the design of the MWFRS for an enclosed or partially enclosed building shall not be less than 16 lb/ft^2 (0.77 kN/m^2) multiplied by the wall area of the building, and 8 lb/ft^2 (0.38 kN/m^2) multiplied by the roof area of the building projected onto a vertical plane normal to the assumed wind direction. Wall and roof loads shall be applied simultaneously. The design wind force for open buildings shall be not less than 16 lb/ft^2 (0.77 kN/m^2) multiplied by the area, A_f.

PART 1: ENCLOSED, PARTIALLY ENCLOSED, AND OPEN BUILDINGS OF ALL HEIGHTS

User Note: Use Part 1 of Chapter 27 to determine wind pressures on the MWFRS of enclosed, partially enclosed, or open buildings with any general plan shape, building height, or roof geometry that matches the figures provided. These provisions use the traditional "all heights" method (Directional Procedure) by calculating wind pressures using *specific wind pressure equations* applicable to each building surface.

27.2 GENERAL REQUIREMENTS

The steps to determine the wind loads on the MWFRS for enclosed, partially enclosed, and open buildings of all heights are provided in Table 27.2-1.

Table 27.2-1 Steps to Determine MWFRS Wind Loads for Enclosed, Partially Enclosed, and Open Buildings of All Heights

Step 1: Determine Risk Category of building; see Table 1.5-1.
Step 2: Determine the basic wind speed, V, for the applicable Risk Category; see Figs. 26.5-1 and 26.5-2.
Step 3: Determine wind load parameters:
- Wind directionality factor, K_d; see Section 26.6 and Table 26.6-1.
- Exposure category; see Section 26.7.
- Topographic factor, K_{zt}; see Section 26.8 and table in Fig. 26.8-1.
- Ground elevation factor, K_e; see Section 26.9
- Gust-effect factor, G or G_f; see Section 26.11.
- Enclosure classification; see Section 26.12.
- Internal pressure coefficient, (GC_{pi}); see Section 26.13 and Table 26.13-1.

Step 4: Determine velocity pressure exposure coefficient, K_z or K_h; see Table 26.10-1.
Step 5: Determine velocity pressure q_z or q_h, Eq. (26.10-1).
Step 6: Determine external pressure coefficient, C_p or C_N:
- Fig. 27.3-1 for walls and flat, gable, hip, monoslope, or mansard roofs.
- Fig. 27.3-2 for domed roofs.
- Fig. 27.3-3 for arched roofs.
- Fig. 27.3-4 for monoslope roof, open building.
- Fig. 27.3-5 for pitched roof, open building.
- Fig. 27.3-6 for troughed roof, open building.
- Fig. 27.3-7 for along-ridge/valley wind load case for monoslope, pitched, or troughed roof, open building.

Step 7: Calculate wind pressure, p, on each building surface:
- Eq. (27.3-1) for rigid and flexible buildings.
- Eq. (27.3-2) for open buildings.

27.2.1 Wind Load Parameters Specified in Chapter 26. The following wind load parameters shall be determined in accordance with Chapter 26:

- Basic wind speed, V (Section 26.5).
- Wind directionality factor, K_d (Section 26.6).
- Exposure category (Section 26.7).
- Topographic factor, K_{zt} (Section 26.8).
- Ground elevation factor, K_e; see Section 26.9
- Gust-effect factor (Section 26.11).
- Enclosure classification (Section 26.12).
- Internal pressure coefficient, (GC_{pi}) (Section 26.13).

27.3 WIND LOADS: MAIN WIND FORCE RESISTING SYSTEM

27.3.1 Enclosed and Partially Enclosed Rigid and Flexible Buildings. Design wind pressures for the MWFRS of buildings of all heights in lb/ft^2 (N/m^2), shall be determined by the following equation:

$$p = qGC_p - q_i(GC_{pi}) \qquad (27.3\text{-}1)$$

where

$q = q_z$ for windward walls evaluated at height z above the ground.

$q = q_h$ for leeward walls, sidewalls, and roofs evaluated at height h.

$q_i = q_h$ for windward walls, sidewalls, leeward walls, and roofs of enclosed buildings, and for negative internal pressure evaluation in partially enclosed buildings.

$q_i = q_z$ for positive internal pressure evaluation in partially enclosed buildings where height z is defined as the level of the highest opening in the building that could affect the positive internal pressure. For buildings sited in wind-borne debris regions, glazing that is not impact-resistant or protected with an impact-resistant covering shall be treated as an opening in accordance with Section 26.12.3. For positive internal pressure evaluation, q_i may conservatively be evaluated at height $h(q_i = q_h)$.

G = gust-effect factor; see Section 26.11. For flexible buildings, G_f determined in accordance with Section 26.11.5 shall be substituted for G.

C_p = external pressure coefficient from Figs. 27.3-1, 27.3-2, and 27.3-3.

(GC_{pi}) = internal pressure coefficient from Table 26.13-1.

Both q and q_i shall be evaluated using exposure defined in Section 26.7.3. Pressure shall be applied simultaneously on windward and leeward walls and on roof surfaces as defined in Figs. 27.3-1, 27.3-2, and 27.3-3.

27.3.2 Open Buildings with Monoslope, Pitched, or Troughed Free Roofs. The net design pressure for the MWFRS of open buildings with monoslope, pitched, or troughed free roofs in lb/ft^2 (N/m^2), shall be determined by the following equation:

$$p = q_h GC_N \qquad (27.3\text{-}2)$$

where

q_h = velocity pressure evaluated at mean roof height h using the exposure as defined in Section 26.7.3 that results in the highest wind loads for any wind direction at the site.

G = gust-effect factor from Section 26.11.

C_N = net pressure coefficient determined from Figs. 27.3-4 through 27.3-7.

Net pressure coefficients, C_N, include contributions from top and bottom surfaces. All load cases shown for each roof angle shall be investigated. Plus and minus signs signify pressure acting toward and away from the top surface of the roof, respectively.

For free roofs with an angle of plane of roof from horizontal θ less than or equal to 5° and containing fascia panels, the fascia panel shall be considered an inverted parapet. The contribution of loads on the fascia to the MWFRS loads shall be determined using Section 27.3.5, with q_p equal to q_h. For an open or partially enclosed building with transverse frames and a pitched roof ($\theta \le 45°$), an additional horizontal force in the longitudinal direction (parallel to the ridge) that acts in combination with the roof load calculated in Section 27.3.3 shall be determined in accordance with Section 28.3.5.

27.3.3 Roof Overhangs. The positive external pressure on the bottom surface of windward roof overhangs shall be determined using $C_p = 0.8$ and combined with the top surface pressures determined using Fig. 27.3-1.

27.3.4 Parapets. The design wind pressure for the effect of parapets on MWFRS of rigid or flexible buildings with flat, gable, or hip roofs in lb/ft^2 (N/m^2), shall be determined by the following equation:

$$p_p = q_p(GC_{pn})(\text{lb/ft}^2) \qquad (27.3\text{-}3)$$

where

p_p = combined net pressure on the parapet caused by the combination of the net pressures from the front and back parapet surfaces. Plus (and minus) signs signify net pressure acting toward (and away from) the front (exterior) side of the parapet.

q_p = velocity pressure evaluated at the top of the parapet.

(GC_{pn}) = combined net pressure coefficient:
= +1.5 for windward parapet or
= −1.0 for leeward parapet.

27.3.5 Design Wind Load Cases. The MWFRS of buildings of all heights, the wind loads of which have been determined under the provisions of this chapter, shall be designed for the wind load cases as defined in Fig. 27.3-8.

EXCEPTION: Buildings meeting the requirements of Section D1.1 of Appendix D need only be designed for Case 1 and Case 3 of Fig. 27.3-8.

The eccentricity e for rigid buildings shall be measured from the geometric center of the building face and shall be considered for each principal axis (e_X, e_Y). The eccentricity e for flexible buildings shall be determined from the following equation and shall be considered for each principal axis (e_X, e_Y):

$$e = \frac{e_Q + 1.7I_{\bar{z}}\sqrt{(g_Q Q e_Q)^2 + (g_R R e_R)^2}}{1 + 1.7I_{\bar{z}}\sqrt{(g_Q Q)^2 + (g_R R)^2}} \qquad (27.3\text{-}4)$$

where

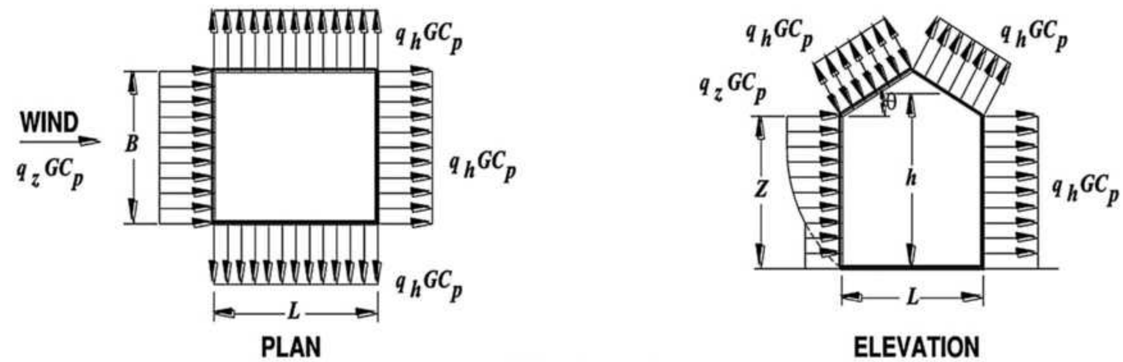
e_Q = eccentricity e as determined for rigid buildings in Fig. 27.3-8.

e_R = distance between the elastic shear center and center of mass of each floor.

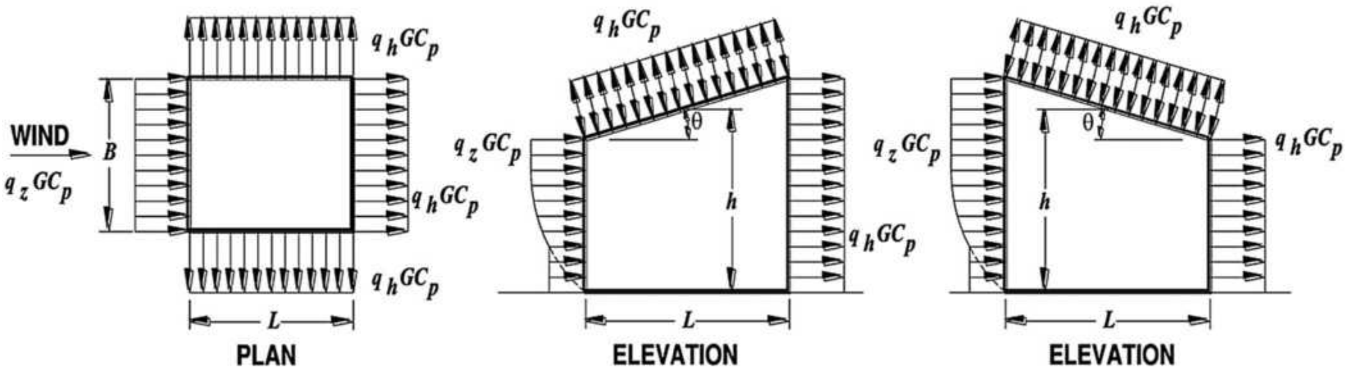
$I_{\bar{z}}$, g_Q, Q, g_R, and R shall be as defined in Section 26.11.

The sign of the eccentricity e shall be plus or minus, whichever causes the more severe load effect.

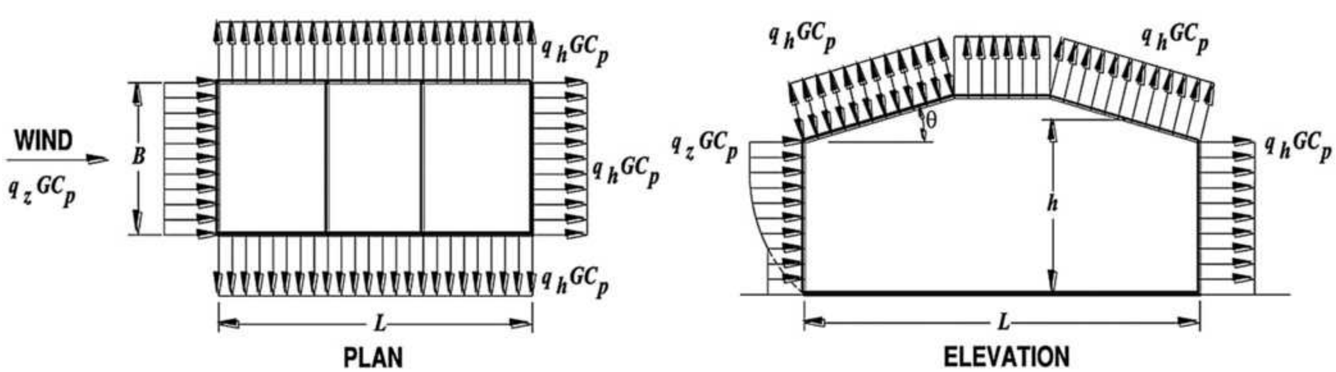
Diagrams



GABLE, HIP ROOF



MONOSLOPE ROOF (NOTE 4)



MANSARD ROOF (NOTE 7)

Notation

B = Horizontal dimension of building, in ft (m), measured normal to wind direction.
L = Horizontal dimension of building, in ft (m), measured parallel to wind direction.
h = Mean roof height, in ft (m), except that eave height shall be used for $\theta \leq 10$ degrees.
z = Height above ground, in ft (m).
G = Gust-effect factor.
q_z, q_h = Velocity pressure, in lb/ft^2 (N/m^2), evaluated at respective height.
θ = Angle of plane of roof from horizontal, in degrees.

FIGURE 27.3-1 Main Wind Force Resisting System, Part 1 (All Heights): External Pressure Coefficients, C_p, for Enclosed and Partially Enclosed Buildings—Walls and Roofs

continues

Wall Pressure Coefficients, C_p

Surface	L/B	C_p	Use With
Windward wall	All values	0.8	q_z
	0–1	−0.5	q_h
Leeward wall	2	−0.3	q_h
	≥4	−0.2	q_h
Sidewall	All values	−0.7	q_h

Roof Pressure Coefficients, C_p, for use with q_h

		Windward								Leeward		
		Angle, θ (degrees)								Angle, θ (degrees)		
Wind Direction	h/L	10	15	20	25	30	35	45	≥60[c]	10	15	≥20
Normal	≤0.25	−0.7	−0.5	−0.3	−0.2	−0.2	0.0[a]					
to Ridge		−0.18	0.0[a]	0.2	0.3	0.3	0.4	0.4	0.01 θ	−0.3	−0.5	−0.6
for	0.5	−0.9	−0.7	−0.4	−0.3	−0.2	−0.2	0.0[a]				
$\theta \geq 10°$		−0.18	−0.18	0.0[a]	0.2	0.2	0.3	0.4	0.01 θ	−0.5	−0.5	−0.6
	≥1.0	−1.3[b]	−1.0	−0.7	−0.5	−0.3	−0.2	0.0[a]				
		−0.18	−0.18	−0.18	0.0[a]	0.2	0.2	0.3	0.01 θ	−0.7	−0.6	−0.6

Wind Direction	h/L	Horizontal Distance from Windward Edge	C_p
Normal to Ridge for	≤0.5	0 to $h/2$	−0.9, −0.18
$\theta < 10°$ and Parallel		$h/2$ to h	−0.9, −0.18
to Ridge for All θ		h to $2h$	−0.5, −0.18
		$>2h$	−0.3, −0.18
	≥1.0	0 to $h/2$	−1.3[b], −0.18
		$>h/2$	−0.7, −0.18

[a]Value is provided for interpolation purposes.
[b]Value can be reduced linearly with area over which it is applicable as follows:
[c]For roof slopes greater than 80°, use $C_p = 0.8$.

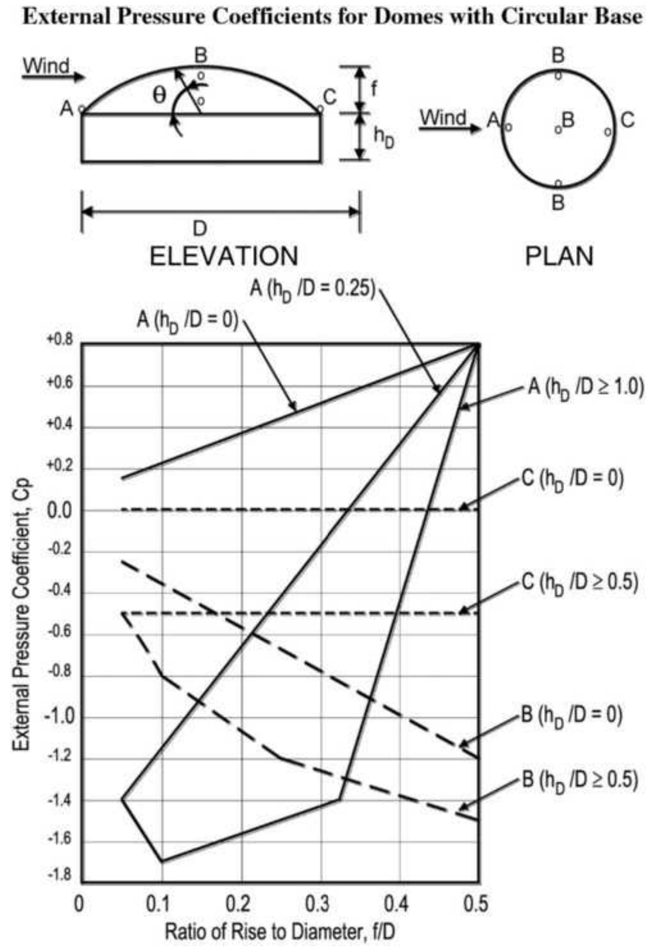
Area, ft^2	Area, m^2	Reduction Factor
≤100	≤9.3	1.0
250	23.2	0.9
≥1,000	≥92.9	0.8

Notes

1. Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
2. Linear interpolation is permitted for values of L/B, h/L, and θ other than shown. Interpolation shall only be carried out between values of the same sign. Where no value of the same sign is given, assume 0.0 for interpolation purposes.
3. Where two values of C_p are listed, this indicates that the windward roof slope is subjected to either positive or negative pressures and the roof structure shall be designed for both conditions. Interpolation for intermediate ratios of h/L in this case shall only be carried out between C_p values of like sign.
4. For monoslope roofs, entire roof surface is either a windward or leeward surface.
5. Refer to Fig. 27.3-2 for domes and Fig. 27.3-3 for arched roofs.
6. For mansard roofs, the top horizontal surface and leeward inclined surface shall be treated as leeward surfaces from the table.
7. Except for MWFRSs at the roof consisting of moment-resisting frames, the total horizontal shear shall not be less than that determined by neglecting wind forces on roof surfaces.

FIGURE 27.3-1 (*Coninued*). Main Wind Force Resisting System, Part 1 (All Heights): External Pressure Coefficients, C_p, for Enclosed and Partially Enclosed Buildings—Walls and Roofs

Diagrams



Notation

f = Dome rise, in ft (m).
h_D = Height to base of dome, in ft (m).
D = Diameter, in ft (m).
θ = Angle of plane of roof from horizontal, in degrees.

Notes

1. Two load cases shall be considered:
   Case A: C_p values between A and B and between B and C shall be determined by linear interpolation along arcs on the dome parallel to the wind direction;
   Case B: C_p shall be the constant value of A for $\theta \leq 25$ degrees and shall be determined by linear interpolation from 25 degrees to B and from B to C.
2. Values denote C_p to be used with $q_{(hD+f)}$ where h_{D+f} is the height at the top of the dome.
3. Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
4. C_p is constant on the dome surface for arcs of circles perpendicular to the wind direction; for example, the arc passing through B–B–B and all arcs parallel to B–B–B.
5. For values of $h_{D/D}$ between those listed on the graph curves, linear interpolation shall be permitted.
6. $\theta = 0$ degrees on dome springline, $\theta = 90$ degrees at dome center top point. f is measured from springline to top.
7. The total horizontal shear shall not be less than that determined by neglecting wind forces on roof surfaces.
8. For f/D values less than 0.05, use Fig. 27.3-1.

FIGURE 27.3-2 Main Wind Force Resisting System, Part 1 (All Heights): External Pressure Coefficients, C_p, for Enclosed and Partially Enclosed Buildings and Structures—Domed Roofs with a Circular Base

External Pressure Coefficient, C_p

Conditions	Rise-to-Span Ratio, r	C_p Windward Quarter	C_p Center Half	C_p Leeward Quarter
Roof on elevated structure	$0 < r < 0.2$	-0.9	$-0.7 - r$	-0.5
	$0.2 \le r < 0.3$[a]	$1.5r - 0.3$	$-0.7 - r$	0.5
	$0.3 \le r \le 0.6$	$2.75r - 0.7$	$-0.7 - r$	0.5
Roof springing from ground level	$0 < r \le 0.6$	$1.4r$	$-0.7 - r$	0.5

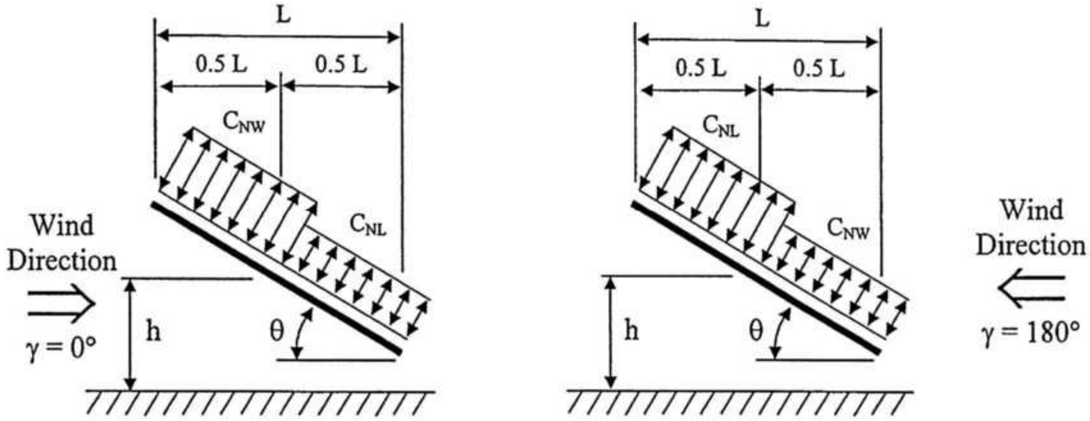
[a]When the rise-to-span ratio is $0.2 \le r \le 0.3$, alternate coefficients given by $6r - 2.1$ shall also be used for the windward quarter.

Notes

1. Values listed are for the determination of average loads on main wind-force resisting systems.
2. Plus and minus signs signify pressures acting toward and away from the surfaces, respectively.
3. For wind directed parallel to the axis of the arch, use pressure coefficients from Fig. 27.3-1 with wind directed parallel to ridge.
4. For components and cladding (1) at roof perimeter, use the external pressure coefficients in Fig. 30.3-2A, B, and C with θ based on springline slope and (2) for remaining roof areas, use external pressure coefficients of this table multiplied by 1.2.

FIGURE 27.3-3 Main Wind Force Resisting System and Components and Cladding, Part 1 (All Heights): External Pressure Coefficients, C_p, for Enclosed and Partially Enclosed Buildings and Structures—Arched Roofs

Diagrams



Notation

L = Horizontal dimension of roof, measured in the along-wind direction, ft (m).
h = Mean roof height, ft (m).
γ = Direction of wind, degrees.
θ = Angle of plane of roof from horizontal, degrees.

Net Pressure Coefficient, C_N

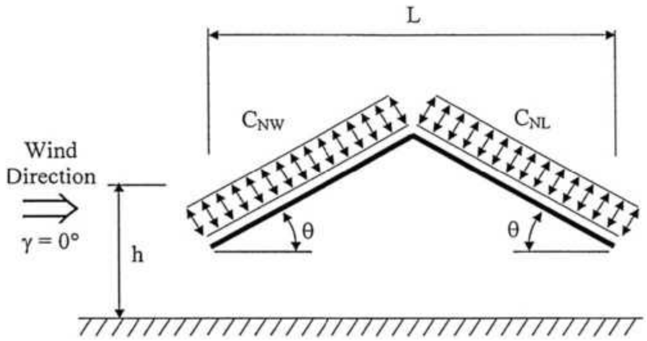
		Wind Direction, $\gamma = 0°$				Wind Direction, $\gamma = 180°$			
		Clear Wind Flow		Obstructed Wind Flow		Clear Wind Flow		Obstructed Wind Flow	
Roof Angle, θ	Load Case	C_{NW}	C_{NL}	C_{NW}	C_{NL}	C_{NW}	C_{NL}	C_{NW}	C_{NL}
0°	A	1.2	0.3	−0.5	−1.2	1.2	0.3	−0.5	−1.2
	B	−1.1	−0.1	−1.1	−0.6	−1.1	−0.1	−1.1	−0.6
7.5°	A	−0.6	−1.0	−1.0	−1.5	0.9	1.5	−0.2	−1.2
	B	−1.4	0.0	−1.7	−0.8	1.6	0.3	0.8	−0.3
15°	A	−0.9	−1.3	−1.1	−1.5	1.3	1.6	0.4	−1.1
	B	−1.9	0.0	−2.1	−0.6	1.8	0.6	1.2	−0.3
22.5°	A	−1.5	−1.6	−1.5	−1.7	1.7	1.8	0.5	−1.0
	B	−2.4	−0.3	−2.3	−0.9	2.2	0.7	1.3	0.0
30°	A	−1.8	−1.8	−1.5	−1.8	2.1	2.1	0.6	−1.0
	B	−2.5	−0.5	−2.3	−1.1	2.6	1.0	1.6	0.1
37.5°	A	−1.8	−1.8	−1.5	−1.8	2.1	2.2	0.7	−0.9
	B	−2.4	−0.6	−2.2	−1.1	2.7	1.1	1.9	0.3
45°	A	−1.6	−1.8	−1.3	−1.8	2.2	2.5	0.8	−0.9
	B	−2.3	−0.7	−1.9	−1.2	2.6	1.4	2.1	0.4

Notes

1. C_{NW} and C_{NL} denote net pressures (contributions from top and bottom surfaces) for windward and leeward half of roof surfaces, respectively.
2. Clear wind flow denotes relatively unobstructed wind flow with blockage less than or equal to 50%. Obstructed wind flow denotes objects below roof inhibiting wind flow (>50% blockage).
3. For values of θ between 7.5° and 45°, linear interpolation is permitted. For values of θ less than 7.5°, use load coefficients for 0°.
4. Plus and minus signs signify pressures acting toward and away from the top roof surface, respectively.
5. All load cases shown for each roof angle shall be investigated.

FIGURE 27.3-4 Main Wind Force Resisting System, Part 1 ($0.25 \leq h/L \leq 1.0$): Net Pressure Coefficient, C_N, for Open Buildings with Monoslope Free Roofs, $\theta \leq 45°$, $\gamma = 0°$, 180°)

Diagram



Notation

L = Horizontal dimension of roof, measured in the along-wind direction, ft (m).
h = Mean roof height, ft (m).
γ = Direction of wind, degrees.
θ = Angle of plane of roof from horizontal, degrees.

Net Pressure Coefficient, C_N

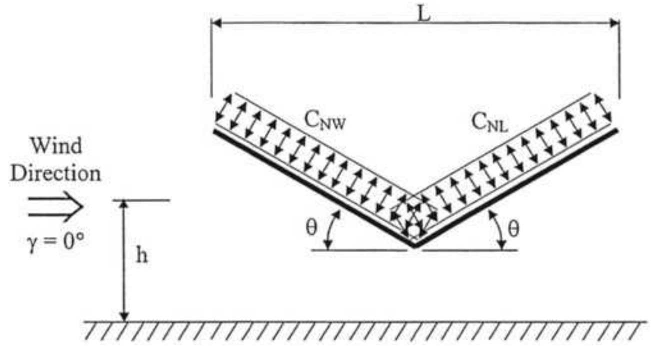
		Wind Direction, $\gamma = 0°$, 180°			
		Clear Wind Flow		Obstructed Wind Flow	
Roof Angle, θ	Load Case	C_{NW}	C_{NL}	C_{NW}	C_{NL}
7.5°	A	1.1	−0.3	−1.6	−1.0
	B	0.2	−1.2	−0.9	−1.7
15°	A	1.1	−0.4	−1.2	−1.0
	B	0.1	−1.1	−0.6	−1.6
22.5°	A	1.1	0.1	−1.2	−1.2
	B	−0.1	−0.8	−0.8	−1.7
30°	A	1.3	0.3	−0.7	−0.7
	B	−0.1	−0.9	−0.2	−1.1
37.5°	A	1.3	0.6	−0.6	−0.6
	B	−0.2	−0.6	−0.3	−0.9
45°	A	1.1	0.9	−0.5	−0.5
	B	−0.3	−0.5	−0.3	−0.7

Notes

1. C_{NW} and C_{NL} denote net pressures (contributions from top and bottom surfaces) for windward and leeward half of roof surfaces, respectively.
2. Clear wind flow denotes relatively unobstructed wind flow with blockage less than or equal to 50%. Obstructed wind flow denotes objects below roof inhibiting wind flow (>50% blockage).
3. For values of θ between 7.5° and 45°, linear interpolation is permitted. For values of θ less than 7.5°, use monoslope roof load coefficients.
4. Plus and minus signs signify pressures acting toward and away from the top roof surface, respectively.
5. All load cases shown for each roof angle shall be investigated.

FIGURE 27.3-5 Main Wind Force Resisting System, Part 1 ($0.25 \leq h/L \leq 1.0$): Net Pressure Coefficient, C_N, for Open Buildings with Pitched Free Roofs, $\theta \leq 45°$, $\gamma = 0°$, 180°

Diagram



Notation

L = Horizontal dimension of roof, measured in the along-wind direction, ft (m).
h = Mean roof height, ft (m).
γ = Direction of wind, degrees.
θ = Angle of plane of roof from horizontal, degrees.

Net Pressure Coefficient, C_N

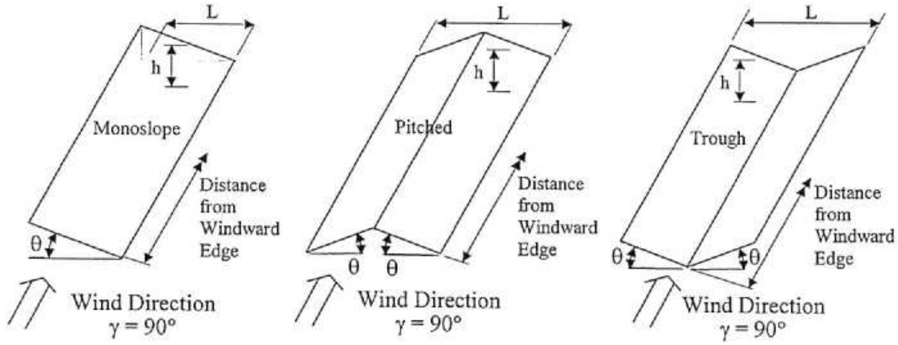
		Wind Direction, $\gamma = 0°$, 180°			
		Clear Wind Flow		Obstructed Wind Flow	
Roof Angle, θ	Load Case	C_{NW}	C_{NL}	C_{NW}	C_{NL}
7.5°	A	−1.1	0.3	−1.6	−0.5
	B	−0.2	1.2	−0.9	−0.8
15°	A	−1.1	0.4	−1.2	−0.5
	B	0.1	1.1	−0.6	−0.8
22.5°	A	−1.1	−0.1	−1.2	−0.6
	B	−0.1	0.8	−0.8	−0.8
30°	A	−1.3	−0.3	−1.4	−0.4
	B	−0.1	0.9	−0.2	−0.5
37.5°	A	−1.3	−0.6	−1.4	−0.3
	B	0.2	0.6	−0.3	−0.4
45°	A	−1.1	−0.9	−1.2	−0.3
	B	0.3	0.5	−0.3	−0.4

Notes

1. C_{NW} and C_{NL} denote net pressures (contributions from top and bottom surfaces) for windward and leeward half of roof surfaces, respectively.
2. Clear wind flow denotes relatively unobstructed wind flow with blockage less than or equal to 50%. Obstructed wind flow denotes objects below roof inhibiting wind flow (>50% blockage).
3. For values of θ between 7.5° and 45°, linear interpolation is permitted. For values of θ less than 7.5°, use monoslope roof load coefficients.
4. Plus and minus signs signify pressures acting toward and away from the top roof surface, respectively.
5. All load cases shown for each roof angle shall be investigated.

FIGURE 27.3-6 Main Wind Force Resisting System, Part 1 ($0.25 \le h/L \le 1.0$): Net Pressure Coefficient, C_N, for Open Buildings with Troughed Free Roofs, $\theta \le 45°$, $\gamma = 0°$, 180°

Diagrams



Notation

L = Horizontal dimension of roof, measured in the along-wind direction, ft (m).
h = Mean roof height, ft (m). See Figs. 27.3-4, 27.3-5, or 27.3-6 for a graphical depiction of this dimension.
γ = Direction of wind, degrees.
θ = Angle of plane of roof from horizontal, degrees.

Net Pressure Coefficient, C_N

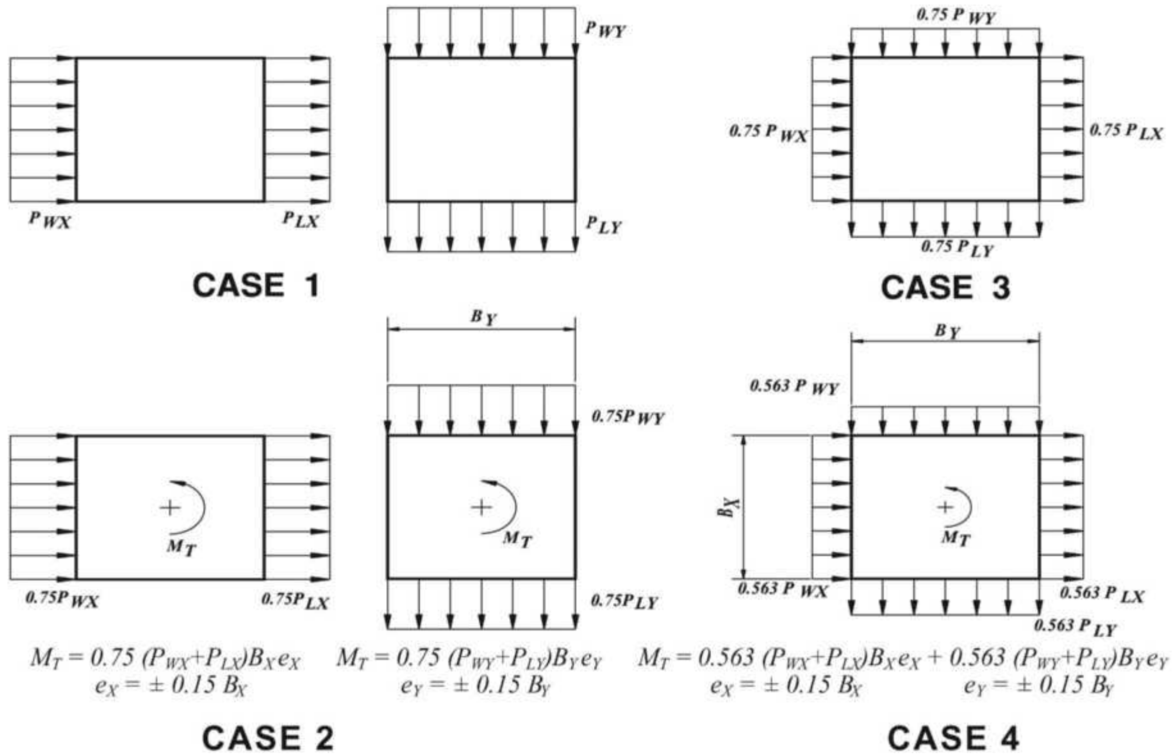
Horizontal Distance from Windward Edge	Roof Angle θ	Load Case	Clear Wind Flow C_N	Obstructed Wind Flow C_N
$<h$	All shapes	A	−0.8	−1.2
	$\theta < 45°$	B	0.8	0.5
$>h, <2h$	All shapes	A	−0.6	−0.9
	$\theta < 45°$	B	0.5	0.5
$>2h$	All shapes	A	−0.3	−0.6
	$\theta < 45°$	B	0.3	0.3

Notes

1. C_N denotes net pressures (contributions from top and bottom surfaces).
2. Clear wind flow denotes relatively unobstructed wind flow with blockage less than or equal to 50%. Obstructed wind flow denotes objects below roof inhibiting wind flow (>50% blockage).
3. Plus and minus signs signify pressures acting toward and away from the top roof surface, respectively.
4. All load cases shown for each roof angle shall be investigated.
5. For monoslope roofs with θ less than 5 degrees, C_N values shown apply also for cases where $\gamma = 0$ degrees and 0.05 less than or equal to h/L less than or equal to 0.25. See Fig. 27.3-4 for other h/L values.

FIGURE 27.3-7 Main Wind Force Resisting System, Part 1 ($0.25 \le h/L \le 1.0$): Net Pressure Coefficient, C_N, for Open Buildings with Free Roofs, $\theta \le 45°$, $\gamma = 90°$, 270°

Diagrams



Notation

P_{WX}, P_{WY} = Windward face design pressure acting in the *x*, *y* principal axis, respectively.
P_{LX}, P_{LY} = Leeward face design pressure acting in the *x*, *y* principal axis, respectively.
$e(e_X, e_Y)$ = Eccentricity for the *x*, *y* principal axis of the structure, respectively.
M_T = Torsional moment per unit height acting about a vertical axis of the building.

Case 1. Full design wind pressure acting on the projected area perpendicular to each principal axis of the structure, considered separately along each principal axis.
Case 2. Three-quarters of the design wind pressure acting on the projected area perpendicular to each principal axis of the structure in conjunction with a torsional moment as shown, considered separately for each principal axis.
Case 3. Wind loading as defined in Case 1, but considered to act simultaneously at 75% of the specified value.
Case 4. Wind loading as defined in Case 2, but considered to act simultaneously at 75% of the specified value.

Notes

1. Design wind pressures for windward and leeward faces shall be determined in accordance with the provisions of Sections 27.3.1 and 27.3.2 as applicable for buildings of all heights.
2. Diagrams show plan views of buildings.

FIGURE 27.3-8 Main Wind Force Resisting System, Part 1 (All Heights): Design Wind Load Cases

PART 2: ENCLOSED SIMPLE DIAPHRAGM BUILDINGS WITH $h \leq 160$ ft ($h \leq 48.8$ m)

User Note: Part 2 of Chapter 27 is a simplified method for determining the wind pressures for the MWFRS of enclosed, simple diaphragm buildings the height h of which is ≤160 ft (48.8 m). The wind pressures are obtained *directly from a table.* The building may be of any general plan shape and roof geometry that matches the specified figures. This method is a simplification of the traditional "all heights" method (Directional Procedure) contained in Part 1 of Chapter 27.

27.4 GENERAL REQUIREMENTS

27.4.1 Design Procedure. The procedure specified herein applies to the determination of MWFRS wind loads of enclosed simple diaphragm buildings, as defined in Section 26.2, with a mean roof height $h \leq 160$ ft ($h \leq 48.8$ m). The steps required for the determination of MWFRS wind loads on enclosed simple diaphragm buildings are shown in Table 27.4-1.

27.4.2 Conditions. In addition to the requirements in Section 27.1.2, a building that has design wind loads determined in accordance with this section shall meet all of the following conditions for either a Class 1 or Class 2 building (Fig. 27.4-1):

Class 1 Buildings:

1. The building shall be an enclosed simple diaphragm building as defined in Section 26.2.
2. The building shall have a mean roof height $h \leq 60$ ft ($h \leq 18.3$ m).
3. The ratio of L/B shall not be less than 0.2 nor more than 5.0 ($0.2 \leq L/B \leq 5.0$).

Class 2 Buildings:

1. The building shall be an enclosed simple diaphragm building as defined in Section 26.2.
2. The building shall have a mean roof height 60 ft $< h \leq$ 160 ft (18.3 m $< h \leq$ 48.8 m).
3. The ratio of L/B shall not be less than 0.5 nor more than 2.0 ($0.5 \leq L/B \leq 2.0$).
4. The fundamental natural frequency (hertz) of the building shall not be less $75/h$ where h is in feet.

27.4.3 Wind Load Parameters Specified in Chapter 26. Refer to Chapter 26 for determination of basic wind speed, V (Section 26.5), exposure category (Section 26.7), and topographic factor K_{zt} (Section 26.10).

27.4.4 Topographic Effects. The wind pressures determined from this section shall be multiplied by K_{zt} as determined from Section 26.10 using one value of K_{zt} for the building calculated at $0.33h$. Alternatively, it shall be permitted to enter Tables 27.5-1 and 27.5-2 with a wind velocity equal to $V\sqrt{K_{zt}}$ where K_{zt} is determined at a height of $0.33h$.

27.4.5 Diaphragm Flexibility. The design procedure specified herein applies to buildings that have either rigid or flexible diaphragms. The structural analysis shall consider the relative stiffness of diaphragms and the vertical elements of the MWFRS.

Diaphragms constructed of wood panels can be idealized as flexible. Diaphragms constructed of untopped metal decks, concrete-filled metal decks, and concrete slabs, each having a span-to-depth ratio of 2 or less, are permitted to be idealized as rigid for consideration of wind loading.

Table 27.4-1 Steps to Determine MWFRS Wind Loads for Enclosed, Simple Diaphragm Buildings, $h \leq 160$ ft ($h \leq 48.8$ m)

Step 1: Determine Risk Category of building; see Table 1.5-1.
Step 2: Determine the basic wind speed, V, for applicable Risk Category; see Figs. 26.5-1 and 26.5-2.
Step 3: Determine wind load parameters:
- Exposure category B, C, or D; see Section 26.7.
- Topographic factor, K_{zt}; see Section 26.8 and Fig. 26.8-1.
- Enclosure classification; see Section 26.12.

Step 4: Enter table to determine net pressures on walls at top and base of building respectively, p_h, p_0, Table 27.5-1.
Step 5: Enter table to determine net roof pressures, p_z, Table 27.5-2.
Step 6: Determine topographic factor, K_{zt}, and apply factor to wall and roof pressures (if applicable); see Section 26.8.
Step 7: Apply loads to walls and roofs simultaneously.

27.5 WIND LOADS: MAIN WIND FORCE RESISTING SYSTEM

27.5.1 Wall and Roof Surfaces: Class 1 and 2 Buildings. Net wind pressures for the walls and roof surfaces shall be determined from Tables 27.5-1 and 27.5-2, respectively, for the applicable exposure category as determined by Section 26.7.

For Class 1 buildings with L/B values less than 0.5, use wind pressures tabulated for $L/B = 0.5$. For Class 1 buildings with L/B values greater than 2.0, use wind pressures tabulated for $L/B = 2.0$.

Net wall pressures shall be applied to the projected area of the building walls in the direction of the wind, and exterior sidewall pressures shall be applied to the projected area of the building walls normal to the direction of the wind acting outward according to Note 3 of Table 27.5-1, simultaneously with the roof pressures from Table 27.5-2, as shown in Fig. 27.5-1.

Where two load cases are shown in the table of roof pressures, the effects of each load case shall be investigated separately. The MWFRS in each direction shall be designed for the wind load cases as defined in Fig. 27.3-8.

EXCEPTION: The torsional load cases in Fig. 27.3-8 (Case 2 and Case 4) need not be considered for buildings that meet the requirements of Appendix D.

27.5.2 Parapets. The effect of horizontal wind loads applied to all vertical surfaces of roof parapets for the design of the MWFRS shall be based on the application of an additional net horizontal wind pressure applied to the projected area of the parapet surface equal to 2.25 times the wall pressures tabulated in Table 27.5-1 for $L/B = 1.0$. The net pressure specified accounts for both the windward and leeward parapet loading on both the windward and leeward building surface. The parapet pressure shall be applied simultaneously with the specified wall and roof pressures shown in the table in

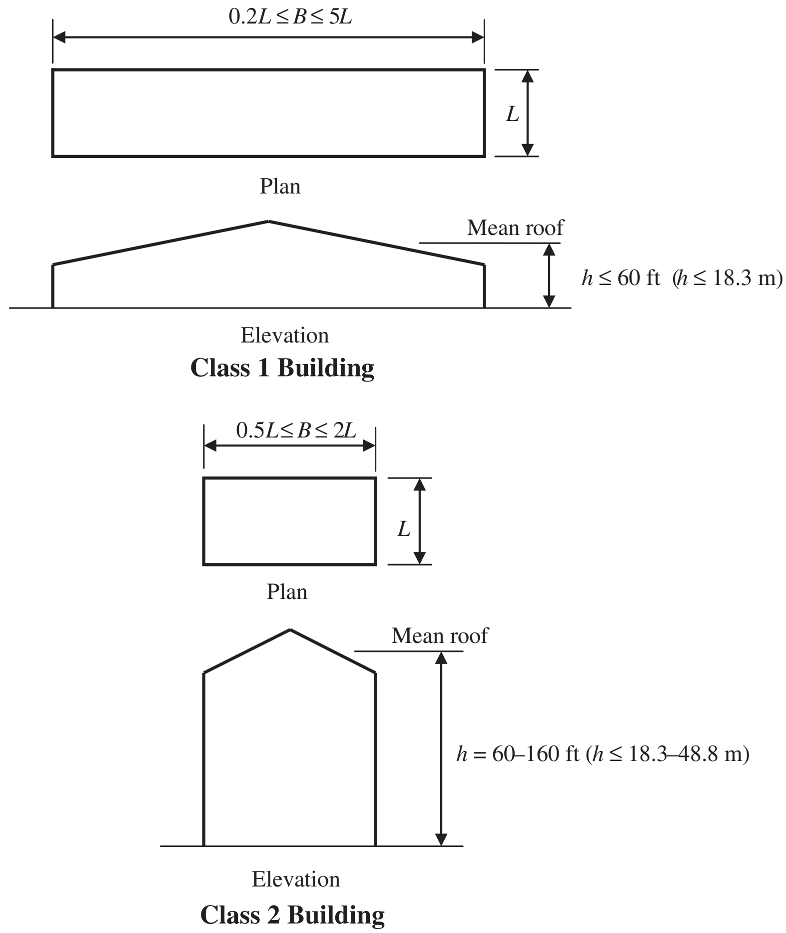
Fig. 27.5-2. The height h used in Table 27.5-1 to determine the parapet pressure shall be the height to the top of the parapet as shown in Fig. 27.5-2 (use $h = h_p$).

27.5.3 Roof Overhangs. The effect of vertical wind loads on any roof overhangs shall be based on the application of a positive wind pressure on the underside of the windward overhang equal to 75% of the roof edge pressure from Table 27.5-2 for Zone 1 or Zone 3, as applicable. This pressure shall be applied to the windward roof overhang only and shall be applied simultaneously with other tabulated wall and roof pressures, as shown in Fig. 27.5-3.

27.6 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

No consensus standards and other documents that shall be considered part of this standard are referenced in this chapter.

Diagrams



Notation

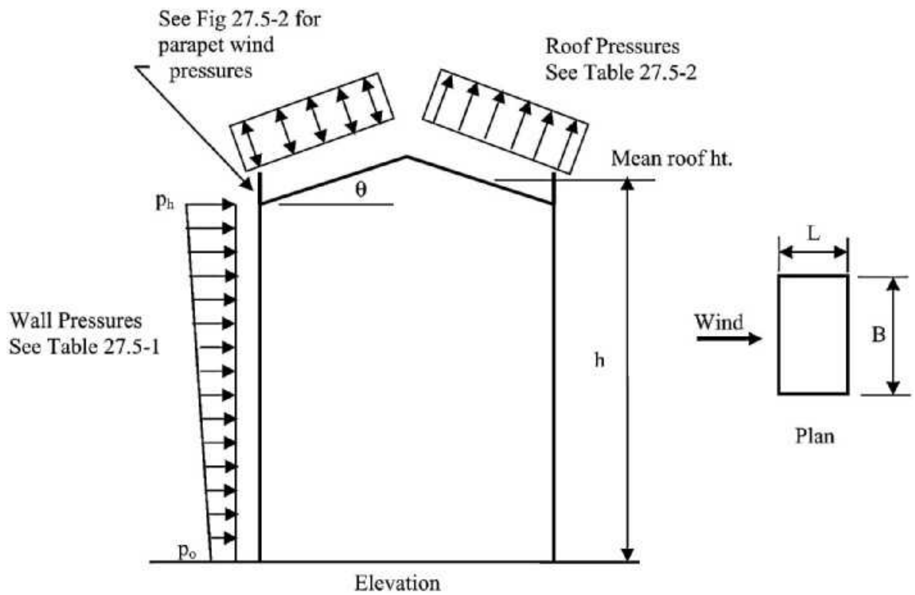
B = Horizontal dimension of building, in ft (m), measured normal to wind direction.
L = Horizontal dimension of building, in ft (m), measured parallel to wind direction.
h = Mean roof height, in ft (m), except that eave height shall be used for $\theta \leq 10$ degrees.

Note

Roof form may be flat, gable, mansard, or hip.

FIGURE 27.4-1 Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Building Class for Enclosed Simple Diaphragm Buildings (Building Geometry Requirements)

Diagram

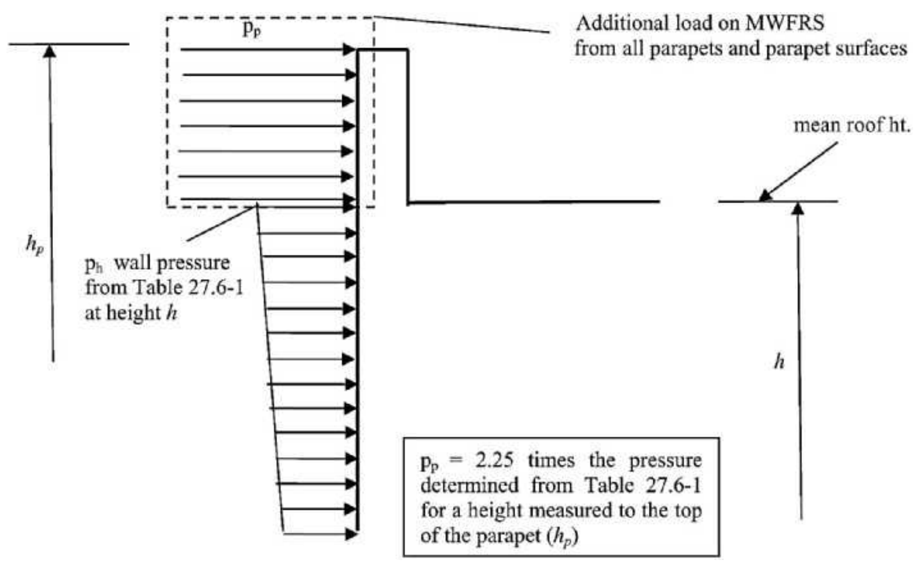


Note

For Application of Wind Pressures, see Tables 27.5-1 and 27.5-2

FIGURE 27.5-1 Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings, Wind Pressures, Walls and Roof

Diagram

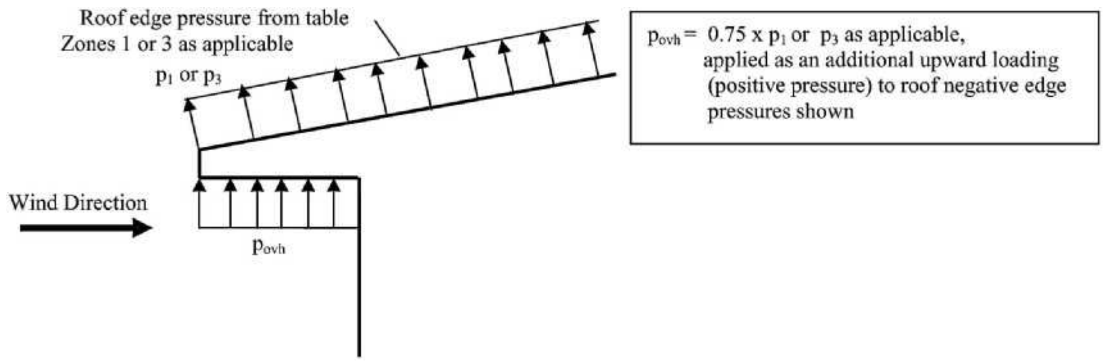


Note

For Application of Parapet Wind Loads, see Table 27.5-1

FIGURE 27.5-2 Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings, Parapet Wind Loads.

Diagram



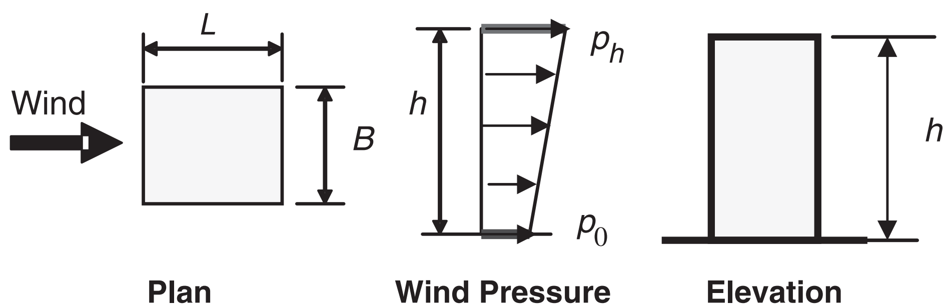
Note

For Application of Roof Overhang Wind Loads, see Table 27.5-1

FIGURE 27.5-3 Main Wind Force Resisting System, Part 2: Enclosed Simple Diaphragm Buildings, Roof Overhang Wind Loads

Table 27.5-1 Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Walls

Parameters for Application of Wall Pressures



Notation

L = Building plan dimension parallel to wind direction, ft (m).
B = Building plan dimension perpendicular to wind direction, ft (m).
h = Mean roof height, ft (m).
p_h, p_0 = Along-wind net wall pressure at top and base of building, respectively, psf (kN/m^2).

Notes for Wall Pressure Tables

1. From table for each Exposure (B, C or D), V, L/B, and h, determine p_h (top number) and p_0 (bottom number) horizontal along-wind net wall pressures.
2. Sidewall external pressures shall be uniform over the wall surface acting outward and shall be taken as 54% of the tabulated p_h pressure for $0.2 \leq L/B \leq 1.0$ and 64% of the tabulated p_h pressure for $2.0 \leq L/B \leq 5.0$. Linear interpolation shall apply for $1.0 < L/B < 2.0$. Sidewall external pressures do not include effect of internal pressure.
3. Apply along-wind net wall pressures as shown above to the projected area of the building walls in the direction of the wind and apply external sidewall pressures to the projected area of the building walls normal to the direction wind, simultaneously with the roof pressures from Table 27.5-2.
4. Distribution of tabulated net wall pressures between windward and leeward wall faces shall be based on the linear distribution of total net pressure with building height as shown above and the leeward external wall pressures assumed uniformly distributed over the leeward wall surface acting outward at 38% of p_h for $0.2 \leq L/B \leq 1.0$ and 27% of p_h for $2.0 \leq L/B \leq 5.0$. Linear interpolation shall be used for $1.0 < L/B < 2.0$. The remaining net pressure shall be applied to the windward walls as an external wall pressure acting toward the wall surface. Windward and leeward wall pressures so determined do not include effect of internal pressure.
5. Interpolation between values of V, h, and L/B is permitted.
6. 1.0 ft = 0.3048 m; 1.0 lb/ft^2 = 0.0479 kN/m^2.

continues

Table 27.5-1 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Walls

Exposure B

		V (mi/h)																							
		110			115			120			130			140			160			180			200		
		L/B			L/B			L/B			L/B			L/B			L/B			L/B			L/B		
h (ft)	Along-wind Net Wall Pressure	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2
160	p_h	38.1	37.7	34.1	42.1	41.7	37.8	46.4	45.9	41.7	55.8	55.1	50.2	66.3	65.4	59.7	91.0	89.4	81.8	120.8	118.3	108.5	156.2	152.4	140.0
	p_0	25.6	25.4	21.0	28.3	28.1	23.3	31.2	30.9	25.7	37.5	37.1	30.9	44.6	44.0	36.8	61.2	60.1	50.4	81.3	79.6	66.9	105.2	102.6	86.2
150	p_h	36.9	36.6	33.0	40.7	40.4	36.5	44.9	44.4	40.3	53.9	53.3	48.5	63.9	63.1	57.6	87.5	86.1	78.9	116.1	113.8	104.5	149.9	146.5	134.7
	p_0	25.1	24.9	20.6	27.7	27.5	22.8	30.5	30.2	25.2	36.7	36.2	30.3	43.5	43.0	36.0	59.6	58.6	49.3	79.0	77.4	65.3	102.0	99.7	84.2
140	p_h	36.6	35.4	31.9	39.3	39.1	35.3	43.3	42.9	38.9	51.9	51.4	46.7	61.5	60.8	55.5	84.0	82.8	75.9	111.2	109.2	100.4	143.5	140.5	129.3
	p_0	24.5	24.4	20.2	27.1	26.9	22.4	29.8	29.6	24.6	35.7	35.4	29.6	42.4	41.9	35.2	57.9	57.0	48.1	76.6	75.2	63.7	98.8	96.7	82.0
130	p_h	34.4	34.2	30.8	37.9	37.7	34.0	41.7	41.4	37.4	49.9	49.5	44.9	59.1	58.5	53.3	80.5	79.5	72.8	106.3	104.6	96.2	136.9	134.3	123.8
	p_0	24.0	23.9	19.8	26.5	26.3	21.9	29.1	28.9	24.1	34.8	34.5	28.9	41.2	40.8	34.3	56.2	55.4	46.9	74.2	73.0	62.0	95.5	93.7	79.8
120	p_h	33.1	33.0	29.6	36.5	36.3	32.7	40.1	39.9	35.9	47.9	47.6	43.1	56.6	56.2	51.0	76.9	76.1	69.6	101.3	99.9	91.8	130.2	128.0	118.0
	p_0	23.4	23.3	19.4	25.8	25.7	21.4	28.4	28.2	23.6	33.9	33.7	28.3	40.1	39.7	33.5	54.4	53.8	45.6	71.7	70.7	60.2	92.2	90.6	77.4
110	p_h	31.8	31.7	28.4	35.1	34.9	31.3	38.5	38.3	34.4	45.9	45.6	41.2	54.1	53.8	48.8	73.3	72.6	66.3	96.3	95.1	87.4	123.5	121.6	112.1
	p_0	22.9	22.8	19.0	25.2	25.1	20.9	27.7	27.5	23.0	33.0	32.8	27.6	38.9	38.7	32.6	52.7	52.2	44.4	69.2	68.4	58.4	88.8	87.4	75.0
100	p_h	30.5	30.4	27.1	33.6	33.5	29.9	36.8	36.7	32.9	43.8	43.6	39.3	51.6	51.3	46.4	69.6	69.1	62.9	91.2	90.3	82.8	116.6	115.1	106.0
	p_0	22.3	22.3	18.5	24.6	24.5	20.4	26.9	26.8	22.5	32.1	31.9	26.8	37.8	37.6	31.7	50.9	50.5	43.0	66.7	66.0	56.6	85.3	84.2	72.5
90	p_h	29.2	29.1	25.9	32.1	32.0	28.5	35.1	35.0	31.2	44.7	41.6	37.3	49.1	48.8	44.0	65.9	65.5	59.5	86.0	85.3	78.0	109.6	108.5	99.8
	p_0	21.8	21.7	18.1	23.9	23.9	19.9	26.2	26.1	21.9	31.1	31.0	26.1	36.6	36.4	30.8	49.2	48.9	41.7	64.2	63.6	54.6	81.8	80.9	69.9
80	p_h	27.8	27.7	24.5	30.5	30.5	27.0	33.4	33.3	29.6	39.6	39.5	35.2	46.4	46.3	41.5	62.2	61.9	55.9	80.8	80.3	73.1	102.6	101.7	93.3
	p_0	21.2	21.2	17.7	23.3	23.2	19.4	25.5	25.4	21.3	30.2	30.1	25.4	35.4	35.3	29.9	47.4	47.2	40.3	61.6	61.2	52.6	78.3	77.6	67.2
70	p_h	26.3	26.3	23.1	28.9	28.8	25.4	31.6	31.5	27.9	37.4	37.3	33.1	43.7	43.6	38.9	58.3	58.1	52.2	75.5	75.1	68.1	95.5	94.9	86.6
	p_0	20.6	20.6	17.2	22.6	22.6	18.9	24.7	24.7	20.7	29.3	29.2	24.6	34.2	34.2	28.9	45.6	45.5	38.8	59.1	58.8	50.6	74.7	74.3	64.3
60	p_h	24.8	24.8	21.7	27.2	27.1	23.8	29.7	29.6	26.1	35.1	35.0	30.9	41.0	40.9	36.2	54.4	54.2	48.4	70.1	69.8	62.8	88.2	87.9	79.6
	p_0	20.0	20.0	16.7	21.9	21.9	18.4	23.9	23.9	20.1	28.3	28.2	23.6	33.0	33.0	27.9	43.9	43.8	37.3	56.5	56.3	48.5	71.2	70.9	61.4
50	p_h	23.1	23.1	20.2	25.3	25.3	22.1	27.6	27.6	24.2	32.6	32.6	28.6	38.0	38.0	33.4	50.3	50.2	44.5	64.5	64.4	57.4	80.9	80.7	72.5
	p_0	19.3	19.3	16.3	21.2	21.2	17.8	23.1	23.1	19.5	27.3	27.3	23.0	31.8	31.8	26.9	42.0	42.0	35.8	54.0	53.8	46.3	67.6	67.5	58.4
40	p_h	21.5	21.5	18.6	23.5	23.5	20.4	25.6	25.6	22.3	30.2	30.2	26.3	35.1	35.1	30.7	46.3	46.2	40.7	59.2	59.1	52.3	73.9	73.8	65.7
	p_0	18.8	18.7	15.8	20.5	20.5	17.4	22.4	22.4	18.9	26.4	26.4	22.4	30.7	30.7	26.1	40.5	40.4	34.6	51.7	51.7	44.5	64.6	64.5	55.8
30	p_h	19.6	19.6	16.9	21.4	21.4	18.5	23.3	23.3	20.2	27.5	27.4	23.8	31.9	31.9	27.7	44.9	41.9	36.6	53.4	53.4	46.8	66.5	66.4	58.5
	p_0	18.1	18.1	15.4	19.8	19.8	16.8	21.5	21.5	18.4	25.3	25.3	21.6	29.5	29.5	25.2	38.7	38.7	33.2	49.3	49.3	42.5	61.4	61.3	53.1
20	p_h	17.5	17.5	15.1	19.2	19.2	16.6	20.9	20.9	18.1	24.5	24.5	21.2	28.5	28.5	24.7	37.3	37.3	32.4	47.4	47.4	41.3	58.8	58.8	51.4
	p_0	17.2	17.2	14.8	18.8	18.8	16.2	20.5	20.5	17.7	24.1	24.1	20.8	28.0	28.0	24.2	36.7	36.7	31.7	46.6	46.6	40.4	57.8	57.7	50.3
15	p_h	16.7	16.7	14.5	18.2	18.2	15.8	19.9	19.9	17.3	23.3	23.3	20.3	27.1	27.1	23.6	35.4	35.4	30.9	44.9	44.9	39.3	55.6	55.6	48.7
	p_0	16.7	16.7	14.5	18.2	18.2	15.8	19.9	19.9	17.3	23.3	23.3	20.3	27.1	27.1	23.6	35.4	35.4	30.9	44.9	44.9	39.3	55.6	55.6	48.7

continues

Table 27.5-1 *(Continued)*. Main Wind Force Resisting System, Part 2 [$h \le 160$ ft ($h \le 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Walls

Exposure C

h (ft)	Along-wind Net Wall Pressure	V (mi/h)																							
		110			115			120			130			140			160			180			200		
		L/B			L/B			L/B			L/B			L/B			L/B			L/B			L/B		
		0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2
160	p_h	49.2	48.7	43.7	54.5	53.8	48.3	60.0	59.3	53.3	72.2	71.1	64.1	85.8	84.3	76.1	117.4	115.0	103.9	155.4	151.8	137.2	200.2	195.0	176.2
	p_0	36.1	35.7	30.0	40.0	39.5	33.2	44.1	43.5	36.6	53.0	52.2	44.0	62.9	61.9	52.3	86.2	84.4	71.5	114.1	111.4	94.3	146.9	143.1	121.1
150	p_h	48.0	47.5	42.6	53.0	52.4	47.1	58.4	57.7	51.9	70.1	69.2	62.3	83.3	82.0	74.0	113.8	111.7	101.0	150.6	147.3	133.3	198.8	189.0	171.0
	p_0	35.5	35.2	29.6	39.3	38.8	32.7	43.3	42.8	36.1	52.0	51.3	43.3	61.7	60.7	51.4	84.3	82.8	70.2	111.5	109.1	92.7	143.5	140.0	118.9
140	p_h	46.6	46.2	41.4	51.5	51.0	45.8	56.7	56.1	50.4	68.1	67.2	60.6	80.7	79.6	71.8	110.2	108.3	98.0	145.6	142.6	129.2	187.2	182.9	165.7
	p_0	34.9	34.6	29.1	38.6	38.2	32.2	42.4	42.0	35.5	50.9	50.3	42.6	60.4	59.5	50.6	82.4	81.0	68.9	108.9	106.7	90.9	140.0	136.8	116.6
130	p_h	45.3	45.0	40.2	50.0	49.6	44.5	55.0	54.5	48.9	65.9	65.2	58.7	78.1	77.1	69.6	106.4	104.7	94.8	140.4	137.7	124.9	180.4	176.5	160.1
	p_0	34.3	34.0	28.7	37.8	37.5	31.7	41.6	41.2	34.9	49.9	49.3	41.9	59.1	58.3	49.6	80.5	79.2	67.6	106.2	104.1	89.1	136.4	133.4	114.2
120	p_h	43.9	43.6	39.0	48.5	48.1	43.1	53.3	52.8	47.4	63.8	63.1	56.8	75.4	74.6	67.3	102.6	101.1	91.5	135.1	132.7	120.5	173.3	169.8	154.3
	p_0	33.6	33.4	28.2	37.1	36.8	31.1	40.7	40.4	34.3	48.8	48.3	41.1	57.7	57.1	48.7	78.5	77.3	66.2	103.3	101.5	87.1	132.6	129.9	111.6
110	p_h	42.5	42.3	37.7	46.9	46.6	41.6	51.5	51.1	45.8	61.5	61.0	54.8	72.7	72.0	64.8	98.6	97.3	88.1	129.6	127.6	115.8	166.0	163.0	148.2
	p_0	32.9	32.8	27.7	36.3	36.1	30.6	39.9	39.6	33.6	47.7	47.3	40.3	56.3	55.8	47.6	76.4	75.4	64.7	100.4	98.8	85.1	128.6	126.3	108.9
100	p_h	41.1	40.9	36.4	45.2	45.0	40.1	49.6	49.3	44.1	59.2	58.8	52.7	69.8	69.3	62.3	94.5	93.5	84.5	123.9	122.2	111.0	158.5	155.9	141.9
	p_0	32.3	32.1	27.2	35.5	35.4	30.0	39.0	38.8	33.0	46.5	46.2	39.4	54.9	54.4	46.6	74.2	73.4	63.2	97.4	96.0	82.9	124.5	122.5	106.1
90	p_h	39.6	39.4	35.0	43.5	43.3	38.5	47.7	47.5	42.3	56.8	56.5	50.6	66.9	66.5	59.7	90.3	89.4	80.8	118.1	116.7	105.9	150.6	148.5	135.2
	p_0	31.6	31.5	26.6	34.7	34.6	29.4	38.1	37.9	32.3	45.4	45.1	38.5	53.4	53.1	45.5	72.1	71.4	61.6	94.2	93.2	80.7	120.3	118.6	103.0
80	p_h	38.0	37.9	33.5	41.8	41.6	36.9	45.8	45.6	40.5	54.4	54.2	48.3	63.9	63.6	56.9	85.9	85.3	76.8	112.0	111.0	100.5	142.6	140.9	128.1
	p_0	30.9	30.8	26.1	33.9	33.8	28.7	37.2	37.1	31.5	44.2	44.0	37.6	52.0	51.7	44.3	69.8	69.3	59.8	91.0	90.2	78.3	115.8	114.5	99.8
70	p_h	36.4	36.3	32.0	39.9	39.9	35.2	43.7	43.6	38.6	51.9	51.7	45.9	60.8	60.6	54.0	81.4	81.0	72.7	105.8	105.0	94.9	134.2	133.0	120.7
	p_0	30.2	30.1	25.5	33.1	33.1	28.1	36.3	36.2	30.8	43.0	42.9	36.6	50.5	50.3	43.1	67.5	67.2	58.0	87.8	87.1	75.7	111.3	110.3	96.3
60	p_h	34.6	34.6	30.3	38.0	38.0	33.3	41.6	41.5	36.5	49.2	49.1	43.4	57.6	57.4	50.9	76.8	76.5	68.3	99.4	98.8	88.9	125.6	124.7	112.8
	p_0	29.4	29.4	24.9	32.3	32.2	27.4	35.3	35.2	30.0	41.8	41.7	35.6	48.9	48.8	41.9	65.2	65.0	56.1	84.4	83.9	73.0	106.7	105.9	92.7
50	p_h	32.8	32.8	28.6	36.0	35.9	31.4	39.3	39.2	34.3	46.4	46.3	40.7	54.2	54.1	47.7	72.0	71.8	63.7	92.7	92.4	82.5	116.7	116.1	104.4
	p_0	28.7	28.6	24.3	31.4	31.4	26.7	34.3	34.3	29.2	40.5	40.5	34.6	47.4	47.3	40.5	62.9	62.7	54.2	81.0	80.7	70.2	101.9	101.4	88.8
40	p_h	30.8	30.8	26.7	33.7	33.7	29.3	36.8	36.8	32.0	43.4	43.4	37.8	50.6	50.5	44.2	66.9	66.8	58.8	85.8	85.6	75.8	107.4	107.1	95.5
	p_0	27.8	27.8	23.6	30.5	30.5	25.9	33.3	33.2	28.3	39.2	39.2	33.5	45.7	45.7	39.2	60.4	60.3	52.1	77.5	77.3	67.2	97.1	96.8	84.6
30	p_h	28.5	28.5	24.6	31.2	31.2	27.0	34.1	34.1	29.5	40.1	40.1	34.8	46.7	46.6	40.5	61.4	61.4	53.6	78.4	78.3	68.8	97.8	97.6	86.1
	p_0	26.9	26.9	22.9	29.4	29.4	25.1	32.1	32.1	27.4	37.8	37.8	32.4	44.0	43.9	37.7	57.9	57.8	49.9	73.9	73.8	64.0	92.1	91.9	80.2
20	p_h	26.2	26.2	22.6	28.6	28.6	24.7	31.2	31.2	26.9	36.7	36.7	31.7	42.6	42.6	36.9	55.9	55.9	48.5	71.1	71.1	61.9	88.2	88.2	77.0
	p_0	25.8	25.8	22.2	28.3	28.3	24.3	30.8	30.8	26.5	36.2	36.2	31.2	42.1	42.1	36.3	55.2	55.1	47.7	70.1	70.1	60.9	87.1	87.0	75.8
15	p_h	25.2	25.2	21.8	27.6	27.6	23.8	30.0	30.0	26.0	35.3	35.3	30.6	41.0	41.0	35.5	53.7	53.7	46.6	68.1	68.1	59.3	84.4	84.4	73.6
	p_0	25.2	25.2	21.8	27.6	27.6	23.8	30.0	30.0	26.0	35.3	35.3	30.6	41.0	41.0	35.5	53.7	53.7	46.6	68.1	68.1	59.3	84.4	84.4	73.6

continues

Table 27.5-1 *(Continued)*. Main Wind Force Resisting System, Part 2 [$h \le 160$ ft ($h \le 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Walls

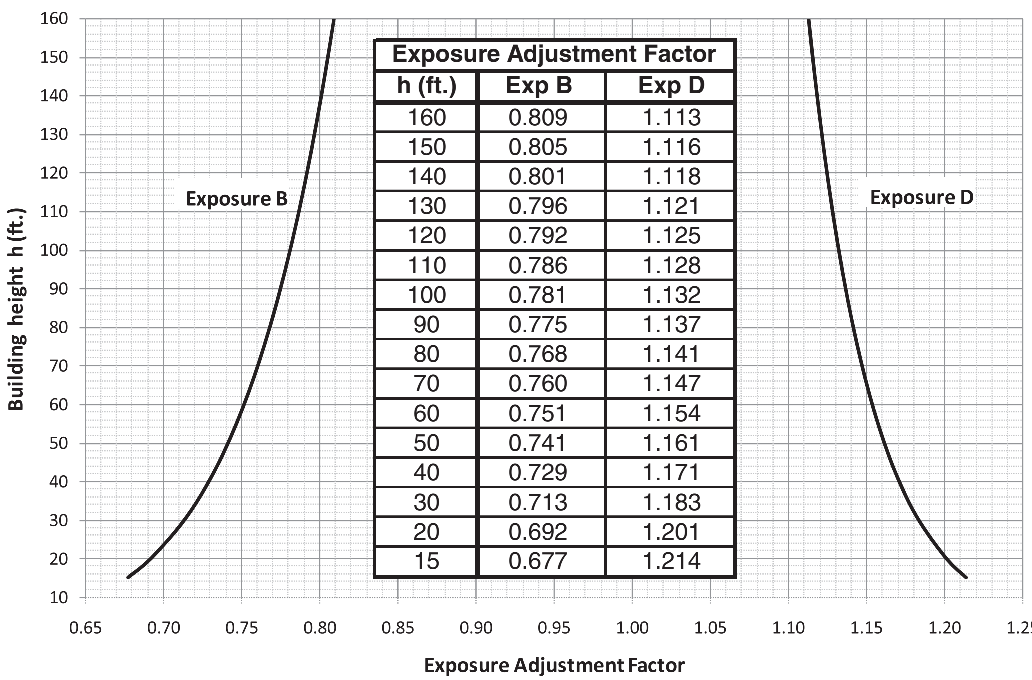
Exposure D

		V (mi/h)																							
	Along-wind Net Wall	110			115			120			130			140			160			180			200		
		L/B			L/B			L/B			L/B			L/B			L/B			L/B			L/B		
h (ft)	Pressure	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2	0.5	1	2
160	p_h	55.7	55.1	49.1	61.6	60.8	54.3	67.9	67.0	59.7	81.5	80.3	71.7	96.7	95.0	85.0	131.9	129.2	115.6	173.9	169.9	152.0	223.0	217.5	194.4
	p_0	42.9	42.4	35.7	47.4	46.8	39.5	52.2	51.5	43.5	62.7	61.7	52.2	74.4	73.1	61.9	101.4	99.4	84.2	133.7	130.7	110.7	171.5	167.2	141.6
150	p_h	54.5	53.9	48.0	60.2	59.5	53.0	66.3	65.4	58.4	79.5	78.4	70.0	94.3	92.8	83.0	128.5	126.0	112.8	169.3	165.6	148.3	217.0	211.8	189.6
	p_0	42.2	41.8	35.3	46.7	46.1	39.0	51.4	50.7	43.0	61.6	60.8	51.5	73.1	71.9	61.0	99.6	97.7	83.0	131.2	128.3	109.1	168.2	164.2	139.4
140	p_h	53.2	52.7	46.9	58.7	58.1	51.8	64.6	63.9	57.0	77.5	76.5	68.3	91.8	90.4	80.9	124.9	122.7	109.9	164.5	161.1	144.4	210.7	205.9	184.5
	p_0	41.6	41.2	34.8	45.9	45.4	38.5	50.5	49.9	42.4	60.6	59.8	50.8	71.7	70.7	60.1	97.7	95.9	81.7	128.6	125.9	107.3	164.7	160.9	137.2
130	p_h	51.8	51.4	45.7	57.2	56.7	50.5	62.9	62.3	55.5	75.4	74.5	66.5	89.2	88.0	78.7	121.2	119.2	106.9	159.5	156.4	140.3	204.2	199.7	179.2
	p_0	40.9	40.5	34.4	45.1	44.7	38.0	49.7	49.1	41.8	59.5	58.8	50.0	70.4	69.4	59.2	95.7	94.1	80.4	125.8	123.4	105.5	161.1	157.6	134.7
120	p_h	50.4	50.1	44.5	55.7	55.2	49.1	61.2	60.6	54.0	73.2	72.4	64.7	86.5	85.5	76.5	117.4	115.6	103.7	154.2	151.5	136.1	197.3	193.3	173.7
	p_0	40.2	39.9	33.9	44.4	44.0	37.4	48.8	48.3	41.1	58.3	57.7	49.2	69.0	68.1	58.2	93.6	92.2	78.9	122.9	120.7	103.5	157.3	154.0	132.2
110	p_h	49.0	48.7	43.2	54.0	53.6	47.7	59.4	58.9	52.4	70.9	70.2	62.7	83.8	82.8	74.1	113.4	111.9	100.4	148.8	146.3	131.6	190.2	186.5	167.9
	p_0	39.5	39.2	33.3	43.5	43.2	36.8	47.8	47.5	40.4	57.2	56.6	48.4	67.5	66.8	57.2	91.4	90.2	77.4	119.9	117.9	101.5	153.2	150.3	129.5
100	p_h	47.5	47.3	41.9	52.4	52.0	46.2	57.5	57.1	50.8	68.6	68.0	60.7	80.9	80.1	71.6	109.3	108.0	96.9	143.1	141.0	126.8	182.7	179.5	161.7
	p_0	38.8	38.6	32.8	42.7	42.5	36.2	46.9	46.6	39.7	55.9	55.5	47.5	66.0	65.4	56.1	89.2	88.1	75.9	116.8	115.0	99.3	149.0	146.4	126.6
90	p_h	46.0	45.8	40.5	50.6	50.4	44.6	55.5	55.2	49.0	66.2	65.7	58.5	77.9	77.3	69.0	105.0	103.9	93.2	137.2	135.4	121.8	174.8	172.1	155.2
	p_0	38.0	37.9	32.2	41.9	41.7	35.5	45.9	45.7	39.0	54.7	54.3	46.6	64.4	63.9	54.9	86.8	85.9	74.2	113.5	112.0	97.0	144.6	142.3	123.5
80	p_h	44.4	44.2	39.0	48.8	48.6	43.0	53.5	53.3	47.2	63.6	63.3	56.2	74.8	74.3	66.2	100.6	99.7	89.3	131.0	129.6	116.5	166.6	164.4	148.2
	p_0	37.3	37.1	31.6	41.0	40.8	34.8	44.9	44.7	38.2	53.4	53.1	45.6	62.8	62.4	53.7	84.4	83.7	72.4	110.0	108.8	94.5	139.9	138.0	120.2
70	p_h	42.7	42.6	37.4	46.9	46.8	41.2	51.4	51.2	45.2	61.0	60.7	53.8	71.6	71.2	63.3	95.9	95.2	85.1	124.6	123.5	110.9	158.0	156.3	140.8
	p_0	36.5	36.4	31.0	40.1	40.0	34.1	43.9	43.8	37.4	52.1	51.9	44.5	61.2	60.9	52.4	81.9	81.4	70.5	106.5	105.5	91.8	135.0	133.5	116.6
60	p_h	40.9	40.9	35.8	44.9	44.8	39.3	49.2	49.0	43.1	58.2	58.1	57.2	68.2	68.0	60.1	91.0	90.6	80.6	117.9	117.1	104.8	149.0	147.7	132.8
	p_0	35.7	35.6	30.3	39.2	39.1	33.4	42.9	42.8	36.6	50.8	50.6	43.4	59.5	59.3	51.0	79.4	79.0	68.4	102.8	102.1	88.9	129.9	128.8	112.7
50	p_h	39.0	39.0	34.0	42.8	42.7	37.3	46.8	46.7	40.8	55.3	55.2	48.4	64.7	64.5	56.8	85.9	85.6	75.9	110.8	110.3	98.3	139.5	138.7	124.2
	p_0	34.9	34.8	29.7	38.2	38.2	32.6	41.8	41.7	35.7	49.4	49.3	42.3	57.7	57.6	49.6	76.7	76.5	66.2	99.0	98.5	85.8	124.6	123.8	108.5
40	p_h	37.0	36.9	32.0	40.5	40.5	35.1	44.2	44.2	38.4	52.2	52.1	45.4	60.9	60.8	53.1	80.5	80.4	70.7	103.4	103.1	91.2	129.6	129.1	114.9
	p_0	34.0	33.9	28.9	37.2	37.2	31.7	40.6	40.6	34.7	47.9	47.9	41.1	55.9	55.8	48.0	74.0	73.8	63.9	95.0	94.7	82.5	119.1	118.7	103.9
30	p_h	34.7	34.6	29.9	37.9	37.9	32.7	41.4	41.4	35.7	48.7	48.7	42.2	56.7	56.7	49.2	74.8	74.7	65.2	95.5	95.4	83.7	119.2	119.0	104.9
	p_0	33.0	33.0	28.2	36.1	36.1	30.9	39.4	39.4	33.7	46.4	46.3	39.8	54.0	54.0	46.4	71.1	71.1	61.4	90.9	90.8	78.9	113.5	113.2	98.9
20	p_h	32.2	32.1	27.6	35.2	35.2	30.3	38.3	38.3	33.0	45.1	45.1	38.8	52.4	52.4	45.2	68.7	68.7	59.5	87.5	87.4	76.0	108.6	108.5	94.7
	p_0	31.8	31.8	27.3	34.8	34.8	29.9	37.9	37.9	32.6	44.6	44.6	38.3	51.8	51.8	44.6	68.0	68.0	58.8	86.5	86.5	75.0	107.5	107.4	93.5
15	p_h	31.1	31.1	26.8	34.0	34.0	29.3	37.0	37.0	31.9	43.5	43.5	37.5	50.5	50.5	43.6	66.2	66.1	57.3	84.0	84.0	73.0	104.1	104.1	90.7
	p_0	31.1	31.1	26.8	34.0	34.0	29.3	37.0	37.0	31.9	43.5	43.5	37.5	50.5	50.5	43.6	66.2	66.1	57.3	84.0	84.0	73.0	104.1	104.1	90.7

Notes for Roof Pressure Tables

1. From table for Exposure C, V, h, and roof slope, determine roof pressure p_h for each roof zone shown in the figures for the applicable roof form. For other exposures B or D, multiply pressures from table by appropriate exposure adjustment factor as determined from figure below.
2. Where two load cases are shown, both load cases shall be investigated. Load Case 2 is required to investigate maximum overturning on the building from roof pressures shown.
3. Apply along-wind net wall pressures to the projected area of the building walls in the direction of the wind and apply exterior sidewall pressures to the projected area of the building walls normal to the direction of the wind acting outward, simultaneously with the roof pressures from this table.
4. Where a value of zero is shown in the tables for the flat roof case, it is provided for the purpose of interpolation.
5. Interpolation between V, h, and roof slope is permitted.
6. 1.0 ft = 0.3048 m; 1.0 lb/ft^2 = 0.0479 kN/m^2.

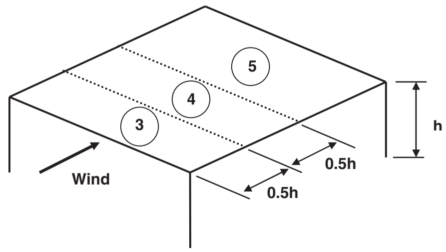
Exposure Adjustment Factor, Exposures B and D



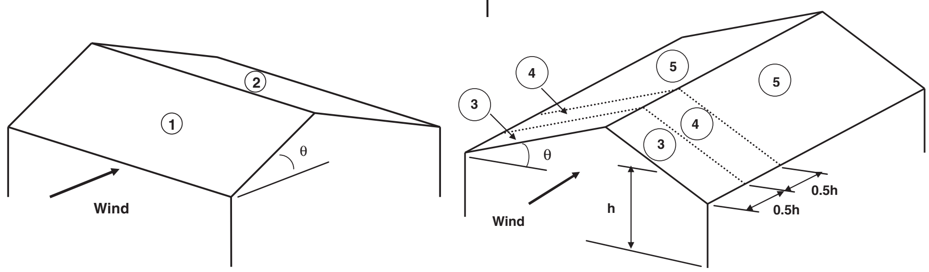
Exposure Adjustment Factor		
h (ft.)	Exp B	Exp D
160	0.809	1.113
150	0.805	1.116
140	0.801	1.118
130	0.796	1.121
120	0.792	1.125
110	0.786	1.128
100	0.781	1.132
90	0.775	1.137
80	0.768	1.141
70	0.760	1.147
60	0.751	1.154
50	0.741	1.161
40	0.729	1.171
30	0.713	1.183
20	0.692	1.201
15	0.677	1.214

continues

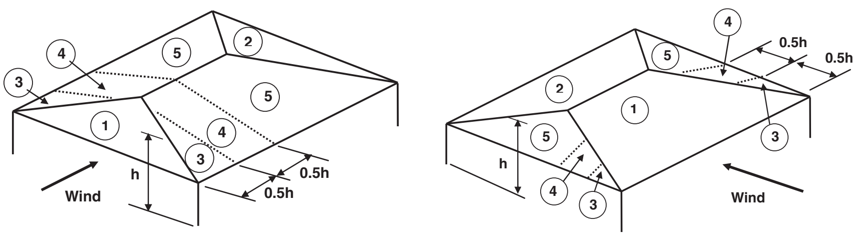
Parameters for Application of Roof Pressures



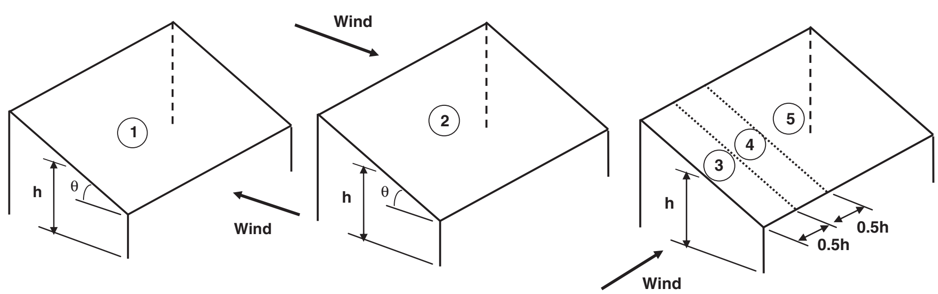
Flat Roof
(θ < 10 deg)



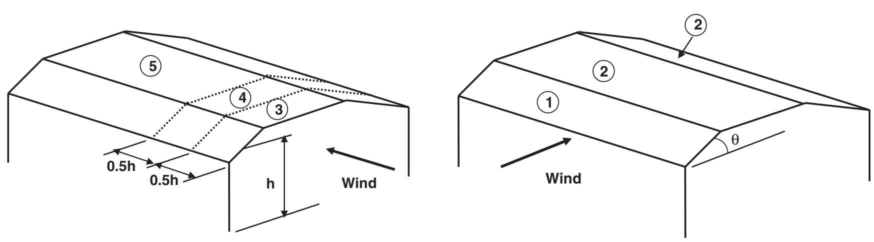
Gable Roof



Hip Roof



Monoslope
Roof



Mansard Roof

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \le 160$ ft ($h \le 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs

Exposure C: $h = 140$–160 ft, $V = 110$–120 mi/h

			V (mi/h)														
			110					115					120				
	Roof Slope	Load Case	Zone					Zone					Zone				
h (ft)			1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
160	Flat < 2:12 (9.46°)	1	NA	NA	−39.0	−34.8	−28.5	NA	NA	−42.6	−38.0	−31.2	NA	NA	−46.4	−41.4	−33.9
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−38.3	−26.0	−39.0	−34.8	−28.5	−41.8	−30.1	−42.6	−38.0	−31.2	−45.5	−31.0	−46.4	−41.4	−33.9
		2	5.5	−7.8	0.0	0.0	0.0	6.0	−8.5	0.0	0.0	0.0	6.6	−9.2	0.0	0.0	0.0
	4:12 (18.4°)	1	−31.5	−25.4	−39.0	−34.8	−28.5	−34.4	−27.7	−42.6	−38.0	−31.2	−37.4	−30.2	−46.4	−41.4	−33.9
		2	10.9	−11.1	0.0	0.0	0.0	11.9	−12.2	0.0	0.0	0.0	13.0	−13.3	0.0	0.0	0.0
	5:12 (22.6°)	1	−25.2	−25.4	−39.0	−34.8	−28.5	−27.6	−27.7	−42.6	−38.0	−31.2	−30.0	−30.2	−46.4	−41.4	−33.9
		2	14.5	−12.1	0.0	0.0	0.0	15.8	−13.3	0.0	0.0	0.0	17.3	−14.4	0.0	0.0	0.0
	6:12 (26.6°)	1	−20.3	−25.4	−39.0	−34.8	−28.5	−22.2	−27.7	−42.6	−38.0	−31.2	−24.1	−30.2	−46.4	−41.4	−33.9
		2	16.0	−12.1	0.0	0.0	0.0	17.5	−13.3	0.0	0.0	0.0	19.0	−14.4	0.0	0.0	0.0
	9:12 (36.9°)	1	−11.7	−25.4	−39.0	−34.8	−28.5	−12.8	−27.7	−42.6	−38.0	−31.2	−14.0	−30.2	−46.4	−41.4	−33.9
		2	19.1	−12.1	0.0	0.0	0.0	20.9	−13.3	0.0	0.0	0.0	22.8	−14.4	0.0	0.0	0.0
	12:12 (45.0°)	1	−6.6	−25.4	−39.0	−34.8	−28.5	−7.2	−27.7	−42.6	−38.0	−31.2	−7.9	−30.2	−46.4	−41.4	−33.9
		2	19.1	−12.1	0.0	0.0	0.0	20.9	−13.3	0.0	0.0	0.0	22.8	−14.4	0.0	0.0	0.0
150	Flat < 2:12 (9.46°)	1	NA	NA	−38.5	−34.3	−28.1	NA	NA	−42.0	−37.5	−30.7	NA	NA	−45.8	−40.8	−33.5
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−37.7	−25.7	−38.5	−34.3	−28.1	−41.3	−29.7	−42.0	−37.5	−30.7	−44.9	−30.5	−45.8	−40.8	−33.5
		2	5.4	−7.7	0.0	0.0	0.0	6.0	−8.4	0.0	0.0	0.0	6.5	−9.1	0.0	0.0	0.0
	4:12 (18.4°)	1	−31.0	−25.0	−38.5	−34.3	−28.1	−33.9	−27.4	−42.0	−37.5	−30.7	−36.9	−29.8	−45.8	−40.8	−33.5
		2	10.7	−11.0	0.0	0.0	0.0	11.7	−12.0	0.0	0.0	0.0	12.8	−13.1	0.0	0.0	0.0
	5:12 (22.6°)	1	−24.9	−25.0	−38.5	−34.3	−28.1	−27.2	−27.4	−42.0	−37.5	−30.7	−29.6	−29.8	−45.8	−40.8	−33.5
		2	14.3	−12.0	0.0	0.0	0.0	15.6	−13.1	0.0	0.0	0.0	17.0	−14.3	0.0	0.0	0.0
	6:12 (26.6°)	1	−20.0	−25.0	−38.5	−34.3	−28.1	−21.9	−27.4	−42.0	−37.5	−30.7	−23.8	−29.8	−45.8	−40.8	−33.5
		2	15.8	−12.0	0.0	0.0	0.0	17.3	−13.1	0.0	0.0	0.0	18.8	−14.3	0.0	0.0	0.0
	9:12 (36.9°)	1	−11.6	−25.0	−38.5	−34.3	−28.1	−12.7	−27.4	−42.0	−37.5	−30.7	−13.8	−29.8	−45.8	−40.8	−33.5
		2	18.9	−12.0	0.0	0.0	0.0	20.6	−13.1	0.0	0.0	0.0	22.5	−14.3	0.0	0.0	0.0
	12:12 (45.0°)	1	−6.5	−25.0	−38.5	−34.3	−28.1	−7.1	−27.4	−42.0	−37.5	−30.7	−7.8	−29.8	−45.8	−40.8	−33.5
		2	18.9	−12.0	0.0	0.0	0.0	20.6	−13.1	0.0	0.0	0.0	22.5	−14.3	0.0	0.0	0.0
140	Flat < 2:12 (9.46°)	1	NA	NA	−37.9	−33.8	−27.7	NA	NA	−41.4	−36.9	−30.3	NA	NA	−45.1	−40.2	−33.0
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−37.2	−25.3	−37.9	−33.8	−27.7	−40.7	−29.3	−41.4	−36.9	−30.3	−44.3	−30.1	−45.1	−40.2	−33.0
		2	5.4	−7.5	0.0	0.0	0.0	5.9	−8.2	0.0	0.0	0.0	6.4	−9.0	0.0	0.0	0.0
	4:12 (18.4°)	1	−30.6	−24.7	−37.9	−33.8	−27.7	−33.4	−27.0	−41.4	−36.9	−30.3	−36.4	−29.4	−45.1	−40.2	−33.0
		2	10.6	−10.8	0.0	0.0	0.0	11.6	−11.8	0.0	0.0	0.0	12.6	−12.9	0.0	0.0	0.0
	5:12 (22.6°)	1	−24.5	−24.7	−37.9	−33.8	−27.7	−26.8	−27.0	−41.4	−36.9	−30.3	−29.2	−29.4	−45.1	−40.2	−33.0
		2	14.1	−11.8	0.0	0.0	0.0	15.4	−12.9	0.0	0.0	0.0	16.8	−14.0	0.0	0.0	0.0
	6:12 (26.6°)	1	−19.7	−24.7	−37.9	−33.8	−27.7	−21.5	−27.0	−41.4	−36.9	−30.3	−23.5	−29.4	−45.1	−40.2	−33.0
		2	15.6	−11.8	0.0	0.0	0.0	17.0	−12.9	0.0	0.0	0.0	18.5	−14.0	0.0	0.0	0.0
	9:12 (36.9°)	1	−11.4	−24.7	−37.9	−33.8	−27.7	−12.5	−27.0	−41.4	−36.9	−30.3	−13.6	−29.4	−45.1	−40.2	−33.0
		2	18.6	−11.8	0.0	0.0	0.0	20.3	−12.9	0.0	0.0	0.0	22.1	−14.0	0.0	0.0	0.0
	12:12 (45.0°)	1	−6.4	−24.7	−37.9	−33.8	−27.7	−7.0	−27.0	−41.4	−36.9	−30.3	−7.7	−29.4	−45.1	−40.2	−33.0
		2	18.6	−11.8	0.0	0.0	0.0	20.3	−12.9	0.0	0.0	0.0	22.1	−14.0	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 140$–160 ft, $V = 130$–150 mi/h

			V (mi/h)														
			130					140					150				
			Zone					Zone					Zone				
h (ft)	Roof Slope	Load Case	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
160	Flat < 2:12 (9.46°)	1	NA	NA	−54.5	−48.6	−39.8	NA	NA	−63.2	−56.3	−46.2	NA	NA	−72.5	−64.6	−53.0
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−53.4	−36.3	−54.5	−48.6	−39.8	−62.0	−42.1	−63.2	−56.3	−46.2	−71.1	−48.4	−72.5	−64.6	−53.0
		2	7.7	−10.8	0.0	0.0	0.0	8.9	−12.6	0.0	0.0	0.0	10.3	−14.4	0.0	0.0	0.0
	4:12 (18.4°)	1	−43.9	−35.5	−54.5	−48.6	−39.8	−51.0	−41.1	−63.2	−56.3	−46.2	−58.5	−47.2	−72.5	−64.6	−53.0
		2	15.2	−15.6	0.0	0.0	0.0	17.6	−18.1	0.0	0.0	0.0	20.2	−20.7	0.0	0.0	0.0
	5:12 (22.6°)	1	−35.2	−35.5	−54.5	−48.6	−39.8	−40.9	−41.1	−63.2	−56.3	−46.2	−46.9	−47.2	−72.5	−64.6	−53.0
		2	20.2	−17.0	0.0	0.0	0.0	23.5	−19.7	0.0	0.0	0.0	27.0	−22.6	0.0	0.0	0.0
	6:12 (26.6°)	1	−28.3	−35.5	−54.5	−48.6	−39.8	−32.8	−41.1	−63.2	−56.3	−46.2	−37.7	−47.2	−72.5	−64.6	−53.0
		2	22.4	−17.0	0.0	0.0	0.0	25.9	−19.7	0.0	0.0	0.0	29.8	−22.6	0.0	0.0	0.0
	9:12 (36.9°)	1	−16.4	−35.5	−54.5	−48.6	−39.8	−19.0	−41.1	−63.2	−56.3	−46.2	−21.8	−47.2	−72.5	−64.6	−53.0
		2	26.7	−17.0	0.0	0.0	0.0	31.0	−19.7	0.0	0.0	0.0	11.4	−22.6	0.0	0.0	0.0
	12:12 (45.0°)	1	−9.2	−35.5	−54.5	−48.6	−39.8	−10.7	−41.1	−63.2	−56.3	−46.2	−12.3	−47.2	−72.5	−64.6	−53.0
		2	26.7	−17.0	0.0	0.0	0.0	31.0	−19.7	0.0	0.0	0.0	35.6	−22.6	0.0	0.0	0.0
150	Flat < 2:12 (9.46°)	1	NA	NA	−53.7	−47.9	−39.3	NA	NA	−62.3	−55.6	−45.6	NA	NA	−71.5	−63.8	−52.3
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−52.7	−35.8	−53.7	−47.9	−39.3	−61.1	−41.6	−62.3	−55.6	−45.6	−70.2	−47.7	−71.5	−63.8	−52.3
		2	7.6	−10.7	0.0	0.0	0.0	8.8	−12.4	0.0	0.0	0.0	10.1	−14.2	0.0	0.0	0.0
	4:12 (18.4°)	1	−43.3	−35.0	−53.7	−47.9	−39.3	−50.3	−40.6	−62.3	−55.6	−45.6	−57.7	−46.6	−71.5	−63.8	−52.3
		2	15.0	−15.4	0.0	0.0	0.0	17.4	−17.8	0.0	0.0	0.0	20.0	−20.4	0.0	0.0	0.0
	5:12 (22.6°)	1	−34.8	−35.0	−53.7	−47.9	−39.3	−40.3	−40.6	−62.3	−55.6	−45.6	−46.3	−46.6	−71.5	−63.8	−52.3
		2	20.0	−16.7	0.0	0.0	0.0	23.2	−19.4	0.0	0.0	0.0	26.6	−22.3	0.0	0.0	0.0
	6:12 (26.6°)	1	−27.9	−35.0	−53.7	−47.9	−39.3	−32.4	−40.6	−62.3	−55.6	−45.6	−37.2	−46.6	−71.5	−63.8	−52.3
		2	22.1	−16.7	0.0	0.0	0.0	25.6	−19.4	0.0	0.0	0.0	29.4	−22.3	0.0	0.0	0.0
	9:12 (36.9°)	1	−16.2	−35.0	−53.7	−47.9	−39.3	−18.8	−40.6	−62.3	−55.6	−45.6	−21.5	−46.6	−71.5	−63.8	−52.3
		2	26.4	−16.7	0.0	0.0	0.0	30.6	−19.4	0.0	0.0	0.0	11.3	−22.3	0.0	0.0	0.0
	12:12 (45.0°)	1	−9.1	−35.0	−53.7	−47.9	−39.3	−10.6	−40.6	−62.3	−55.6	−45.6	−12.1	−46.6	−71.5	−63.8	−52.3
		2	26.4	−16.7	0.0	0.0	0.0	30.6	−19.4	0.0	0.0	0.0	35.1	−22.3	0.0	0.0	0.0
140	Flat < 2:12 (9.46°)	1	NA	NA	−53.0	−47.2	−38.7	NA	NA	−61.4	−54.8	−44.9	NA	NA	−70.5	−62.9	−51.5
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−52.0	−35.3	−53.0	−47.2	−38.7	−60.3	−41.0	−61.4	−54.8	−44.9	−69.2	−47.0	−70.5	−62.9	−51.5
		2	7.5	−10.5	0.0	0.0	0.0	8.7	−12.2	0.0	0.0	0.0	10.0	−14.0	0.0	0.0	0.0
	4:12 (18.4°)	1	−42.7	−34.5	−53.0	−47.2	−38.7	−49.5	−40.0	−61.4	−54.8	−44.9	−56.9	−45.9	−70.5	−62.9	−51.5
		2	14.8	−15.1	0.0	0.0	0.0	17.2	−17.6	0.0	0.0	0.0	19.7	−20.2	0.0	0.0	0.0
	5:12 (22.6°)	1	−34.3	−34.5	−53.0	−47.2	−38.7	−39.7	−40.0	−61.4	−54.8	−44.9	−45.6	−45.9	−70.5	−62.9	−51.5
		2	19.7	−16.5	0.0	0.0	0.0	22.8	−19.1	0.0	0.0	0.0	26.2	−21.9	0.0	0.0	0.0
	6:12 (26.6°)	1	−27.5	−34.5	−53.0	−47.2	−38.7	−31.9	−40.0	−61.4	−54.8	−44.9	−36.6	−45.9	−70.5	−62.9	−51.5
		2	21.7	−16.5	0.0	0.0	0.0	25.2	−19.1	0.0	0.0	0.0	28.9	−21.9	0.0	0.0	0.0
	9:12 (36.9°)	1	−15.9	−34.5	−53.0	−47.2	−38.7	−18.5	−40.0	−61.4	−54.8	−44.9	−21.2	−45.9	−70.5	−62.9	−51.5
		2	26.0	−16.5	0.0	0.0	0.0	30.1	−19.1	0.0	0.0	0.0	11.1	−21.9	0.0	0.0	0.0
	12:12 (45.0°)	1	−9.0	−34.5	−53.0	−47.2	−38.7	−10.4	−40.0	−61.4	−54.8	−44.9	−12.0	−45.9	−70.5	−62.9	−51.5
		2	26.0	−16.5	0.0	0.0	0.0	30.1	−19.1	0.0	0.0	0.0	34.6	−21.9	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 140–160$ ft, $V = 160–200$ mi/h

			V (mi/h)														
			160					180					200				
h	Roof Slope	Load	Zone					Zone					Zone				
(ft)		Case	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
160	Flat < 2:12 (9.46°)	1	NA	NA	−82.5	−73.6	−60.3	NA	NA	−104.4	−93.1	−76.3	NA	NA	−128.9	−114.9	−94.3
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−80.9	−58.3	−82.5	−73.6	−60.3	−102.5	−69.6	−104.4	−93.1	−76.3	−126.5	−86.0	−128.9	−114.9	−94.3
		2	11.7	−16.4	0.0	0.0	0.0	14.8	−20.8	0.0	0.0	0.0	18.2	−25.7	0.0	0.0	0.0
	4:12 (18.4°)	1	−66.5	−53.7	−82.5	−73.6	−60.3	−84.2	−68.0	−104.4	−93.1	−76.3	−104.0	−83.9	−128.9	−114.9	−94.3
		2	23.0	−23.6	0.0	0.0	0.0	29.2	−29.8	0.0	0.0	0.0	36.0	−36.8	0.0	0.0	0.0
	5:12 (22.6°)	1	−53.4	−53.7	−82.5	−73.6	−60.3	−67.6	−68.0	−104.4	−93.1	−76.3	−83.4	−83.9	−128.9	−114.9	−94.3
		2	30.7	−25.7	0.0	0.0	0.0	38.8	−32.5	0.0	0.0	0.0	47.9	−40.1	0.0	0.0	0.0
	6:12 (26.6°)	1	−42.9	−53.7	−82.5	−73.6	−60.3	−54.3	−68.0	−104.4	−93.1	−76.3	−67.0	−83.9	−128.9	−114.9	−94.3
		2	33.9	−25.7	0.0	0.0	0.0	42.9	−32.5	0.0	0.0	0.0	52.9	−40.1	0.0	0.0	0.0
	9:12 (36.9°)	1	−24.8	−53.7	−82.5	−73.6	−60.3	−31.4	−68.0	−104.4	−93.1	−76.3	−38.8	−83.9	−128.9	−114.9	−94.3
		2	40.5	−25.7	0.0	0.0	0.0	51.2	−32.5	0.0	0.0	0.0	63.2	−40.1	0.0	0.0	0.0
	12:12 (45.0°)	1	−14.0	−53.7	−82.5	−73.6	−60.3	−17.7	−68.0	−104.4	−93.1	−76.3	−21.9	−83.9	−128.9	−114.9	−94.3
		2	40.5	−25.7	0.0	0.0	0.0	51.2	−32.5	0.0	0.0	0.0	63.2	−40.1	0.0	0.0	0.0
150	Flat < 2:12 (9.46°)	1	NA	NA	−81.4	−72.6	−59.5	NA	NA	−103.0	−91.8	−75.3	NA	NA	−127.2	−113.4	−93.0
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−79.9	−57.5	−81.4	−72.6	−59.5	−101.1	−68.7	−103.0	−91.8	−75.3	−124.8	−84.8	−127.2	−113.4	−93.0
		2	11.5	−16.2	0.0	0.0	0.0	14.6	−20.5	0.0	0.0	0.0	18.0	−25.3	0.0	0.0	0.0
	4:12 (18.4°)	1	−65.7	−53.0	−81.4	−72.6	−59.5	−83.1	−67.1	−103.0	−91.8	−75.3	−102.6	−82.8	−127.2	−113.4	−93.0
		2	22.7	−23.3	0.0	0.0	0.0	28.8	−29.4	0.0	0.0	0.0	35.5	−36.4	0.0	0.0	0.0
	5:12 (22.6°)	1	−52.7	−53.0	−81.4	−72.6	−59.5	−66.7	−67.1	−103.0	−91.8	−75.3	−82.3	−82.8	−127.2	−113.4	−93.0
		2	30.3	−25.3	0.0	0.0	0.0	38.3	−32.1	0.0	0.0	0.0	47.3	−39.6	0.0	0.0	0.0
	6:12 (26.6°)	1	−42.3	−53.0	−81.4	−72.6	−59.5	−53.5	−67.1	−103.0	−91.8	−75.3	−66.1	−82.8	−127.2	−113.4	−93.0
		2	33.4	−25.3	0.0	0.0	0.0	42.3	−32.1	0.0	0.0	0.0	52.2	−39.6	0.0	0.0	0.0
	9:12 (36.9°)	1	−24.5	−53.0	−81.4	−72.6	−59.5	−31.0	−67.1	−103.0	−91.8	−75.3	−38.3	−82.8	−127.2	−113.4	−93.0
		2	39.9	−25.3	0.0	0.0	0.0	50.5	−32.1	0.0	0.0	0.0	62.4	−39.6	0.0	0.0	0.0
	12:12 (45.0°)	1	−13.8	−53.0	−81.4	−72.6	−59.5	−17.5	−67.1	−103.0	−91.8	−75.3	−21.6	−82.8	−127.2	−113.4	−93.0
		2	39.9	−25.3	0.0	0.0	0.0	50.5	−32.1	0.0	0.0	0.0	62.4	−39.6	0.0	0.0	0.0
140	Flat < 2:12 (9.46°)	1	NA	NA	−80.2	−71.5	−58.6	NA	NA	−101.5	−90.5	−74.2	NA	NA	−125.3	−111.7	−91.6
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−78.7	−56.7	−80.2	−71.5	−58.6	−99.6	−67.7	−101.5	−90.5	−74.2	−123.0	−83.6	−125.3	−111.7	−91.6
		2	11.4	−16.0	0.0	0.0	0.0	14.4	−20.2	0.0	0.0	0.0	17.7	−24.9	0.0	0.0	0.0
	4:12 (18.4°)	1	−64.7	−52.2	−80.2	−71.5	−58.6	−81.9	−66.1	−101.5	−90.5	−74.2	−101.1	−81.6	−125.3	−111.7	−91.6
		2	22.4	−22.9	0.0	0.0	0.0	28.4	−29.0	0.0	0.0	0.0	35.0	−35.8	0.0	0.0	0.0
	5:12 (22.6°)	1	−51.9	−52.2	−80.2	−71.5	−58.6	−65.7	−66.1	−101.5	−90.5	−74.2	−81.1	−81.6	−125.3	−111.7	−91.6
		2	29.8	−25.0	0.0	0.0	0.0	37.7	−31.6	0.0	0.0	0.0	46.6	−39.0	0.0	0.0	0.0
	6:12 (26.6°)	1	−41.7	−52.2	−80.2	−71.5	−58.6	−52.8	−66.1	−101.5	−90.5	−74.2	−65.2	−81.6	−125.3	−111.7	−91.6
		2	32.9	−25.0	0.0	0.0	0.0	41.7	−31.6	0.0	0.0	0.0	51.4	−39.0	0.0	0.0	0.0
	9:12 (36.9°)	1	−24.1	−52.2	−80.2	−71.5	−58.6	−30.6	−66.1	−101.5	−90.5	−74.2	−37.7	−81.6	−125.3	−111.7	−91.6
		2	39.4	−25.0	0.0	0.0	0.0	49.8	−31.6	0.0	0.0	0.0	61.5	−39.0	0.0	0.0	0.0
	12:12 (45.0°)	1	−13.6	−52.2	−80.2	−71.5	−58.6	−17.2	−66.1	−101.5	−90.5	−74.2	−21.3	−81.6	−125.3	−111.7	−91.6
		2	39.4	−25.0	0.0	0.0	0.0	49.8	−31.6	0.0	0.0	0.0	61.5	−39.0	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 110–130$ ft, $V = 110–120$ mi/h

			V (mi/h)														
			110					115					120				
			Zone					Zone					Zone				
h (ft)	Roof Slope	Load Case	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
130	Flat < 2:12 (9.46°)	1	NA	NA	−37.3	−33.3	−27.3	NA	NA	−40.8	−36.4	−29.8	NA	NA	−44.4	−39.6	−32.5
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−36.6	−24.9	−37.3	−33.3	−27.3	−40.0	−28.8	−40.8	−36.4	−29.8	−43.6	−29.6	−44.4	−39.6	−32.5
		2	5.3	−7.4	0.0	0.0	0.0	5.8	−8.1	0.0	0.0	0.0	6.3	−8.8	0.0	0.0	0.0
	4:12 (18.4°)	1	−30.1	−24.3	−37.3	−33.3	−27.3	−32.9	−26.6	−40.8	−36.4	−29.8	−35.8	−28.9	−44.4	−39.6	−32.5
		2	10.4	−10.7	0.0	0.0	0.0	11.4	−11.7	0.0	0.0	0.0	12.4	−12.7	0.0	0.0	0.0
	5:12 (22.6°)	1	−24.2	−24.3	−37.3	−33.3	−27.3	−26.4	−26.6	−40.8	−36.4	−29.8	−28.7	−28.9	−44.4	−39.6	−32.5
		2	13.9	−11.6	0.0	0.0	0.0	15.2	−12.7	0.0	0.0	0.0	16.5	−13.8	0.0	0.0	0.0
	6:12 (26.6°)	1	−19.4	−24.3	−37.3	−33.3	−27.3	−21.2	−26.6	−40.8	−36.4	−29.8	−23.1	−28.9	−44.4	−39.6	−32.5
		2	15.3	−11.6	0.0	0.0	0.0	16.7	−12.7	0.0	0.0	0.0	18.2	−13.8	0.0	0.0	0.0
	9:12 (36.9°)	1	−11.2	−24.3	−37.3	−33.3	−27.3	−12.3	−26.6	−40.8	−36.4	−29.8	−13.4	−28.9	−44.4	−39.6	−32.5
		2	18.3	−11.6	0.0	0.0	0.0	20.0	−12.7	0.0	0.0	0.0	21.8	−13.8	0.0	0.0	0.0
	12:12 (45.0°)	1	−6.3	−24.3	−37.3	−33.3	−27.3	−6.9	−26.6	−40.8	−36.4	−29.8	−7.5	−28.9	−44.4	−39.6	−32.5
		2	18.3	−11.6	0.0	0.0	0.0	20.0	−12.7	0.0	0.0	0.0	21.8	−13.8	0.0	0.0	0.0
120	Flat < 2:12 (9.46°)	1	NA	NA	−36.7	−32.7	−26.8	NA	NA	−40.1	−35.8	−29.3	NA	NA	−43.7	−38.9	−31.9
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−36.0	−24.5	−36.7	−32.7	−26.8	−39.4	−28.3	−40.1	−35.8	−29.3	−42.9	−29.1	−43.7	−38.9	−31.9
		2	5.2	−7.3	0.0	0.0	0.0	5.7	−8.0	0.0	0.0	0.0	6.2	−8.7	0.0	0.0	0.0
	4:12 (18.4°)	1	−29.6	−23.9	−36.7	−32.7	−26.8	−32.4	−26.1	−40.1	−35.8	−29.3	−35.2	−28.4	−43.7	−38.9	−31.9
		2	10.2	−10.5	0.0	0.0	0.0	11.2	−11.5	0.0	0.0	0.0	12.2	−12.5	0.0	0.0	0.0
	5:12 (22.6°)	1	−23.8	−23.9	−36.7	−32.7	−26.8	−26.0	−26.1	−40.1	−35.8	−29.3	−28.3	−28.4	−43.7	−38.9	−31.9
		2	13.6	−11.4	0.0	0.0	0.0	14.9	−12.5	0.0	0.0	0.0	16.2	−13.6	0.0	0.0	0.0
	6:12 (26.6°)	1	−19.1	−23.9	−36.7	−32.7	−26.8	−20.9	−26.1	−40.1	−35.8	−29.3	−22.7	−28.4	−43.7	−38.9	−31.9
		2	15.1	−11.4	0.0	0.0	0.0	16.5	−12.5	0.0	0.0	0.0	17.9	−13.6	0.0	0.0	0.0
	9:12 (36.9°)	1	−11.0	−23.9	−36.7	−32.7	−26.8	−12.1	−26.1	−40.1	−35.8	−29.3	−13.1	−28.4	−43.7	−38.9	−31.9
		2	18.0	−11.4	0.0	0.0	0.0	19.7	−12.5	0.0	0.0	0.0	21.4	−13.6	0.0	0.0	0.0
	12:12 (45.0°)	1	−6.2	−23.9	−36.7	−32.7	−26.8	−6.8	−26.1	−40.1	−35.8	−29.3	−7.4	−28.4	−43.7	−38.9	−31.9
		2	18.0	−11.4	0.0	0.0	0.0	19.7	−12.5	0.0	0.0	0.0	21.4	−13.6	0.0	0.0	0.0
110	Flat < 2:12 (9.46°)	1	NA	NA	−36.0	−32.1	−26.3	NA	NA	−39.4	−35.1	−28.8	NA	NA	−42.9	−38.2	−31.4
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−35.4	−24.0	−36.0	−32.1	−26.3	−38.6	−27.8	−39.4	−35.1	−28.8	−42.1	−28.6	−42.9	−38.2	−31.4
		2	5.1	−7.2	0.0	0.0	0.0	5.6	−7.8	0.0	0.0	0.0	6.1	−8.5	0.0	0.0	0.0
	4:12 (18.4°)	1	−29.1	−23.5	−36.0	−32.1	−26.3	−31.8	−25.6	−39.4	−35.1	−28.8	−34.6	−27.9	−42.9	−38.2	−31.4
		2	10.1	−10.3	0.0	0.0	0.0	11.0	−11.3	0.0	0.0	0.0	12.0	−12.3	0.0	0.0	0.0
	5:12 (22.6°)	1	−23.3	−23.5	−36.0	−32.1	−26.3	−25.5	−25.6	−39.4	−35.1	−28.8	−27.8	−27.9	−42.9	−38.2	−31.4
		2	13.4	−11.2	0.0	0.0	0.0	14.6	−12.3	0.0	0.0	0.0	15.9	−13.4	0.0	0.0	0.0
	6:12 (26.6°)	1	−18.7	−23.5	−36.0	−32.1	−26.3	−20.5	−25.6	−39.4	−35.1	−28.8	−22.3	−27.9	−42.9	−38.2	−31.4
		2	14.8	−11.2	0.0	0.0	0.0	16.2	−12.3	0.0	0.0	0.0	17.6	−13.4	0.0	0.0	0.0
	9:12 (36.9°)	1	−10.8	−23.5	−36.0	−32.1	−26.3	−11.9	−25.6	−39.4	−35.1	−28.8	−12.9	−27.9	−42.9	−38.2	−31.4
		2	17.7	−11.2	0.0	0.0	0.0	19.3	−12.3	0.0	0.0	0.0	21.0	−13.4	0.0	0.0	0.0
	12:12 (45.0°)	1	−6.1	−23.5	−36.0	−32.1	−26.3	−6.7	−25.6	−39.4	−35.1	−28.8	−7.3	−27.9	−42.9	−38.2	−31.4
		2	17.7	−11.2	0.0	0.0	0.0	19.3	−12.3	0.0	0.0	0.0	21.0	−13.4	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 110–130$ ft, $V = 130–150$ mi/h

			V (mi/h)														
			130					140					150				
			Zone					Zone					Zone				
h (ft)	Roof Slope	Load Case	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
130	Flat < 2:12 (9.46°)	1	NA	NA	–52.1	–46.5	–38.1	NA	NA	–60.5	–53.9	–44.2	NA	NA	–69.4	–61.9	–50.7
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–51.2	–34.8	–52.1	–46.5	–38.1	–59.3	–40.3	–60.5	–53.9	–44.2	–68.1	–46.3	–69.4	–61.9	–50.7
		2	7.4	–10.4	0.0	0.0	0.0	8.6	–12.0	0.0	0.0	0.0	9.8	–13.8	0.0	0.0	0.0
	4:12 (18.4°)	1	–42.1	–33.9	–52.1	–46.5	–38.1	–48.8	–39.4	–60.5	–53.9	–44.2	–56.0	–45.2	–69.4	–61.9	–50.7
		2	14.6	–14.9	0.0	0.0	0.0	16.9	–17.3	0.0	0.0	0.0	19.4	–19.8	0.0	0.0	0.0
	5:12 (22.6°)	1	–33.7	–33.9	–52.1	–46.5	–38.1	–39.1	–39.4	–60.5	–53.9	–44.2	–44.9	–45.2	–69.4	–61.9	–50.7
		2	19.4	–16.2	0.0	0.0	0.0	22.5	–18.8	0.0	0.0	0.0	25.8	–21.6	0.0	0.0	0.0
	6:12 (26.6°)	1	–27.1	–33.9	–52.1	–46.5	–38.1	–31.4	–39.4	–60.5	–53.9	–44.2	–36.1	–45.2	–69.4	–61.9	–50.7
		2	21.4	–16.2	0.0	0.0	0.0	24.8	–18.8	0.0	0.0	0.0	28.5	–21.6	0.0	0.0	0.0
	9:12 (36.9°)	1	–15.7	–33.9	–52.1	–46.5	–38.1	–18.2	–39.4	–60.5	–53.9	–44.2	–20.9	–45.2	–69.4	–61.9	–50.7
		2	25.6	–16.2	0.0	0.0	0.0	29.7	–18.8	0.0	0.0	0.0	10.9	–21.6	0.0	0.0	0.0
	12:12 (45.0°)	1	–8.9	–33.9	–52.1	–46.5	–38.1	–10.3	–39.4	–60.5	–53.9	–44.2	–11.8	–45.2	–69.4	–61.9	–50.7
		2	15.0	–9.5	0.0	0.0	0.0	16.4	–10.4	0.0	0.0	0.0	34.1	–21.6	0.0	0.0	0.0
120	Flat < 2:12 (9.46°)	1	NA	NA	–51.3	–45.7	–37.5	NA	NA	–59.5	–53.0	–43.5	NA	NA	–59.5	–53.0	–43.5
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–50.3	–34.2	–51.3	–45.7	–37.5	–58.3	–39.7	–59.5	–53.0	–43.5	–66.9	–45.6	–68.3	–60.8	–49.9
		2	7.3	–10.2	0.0	0.0	0.0	8.4	–11.8	0.0	0.0	0.0	9.6	–13.5	0.0	0.0	0.0
	4:12 (18.4°)	1	–41.4	–33.4	–51.3	–45.7	–37.5	–48.0	–38.7	–59.5	–53.0	–43.5	–55.1	–44.4	–68.3	–60.8	–49.9
		2	14.3	–14.7	0.0	0.0	0.0	16.6	–17.0	0.0	0.0	0.0	19.1	–19.5	0.0	0.0	0.0
	5:12 (22.6°)	1	–33.2	–33.4	–51.3	–45.7	–37.5	–38.5	–38.7	–59.5	–53.0	–43.5	–44.2	–44.4	–68.3	–60.8	–49.9
		2	19.1	–16.0	0.0	0.0	0.0	22.1	–18.5	0.0	0.0	0.0	25.4	–21.2	0.0	0.0	0.0
	6:12 (26.6°)	1	–26.6	–33.4	–51.3	–45.7	–37.5	–30.9	–38.7	–59.5	–53.0	–43.5	–35.5	–44.4	–68.3	–60.8	–49.9
		2	21.0	–16.0	0.0	0.0	0.0	24.4	–18.5	0.0	0.0	0.0	28.0	–21.2	0.0	0.0	0.0
	9:12 (36.9°)	1	–15.4	–33.4	–51.3	–45.7	–37.5	–17.9	–38.7	–59.5	–53.0	–43.5	–20.5	–44.4	–68.3	–60.8	–49.9
		2	25.1	–16.0	0.0	0.0	0.0	29.2	–18.5	0.0	0.0	0.0	33.5	–21.2	0.0	0.0	0.0
	12:12 (45.0°)	1	–8.7	–33.4	–51.3	–45.7	–37.5	–10.1	–38.7	–59.5	–53.0	–43.5	–11.6	–44.4	–68.3	–60.8	–49.9
		2	25.1	–16.0	0.0	0.0	0.0	29.2	–18.5	0.0	0.0	0.0	33.5	–21.2	0.0	0.0	0.0
110	Flat < 2:12 (9.46°)	1	NA	NA	–50.3	–44.9	–36.8	NA	NA	–58.4	–52.0	–42.7	NA	NA	–67.0	–59.7	–49.0
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–49.4	–33.6	–50.3	–44.9	–36.8	–57.3	–38.9	–58.4	–52.0	–42.7	–65.8	–44.7	–67.0	–59.7	–49.0
		2	7.1	–10.0	0.0	0.0	0.0	8.3	–11.6	0.0	0.0	0.0	9.5	–13.3	0.0	0.0	0.0
	4:12 (18.4°)	1	–40.6	–32.8	–50.3	–44.9	–36.8	–47.1	–38.0	–58.4	–52.0	–42.7	–54.1	–43.6	–67.0	–59.7	–49.0
		2	14.1	–14.4	0.0	0.0	0.0	16.3	–16.7	0.0	0.0	0.0	18.7	–19.2	0.0	0.0	0.0
	5:12 (22.6°)	1	–32.6	–32.8	–50.3	–44.9	–36.8	–37.8	–38.0	–58.4	–52.0	–42.7	–43.4	–43.6	–67.0	–59.7	–49.0
		2	18.7	–15.7	0.0	0.0	0.0	21.7	–18.2	0.0	0.0	0.0	24.9	–20.9	0.0	0.0	0.0
	6:12 (26.6°)	1	–26.2	–32.8	–50.3	–44.9	–36.8	–30.3	–38.0	–58.4	–52.0	–42.7	–34.8	–43.6	–67.0	–59.7	–49.0
		2	20.7	–15.7	0.0	0.0	0.0	24.0	–18.2	0.0	0.0	0.0	27.6	–20.9	0.0	0.0	0.0
	9:12 (36.9°)	1	–15.1	–32.8	–50.3	–44.9	–36.8	–17.6	–38.0	–58.4	–52.0	–42.7	–20.2	–43.6	–67.0	–59.7	–49.0
		2	24.7	–15.7	0.0	0.0	0.0	28.6	–18.2	0.0	0.0	0.0	32.8	–20.9	0.0	0.0	0.0
	12:12 (45.0°)	1	–8.5	–32.8	–50.3	–44.9	–36.8	–9.9	–38.0	–58.4	–52.0	–42.7	–11.4	–43.6	–67.0	–59.7	–49.0
		2	24.7	–15.7	0.0	0.0	0.0	28.6	–18.2	0.0	0.0	0.0	32.8	–20.9	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 110–130$ ft, $V = 160–200$ mi/h

			V (mi/h)														
			160					180					200				
h (ft)	Roof Slope	Load Case	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
130	Flat < 2:12 (9.46°)	1	NA	NA	−79.0	−70.4	−57.7	NA	NA	−100.0	−89.1	−73.1	NA	NA	−123.4	−110.0	−90.2
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−77.5	−55.8	−79.0	−70.4	−57.7	−98.1	−66.7	−100.0	−89.1	−73.1	−121.1	−82.3	−123.4	−110.0	−90.2
		2	11.2	−15.7	0.0	0.0	0.0	14.1	−19.9	0.0	0.0	0.0	17.5	−24.6	0.0	0.0	0.0
	4:12 (18.4°)	1	−63.7	−51.4	−79.0	−70.4	−57.7	−80.6	−65.1	−100.0	−89.1	−73.1	−99.5	−80.3	−123.4	−110.0	−90.2
		2	22.1	−22.6	0.0	0.0	0.0	27.9	−28.6	0.0	0.0	0.0	34.5	−35.3	0.0	0.0	0.0
	5:12 (22.6°)	1	−51.1	−51.4	−79.0	−70.4	−57.7	−64.7	−65.1	−100.0	−89.1	−73.1	−79.9	−80.3	−123.4	−110.0	−90.2
		2	29.4	−24.6	0.0	0.0	0.0	37.2	−31.1	0.0	0.0	0.0	45.9	−38.4	0.0	0.0	0.0
	6:12 (26.6°)	1	−41.1	−51.4	−79.0	−70.4	−57.7	−52.0	−65.1	−100.0	−89.1	−73.1	−64.1	−80.3	−123.4	−110.0	−90.2
		2	32.4	−24.6	0.0	0.0	0.0	41.0	−31.1	0.0	0.0	0.0	50.6	−38.4	0.0	0.0	0.0
	9:12 (36.9°)	1	−23.8	−51.4	−79.0	−70.4	−57.7	−30.1	−65.1	−100.0	−89.1	−73.1	−37.1	−80.3	−123.4	−110.0	−90.2
		2	38.7	−24.6	0.0	0.0	0.0	49.0	−31.1	0.0	0.0	0.0	60.5	−38.4	0.0	0.0	0.0
	12:12 (45.0°)	1	−13.4	−51.4	−79.0	−70.4	−57.7	−17.0	−65.1	−100.0	−89.1	−73.1	−21.0	−80.3	−123.4	−110.0	−90.2
		2	38.7	−24.6	0.0	0.0	0.0	49.0	−31.1	0.0	0.0	0.0	60.5	−38.4	0.0	0.0	0.0
120	Flat < 2:12 (9.46°)	1	NA	NA	−77.7	−69.2	−56.8	NA	NA	−98.3	−87.6	−71.9	NA	NA	−121.3	−108.2	−88.7
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−76.2	−54.8	−77.7	−69.2	−56.8	−96.4	−65.6	−98.3	−87.6	−71.9	−119.0	−80.9	−121.3	−108.2	−88.7
		2	11.0	−15.5	0.0	0.0	0.0	13.9	−19.6	0.0	0.0	0.0	17.2	−24.2	0.0	0.0	0.0
	4:12 (18.4°)	1	−62.6	−50.5	−77.7	−69.2	−56.8	−79.3	−64.0	−98.3	−87.6	−71.9	−97.9	−79.0	−121.3	−108.2	−88.7
		2	21.7	−22.2	0.0	0.0	0.0	27.4	−28.1	0.0	0.0	0.0	33.9	−34.7	0.0	0.0	0.0
	5:12 (22.6°)	1	−50.3	−50.5	−77.7	−69.2	−56.8	−63.6	−64.0	−98.3	−87.6	−71.9	−78.5	−79.0	−121.3	−108.2	−88.7
		2	28.9	−24.2	0.0	0.0	0.0	36.5	−30.6	0.0	0.0	0.0	45.1	−37.8	0.0	0.0	0.0
	6:12 (26.6°)	1	−40.4	−50.5	−77.7	−69.2	−56.8	−51.1	−64.0	−98.3	−87.6	−71.9	−63.1	−79.0	−121.3	−108.2	−88.7
		2	31.9	−24.2	0.0	0.0	0.0	40.3	−30.6	0.0	0.0	0.0	49.8	−37.8	0.0	0.0	0.0
	9:12 (36.9°)	1	−23.4	−50.5	−77.7	−69.2	−56.8	−29.6	−64.0	−98.3	−87.6	−71.9	−36.5	−79.0	−121.3	−108.2	−88.7
		2	38.1	−24.2	0.0	0.0	0.0	48.2	−30.6	0.0	0.0	0.0	59.5	−37.8	0.0	0.0	0.0
	12:12 (45.0°)	1	−13.2	−50.5	−77.7	−69.2	−56.8	−16.7	−64.0	−98.3	−87.6	−71.9	−20.6	−79.0	−121.3	−108.2	−88.7
		2	38.1	−24.2	0.0	0.0	0.0	48.2	−30.6	0.0	0.0	0.0	59.5	−37.8	0.0	0.0	0.0
110	Flat < 2:12 (9.46°)	1	NA	NA	−76.2	−68.0	−55.7	NA	NA	−96.5	−86.0	−70.6	NA	NA	−119.1	−106.2	−87.1
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−74.8	−53.8	−76.2	−68.0	−55.7	−94.7	−64.4	−96.5	−86.0	−70.6	−116.9	−79.5	−119.1	−106.2	−87.1
		2	10.8	−15.2	0.0	0.0	0.0	13.7	−19.2	0.0	0.0	0.0	16.9	−23.7	0.0	0.0	0.0
	4:12 (18.4°)	1	−61.5	−49.6	−76.2	−68.0	−55.7	−77.8	−62.8	−96.5	−86.0	−70.6	−96.1	−77.6	−119.1	−106.2	−87.1
		2	21.3	−21.8	0.0	0.0	0.0	26.9	−27.6	0.0	0.0	0.0	33.3	−34.1	0.0	0.0	0.0
	5:12 (22.6°)	1	−49.3	−49.6	−76.2	−68.0	−55.7	−62.5	−62.8	−96.5	−86.0	−70.6	−77.1	−77.6	−119.1	−106.2	−87.1
		2	28.3	−23.7	0.0	0.0	0.0	35.9	−30.0	0.0	0.0	0.0	44.3	−37.1	0.0	0.0	0.0
	6:12 (26.6°)	1	−39.6	−49.6	−76.2	−68.0	−55.7	−50.2	−62.8	−96.5	−86.0	−70.6	−61.9	−77.6	−119.1	−106.2	−87.1
		2	31.3	−23.7	0.0	0.0	0.0	39.6	−30.0	0.0	0.0	0.0	48.9	−37.1	0.0	0.0	0.0
	9:12 (36.9°)	1	−22.9	−49.6	−76.2	−68.0	−55.7	−29.0	−62.8	−96.5	−86.0	−70.6	−35.9	−77.6	−119.1	−106.2	−87.1
		2	37.4	−23.7	0.0	0.0	0.0	47.3	−30.0	0.0	0.0	0.0	58.4	−37.1	0.0	0.0	0.0
	12:12 (45.0°)	1	−12.9	−49.6	−76.2	−68.0	−55.7	−16.4	−62.8	−96.5	−86.0	−70.6	−20.2	−77.6	−119.1	−106.2	−87.1
		2	37.4	−23.7	0.0	0.0	0.0	47.3	−30.0	0.0	0.0	0.0	58.4	−37.1	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 80$–100 ft, $V = 110$–120 mi/h

			V (mi/h)														
			110					115					120				
			Zone					Zone					Zone				
h (ft)	Roof Slope	Load Case	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
100	Flat < 2:12 (9.46°)	1	NA	NA	−35.3	−31.5	−25.8	NA	NA	−38.6	−34.4	−28.2	NA	NA	−42.0	−37.5	−30.7
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−34.7	−23.6	−35.3	−31.5	−25.8	−37.9	−27.3	−38.6	−34.4	−28.2	−41.2	−28.0	−42.0	−37.5	−30.7
		2	5.0	−7.0	0.0	0.0	0.0	5.5	−7.7	0.0	0.0	0.0	5.9	−8.4	0.0	0.0	0.0
	4:12 (18.4°)	1	−28.5	−23.0	−35.3	−31.5	−25.8	−31.1	−25.1	−38.6	−34.4	−28.2	−33.9	−27.4	−42.0	−37.5	−30.7
		2	9.9	−10.1	0.0	0.0	0.0	10.8	−11.0	0.0	0.0	0.0	11.7	−12.0	0.0	0.0	0.0
	5:12 (22.6°)	1	−22.9	−23.0	−35.3	−31.5	−25.8	−25.0	−25.1	−38.6	−34.4	−28.2	−27.2	−27.4	−42.0	−37.5	−30.7
		2	13.1	−11.0	0.0	0.0	0.0	14.4	−12.0	0.0	0.0	0.0	15.6	−13.1	0.0	0.0	0.0
	6:12 (26.6°)	1	−18.4	−23.0	−35.3	−31.5	−25.8	−20.1	−25.1	−38.6	−34.4	−28.2	−21.9	−27.4	−42.0	−37.5	−30.7
		2	14.5	−11.0	0.0	0.0	0.0	15.8	−12.0	0.0	0.0	0.0	17.3	−13.1	0.0	0.0	0.0
	9:12 (36.9°)	1	−10.6	−23.0	−35.3	−31.5	−25.8	−11.6	−25.1	−38.6	−34.4	−28.2	−12.7	−27.4	−42.0	−37.5	−30.7
		2	17.3	−11.0	0.0	0.0	0.0	18.9	−12.0	0.0	0.0	0.0	20.6	−13.1	0.0	0.0	0.0
	12:12 (45.0°)	1	−6.0	−23.0	−35.3	−31.5	−25.8	−6.6	−25.1	−38.6	−34.4	−28.2	−7.1	−27.4	−42.0	−37.5	−30.7
		2	17.3	−11.0	0.0	0.0	0.0	18.9	−12.0	0.0	0.0	0.0	20.6	−13.1	0.0	0.0	0.0
90	Flat < 2:12 (9.46°)	1	NA	NA	−34.5	−30.8	−25.3	NA	NA	−37.8	−33.7	−27.6	NA	NA	−41.1	−36.7	−30.1
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−33.9	−23.0	−34.5	−30.8	−25.3	−37.0	−26.7	−37.8	−33.7	−27.6	−40.3	−27.4	−41.1	−36.7	−30.1
		2	4.9	−6.9	0.0	0.0	0.0	5.3	−7.5	0.0	0.0	0.0	5.8	−8.2	0.0	0.0	0.0
	4:12 (18.4°)	1	−27.9	−22.5	−34.5	−30.8	−25.3	−30.5	−24.6	−37.8	−33.7	−27.6	−33.2	−26.8	−41.1	−36.7	−30.1
		2	9.6	−9.9	0.0	0.0	0.0	10.5	−10.8	0.0	0.0	0.0	11.5	−11.8	0.0	0.0	0.0
	5:12 (22.6°)	1	−22.4	−22.5	−34.5	−30.8	−25.3	−24.4	−24.6	−37.8	−33.7	−27.6	−26.6	−26.8	−41.1	−36.7	−30.1
		2	12.8	−10.8	0.0	0.0	0.0	14.0	−11.8	0.0	0.0	0.0	15.3	−12.8	0.0	0.0	0.0
	6:12 (26.6°)	1	−18.0	−22.5	−34.5	−30.8	−25.3	−19.6	−24.6	−37.8	−33.7	−27.6	−21.4	−26.8	−41.1	−36.7	−30.1
		2	14.2	−10.8	0.0	0.0	0.0	15.5	−11.8	0.0	0.0	0.0	16.9	−12.8	0.0	0.0	0.0
	9:12 (36.9°)	1	−10.4	−22.5	−34.5	−30.8	−25.3	−11.4	−24.6	−37.8	−33.7	−27.6	−12.4	−26.8	−41.1	−36.7	−30.1
		2	16.9	−10.8	0.0	0.0	0.0	18.5	−11.8	0.0	0.0	0.0	20.2	−12.8	0.0	0.0	0.0
	12:12 (45.0°)	1	−5.9	−22.5	−34.5	−30.8	−25.3	−6.4	−24.6	−37.8	−33.7	−27.6	−7.0	−26.8	−41.1	−36.7	−30.1
		2	16.9	−10.8	0.0	0.0	0.0	18.5	−11.8	0.0	0.0	0.0	20.2	−12.8	0.0	0.0	0.0
80	Flat < 2:12 (9.46°)	1	NA	NA	−33.7	−30.0	−24.6	NA	NA	−36.8	−32.8	−26.9	NA	NA	−40.1	−35.8	−29.3
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−33.1	−22.5	−33.7	−30.0	−24.6	−36.1	−26.0	−36.8	−32.8	−26.9	−39.4	−26.8	−40.1	−35.8	−29.3
		2	4.8	−6.7	0.0	0.0	0.0	5.2	−7.3	0.0	0.0	0.0	5.7	−8.0	0.0	0.0	0.0
	4:12 (18.4°)	1	−27.2	−21.9	−33.7	−30.0	−24.6	−29.7	−24.0	−36.8	−32.8	−26.9	−32.4	−26.1	−40.1	−35.8	−29.3
		2	9.4	−9.6	0.0	0.0	0.0	10.3	−10.5	0.0	0.0	0.0	11.2	−11.5	0.0	0.0	0.0
	5:12 (22.6°)	1	−21.8	−21.9	−33.7	−30.0	−24.6	−23.8	−24.0	−36.8	−32.8	−26.9	−26.0	−26.1	−40.1	−35.8	−29.3
		2	12.5	−10.5	0.0	0.0	0.0	13.7	−11.5	0.0	0.0	0.0	14.9	−12.5	0.0	0.0	0.0
	6:12 (26.6°)	1	−17.5	−21.9	−33.7	−30.0	−24.6	−19.1	−24.0	−36.8	−32.8	−26.9	−20.8	−26.1	−40.1	−35.8	−29.3
		2	13.8	−10.5	0.0	0.0	0.0	15.1	−11.5	0.0	0.0	0.0	16.5	−12.5	0.0	0.0	0.0
	9:12 (36.9°)	1	−10.1	−21.9	−33.7	−30.0	−24.6	−11.1	−24.0	−36.8	−32.8	−26.9	−12.1	−26.1	−40.1	−35.8	−29.3
		2	16.5	−10.5	0.0	0.0	0.0	18.1	−11.5	0.0	0.0	0.0	19.7	−12.5	0.0	0.0	0.0
	12:12 (45.0°)	1	−5.7	−21.9	−33.7	−30.0	−24.6	−6.3	−24.0	−36.8	−32.8	−26.9	−6.8	−26.1	−40.1	−35.8	−29.3
		2	16.5	−10.5	0.0	0.0	0.0	18.1	−11.5	0.0	0.0	0.0	19.7	−12.5	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 80$–100 ft, $V = 130$–150 mi/h

			V (mi/h)														
			130					140					150				
h (ft)	Roof Slope	Load Case	Zone 1	2	3	4	5	Zone 1	2	3	4	5	Zone 1	2	3	4	5
100	**Flat < 2:12 (9.46°)**	**1**	NA	NA	−49.3	−44.0	−36.1	NA	NA	−57.2	−51.0	−41.8	NA	NA	−65.7	−58.5	−48.0
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	**1**	−48.4	−32.9	−49.3	−44.0	−36.1	−56.1	−38.2	−57.2	−51.0	−41.8	−64.4	−43.9	−65.7	−58.5	−48.0
		2	7.0	−9.8	0.0	0.0	0.0	8.1	−11.4	0.0	0.0	0.0	9.3	−13.1	0.0	0.0	0.0
	4:12 (18.4°)	**1**	−39.8	−32.1	−49.3	−44.0	−36.1	−46.2	−37.2	−57.2	−51.0	−41.8	−53.0	−42.7	−65.7	−58.5	−48.0
		2	13.8	−14.1	0.0	0.0	0.0	16.0	−16.4	0.0	0.0	0.0	18.4	−18.8	0.0	0.0	0.0
	5:12 (22.6°)	**1**	−31.9	−32.1	−49.3	−44.0	−36.1	−37.0	−37.2	−57.2	−51.0	−41.8	−42.5	−42.7	−65.7	−58.5	−48.0
		2	18.3	−15.4	0.0	0.0	0.0	21.3	−17.8	0.0	0.0	0.0	24.5	−20.4	0.0	0.0	0.0
	6:12 (26.6°)	**1**	−25.6	−32.1	−49.3	−44.0	−36.1	−29.7	−37.2	−57.2	−51.0	−41.8	−34.1	−42.7	−65.7	−58.5	−48.0
		2	20.2	−15.4	0.0	0.0	0.0	23.5	−17.8	0.0	0.0	0.0	27.0	−20.4	0.0	0.0	0.0
	9:12 (36.9°)	**1**	−14.8	−32.1	−49.3	−44.0	−36.1	−17.2	−37.2	−57.2	−51.0	−41.8	−19.7	−42.7	−65.7	−58.5	−48.0
		2	24.2	−15.4	0.0	0.0	0.0	28.1	−17.8	0.0	0.0	0.0	32.3	−20.4	0.0	0.0	0.0
	12:12 (45.0°)	**1**	−8.4	−32.1	−49.3	−44.0	−36.1	−9.7	−37.2	−57.2	−51.0	−41.8	−11.1	−42.7	−65.7	−58.5	−48.0
		2	24.2	−15.4	0.0	0.0	0.0	28.1	−17.8	0.0	0.0	0.0	32.3	−20.4	0.0	0.0	0.0
90	**Flat < 2:12 (9.46°)**	**1**	NA	NA	−48.3	−43.0	−35.3	NA	NA	−56.0	−49.9	−40.9	NA	NA	−64.3	−57.3	−47.0
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	**1**	−47.3	−32.2	−48.3	−43.0	−35.3	−54.9	−37.3	−56.0	−49.9	−40.9	−63.0	−42.8	−64.3	−57.3	−47.0
		2	6.8	−9.6	0.0	0.0	0.0	7.9	−11.1	0.0	0.0	0.0	9.1	−12.7	0.0	0.0	0.0
	4:12 (18.4°)	**1**	−38.9	−31.4	−48.3	−43.0	−35.3	−45.1	−36.4	−56.0	−49.9	−40.9	−51.8	−41.8	−64.3	−57.3	−47.0
		2	13.5	−13.8	0.0	0.0	0.0	15.6	−16.0	0.0	0.0	0.0	17.9	−18.4	0.0	0.0	0.0
	5:12 (22.6°)	**1**	−31.2	−31.4	−48.3	−43.0	−35.3	−36.2	−36.4	−56.0	−49.9	−40.9	−41.6	−41.8	−64.3	−57.3	−47.0
		2	17.9	−15.0	0.0	0.0	0.0	20.8	−17.4	0.0	0.0	0.0	23.9	−20.0	0.0	0.0	0.0
	6:12 (26.6°)	**1**	−25.1	−31.4	−48.3	−43.0	−35.3	−29.1	−36.4	−56.0	−49.9	−40.9	−33.4	−41.8	−64.3	−57.3	−47.0
		2	19.8	−15.0	0.0	0.0	0.0	23.0	−17.4	0.0	0.0	0.0	26.4	−20.0	0.0	0.0	0.0
	9:12 (36.9°)	**1**	−14.5	−31.4	−48.3	−43.0	−35.3	−16.8	−36.4	−56.0	−49.9	−40.9	−19.3	−41.8	−64.3	−57.3	−47.0
		2	23.7	−15.0	0.0	0.0	0.0	27.5	−17.4	0.0	0.0	0.0	31.6	−20.0	0.0	0.0	0.0
	12:12 (45.0°)	**1**	−8.2	−31.4	−48.3	−43.0	−35.3	−9.5	−36.4	−56.0	−49.9	−40.9	−10.9	−41.8	−64.3	−57.3	−47.0
		2	23.7	−15.0	0.0	0.0	0.0	27.5	−17.4	0.0	0.0	0.0	31.6	−20.0	0.0	0.0	0.0
80	**Flat < 2:12 (9.46°)**	**1**	NA	NA	−47.1	−42.0	−34.4	NA	NA	−54.6	−48.7	−39.9	NA	NA	−62.7	−55.9	−45.8
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	**1**	−46.2	−31.4	−47.1	−42.0	−34.4	−53.6	−36.4	−54.6	−48.7	−39.9	−61.5	−41.8	−62.7	−55.9	−45.8
		2	6.7	−9.4	0.0	0.0	0.0	7.7	−10.9	0.0	0.0	0.0	8.9	−12.5	0.0	0.0	0.0
	4:12 (18.4°)	**1**	−38.0	−30.6	−47.1	−42.0	−34.4	−44.0	−35.5	−54.6	−48.7	−39.9	−50.5	−40.8	−62.7	−55.9	−45.8
		2	13.1	−13.5	0.0	0.0	0.0	15.2	−15.6	0.0	0.0	0.0	17.5	−17.9	0.0	0.0	0.0
	5:12 (22.6°)	**1**	−30.5	−30.6	−47.1	−42.0	−34.4	−35.3	−35.5	−54.6	−48.7	−39.9	−40.6	−40.8	−62.7	−55.9	−45.8
		2	17.5	−14.7	0.0	0.0	0.0	20.3	−17.0	0.0	0.0	0.0	23.3	−19.5	0.0	0.0	0.0
	6:12 (26.6°)	**1**	−24.5	−30.6	−47.1	−42.0	−34.4	−28.4	−35.5	−54.6	−48.7	−39.9	−32.6	−40.8	−62.7	−55.9	−45.8
		2	19.3	−14.7	0.0	0.0	0.0	22.4	−17.0	0.0	0.0	0.0	25.7	−19.5	0.0	0.0	0.0
	9:12 (36.9°)	**1**	−14.2	−30.6	−47.1	−42.0	−34.4	−16.4	−35.5	−54.6	−48.7	−39.9	−18.9	−40.8	−62.7	−55.9	−45.8
		2	23.1	−14.7	0.0	0.0	0.0	26.8	−17.0	0.0	0.0	0.0	9.9	−19.5	0.0	0.0	0.0
	12:12 (45.0°)	**1**	−8.0	−30.6	−47.1	−42.0	−34.4	−9.3	−35.5	−54.6	−48.7	−39.9	−10.6	−40.8	−62.7	−55.9	−45.8
		2	23.1	−14.7	0.0	0.0	0.0	26.8	−17.0	0.0	0.0	0.0	30.7	−19.5	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 80$–100 ft, $V = 160$–200 mi/h

			V (mi/h)														
			160					180					200				
h	Roof Slope	Load	Zone					Zone					Zone				
(ft)		Case	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
100	Flat < 2:12 (9.46°)	1	NA	NA	−74.7	−66.6	−54.6	NA	NA	−94.6	−84.3	−69.2	NA	NA	−116.8	−104.1	−85.4
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−73.3	−52.8	−74.7	−66.6	−54.6	−92.8	−63.1	−94.6	−84.3	−69.2	−114.6	−77.9	−116.8	−104.1	−85.4
		2	10.6	−14.9	0.0	0.0	0.0	13.4	−18.8	0.0	0.0	0.0	16.5	−23.2	0.0	0.0	0.0
	4:12 (18.4°)	1	−60.3	−48.6	−74.7	−66.6	−54.6	−76.3	−61.6	−94.6	−84.3	−69.2	−94.2	−76.0	−116.8	−104.1	−85.4
		2	20.9	−21.4	0.0	0.0	0.0	26.4	−27.0	0.0	0.0	0.0	32.6	−33.4	0.0	0.0	0.0
	5:12 (22.6°)	1	−48.4	−48.6	−74.7	−66.6	−54.6	−61.2	−61.6	−94.6	−84.3	−69.2	−75.6	−76.0	−116.8	−104.1	−85.4
		2	27.8	−23.3	0.0	0.0	0.0	35.2	−29.4	0.0	0.0	0.0	43.4	−36.4	0.0	0.0	0.0
	6:12 (26.6°)	1	−38.8	−48.6	−74.7	−66.6	−54.6	−49.2	−61.6	−94.6	−84.3	−69.2	−60.7	−76.0	−116.8	−104.1	−85.4
		2	30.7	−23.3	0.0	0.0	0.0	38.8	−29.4	0.0	0.0	0.0	47.9	−36.4	0.0	0.0	0.0
	9:12 (36.9°)	1	−22.5	−48.6	−74.7	−66.6	−54.6	−28.5	−61.6	−94.6	−84.3	−69.2	−35.1	−76.0	−116.8	−104.1	−85.4
		2	36.7	−23.3	0.0	0.0	0.0	46.4	−29.4	0.0	0.0	0.0	57.3	−36.4	0.0	0.0	0.0
	12:12 (45.0°)	1	−12.7	−48.6	−74.7	−66.6	−54.6	−16.1	−61.6	−94.6	−84.3	−69.2	−19.8	−76.0	−116.8	−104.1	−85.4
		2	36.7	−23.3	0.0	0.0	0.0	46.4	−29.4	0.0	0.0	0.0	57.3	−36.4	0.0	0.0	0.0
90	Flat < 2:12 (9.46°)	1	NA	NA	−73.1	−65.2	−53.4	NA	NA	−92.5	−82.5	−67.6	NA	NA	−114.2	−101.8	−83.5
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−71.7	−51.6	−73.1	−65.2	−53.4	−90.8	−61.7	−92.5	−82.5	−67.6	−112.1	−76.2	−114.2	−101.8	−83.5
		2	10.3	−14.5	0.0	0.0	0.0	13.1	−18.4	0.0	0.0	0.0	16.2	−22.7	0.0	0.0	0.0
	4:12 (18.4°)	1	−59.0	−47.6	−73.1	−65.2	−53.4	−74.6	−60.2	−92.5	−82.5	−67.6	−92.1	−74.3	−114.2	−101.8	−83.5
		2	20.4	−20.9	0.0	0.0	0.0	25.8	−26.4	0.0	0.0	0.0	31.9	−32.6	0.0	0.0	0.0
	5:12 (22.6°)	1	−47.3	−47.6	−73.1	−65.2	−53.4	−59.9	−60.2	−92.5	−82.5	−67.6	−73.9	−74.3	−114.2	−101.8	−83.5
		2	27.2	−22.8	0.0	0.0	0.0	34.4	−28.8	0.0	0.0	0.0	42.5	−35.6	0.0	0.0	0.0
	6:12 (26.6°)	1	−38.0	−47.6	−73.1	−65.2	−53.4	−48.1	−60.2	−92.5	−82.5	−67.6	−59.4	−74.3	−114.2	−101.8	−83.5
		2	30.0	−22.8	0.0	0.0	0.0	38.0	−28.8	0.0	0.0	0.0	46.9	−35.6	0.0	0.0	0.0
	9:12 (36.9°)	1	−22.0	−47.6	−73.1	−65.2	−53.4	−27.8	−60.2	−92.5	−82.5	−67.6	−34.4	−74.3	−114.2	−101.8	−83.5
		2	35.9	−22.8	0.0	0.0	0.0	45.4	−28.8	0.0	0.0	0.0	56.0	−35.6	0.0	0.0	0.0
	12:12 (45.0°)	1	−12.4	−47.6	−73.1	−65.2	−53.4	−15.7	−60.2	−92.5	−82.5	−67.6	−19.4	−74.3	−114.2	−101.8	−83.5
		2	35.9	−22.8	0.0	0.0	0.0	45.4	−28.8	0.0	0.0	0.0	56.0	−35.6	0.0	0.0	0.0
80	Flat < 2:12 (9.46°)	1	NA	NA	−71.3	−63.6	−52.1	NA	NA	−90.2	−80.5	−66.0	NA	NA	−111.4	−99.3	−81.5
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−70.0	−50.4	−71.3	−63.6	−52.1	−88.5	−60.2	−90.2	−80.5	−66.0	−109.3	−74.3	−111.4	−99.3	−81.5
		2	10.1	−14.2	0.0	0.0	0.0	12.8	−18.0	0.0	0.0	0.0	15.8	−22.2	0.0	0.0	0.0
	4:12 (18.4°)	1	−57.5	−46.4	−71.3	−63.6	−52.1	−72.8	−58.7	−90.2	−80.5	−66.0	−89.9	−72.5	−111.4	−99.3	−81.5
		2	19.9	−20.4	0.0	0.0	0.0	25.2	−25.8	0.0	0.0	0.0	31.1	−31.8	0.0	0.0	0.0
	5:12 (22.6°)	1	−46.1	−46.4	−71.3	−63.6	−52.1	−58.4	−58.7	−90.2	−80.5	−66.0	−72.1	−72.5	−111.4	−99.3	−81.5
		2	26.5	−22.2	0.0	0.0	0.0	33.5	−28.1	0.0	0.0	0.0	41.4	−34.7	0.0	0.0	0.0
	6:12 (26.6°)	1	−37.1	−46.4	−71.3	−63.6	−52.1	−46.9	−58.7	−90.2	−80.5	−66.0	−57.9	−72.5	−111.4	−99.3	−81.5
		2	29.3	−22.2	0.0	0.0	0.0	37.0	−28.1	0.0	0.0	0.0	45.7	−34.7	0.0	0.0	0.0
	9:12 (36.9°)	1	−21.5	−46.4	−71.3	−63.6	−52.1	−27.2	−58.7	−90.2	−80.5	−66.0	−33.5	−72.5	−111.4	−99.3	−81.5
		2	35.0	−22.2	0.0	0.0	0.0	44.3	−28.1	0.0	0.0	0.0	54.7	−34.7	0.0	0.0	0.0
	12:12 (45.0°)	1	−12.1	−46.4	−71.3	−63.6	−52.1	−15.3	−58.7	−90.2	−80.5	−66.0	−18.9	−72.5	−111.4	−99.3	−81.5
		2	35.0	−22.2	0.0	0.0	0.0	44.3	−28.1	0.0	0.0	0.0	54.7	−34.7	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: h = 50–70 ft, V = 110–120 mi/h

			V (mi/h)														
			110					115					120				
	Roof Slope	Load	Zone					Zone					Zone				
h (ft)		Case	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3
70	Flat < 2:12 (9.46°)	1	NA	NA	−32.8	−29.2	−24.0	NA	NA	−35.8	−31.9	−26.2	NA	NA	−39.0	−34.8	−28.5
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−32.1	−21.9	−32.8	−29.2	−24.0	−35.1	−25.3	−35.8	−31.9	−26.2	−38.3	−26.0	−39.0	−34.8	−28.5
		2	4.6	−6.5	0.0	0.0	0.0	5.1	−7.1	0.0	0.0	0.0	5.5	−7.8	0.0	0.0	0.0
	4:12 (18.4°)	1	−26.4	−21.3	−32.8	−29.2	−24.0	−28.9	−23.3	−35.8	−31.9	−26.2	−31.5	−25.4	−39.0	−34.8	−28.5
		2	9.2	−9.4	0.0	0.0	0.0	10.0	−10.2	0.0	0.0	0.0	10.9	−11.1	0.0	0.0	0.0
	5:12 (22.6°)	1	−21.2	−21.3	−32.8	−29.2	−24.0	−23.2	−23.3	−35.8	−31.9	−26.2	−25.2	−25.4	−39.0	−34.8	−28.5
		2	12.2	−10.2	0.0	0.0	0.0	13.3	−11.1	0.0	0.0	0.0	14.5	−12.1	0.0	0.0	0.0
	6:12 (26.6°)	1	−17.0	−21.3	−32.8	−29.2	−24.0	−18.6	−23.3	−35.8	−31.9	−26.2	−20.3	−25.4	−39.0	−34.8	−28.5
		2	13.4	−10.2	0.0	0.0	0.0	14.7	−11.1	0.0	0.0	0.0	16.0	−12.1	0.0	0.0	0.0
	9:12 (36.9°)	1	−9.9	−21.3	−32.8	−29.2	−24.0	−10.8	−23.3	−35.8	−31.9	−26.2	−11.7	−25.4	−39.0	−34.8	−28.5
		2	16.1	−10.2	0.0	0.0	0.0	17.6	−11.1	0.0	0.0	0.0	19.1	−12.1	0.0	0.0	0.0
	12:12 (45.0°)	1	−5.6	−21.3	−32.8	−29.2	−24.0	−6.1	−23.3	−35.8	−31.9	−26.2	−6.6	−25.4	−39.0	−34.8	−28.5
		2	16.1	−10.2	0.0	0.0	0.0	17.6	−11.1	0.0	0.0	0.0	19.1	−12.1	0.0	0.0	0.0
60	Flat < 2:12 (9.46°)	1	NA	NA	−31.7	−28.3	−23.2	NA	NA	−34.7	−30.9	−25.3	NA	NA	−37.8	−33.7	−27.6
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−31.1	−21.2	−31.7	−28.3	−23.2	−34.0	−24.5	−34.7	−30.9	−25.3	−37.0	−25.2	−37.8	−33.7	−27.6
		2	4.5	−6.3	0.0	0.0	0.0	4.9	−6.9	0.0	0.0	0.0	5.3	−7.5	0.0	0.0	0.0
	4:12 (18.4°)	1	−25.6	−20.6	−31.7	−28.3	−23.2	−28.0	−22.6	−34.7	−30.9	−25.3	−30.4	−24.6	−37.8	−33.7	−27.6
		2	8.9	−9.1	0.0	0.0	0.0	9.7	−9.9	0.0	0.0	0.0	10.5	−10.8	0.0	0.0	0.0
	5:12 (22.6°)	1	−20.5	−20.6	−31.7	−28.3	−23.2	−22.4	−22.6	−34.7	−30.9	−25.3	−24.4	−24.6	−37.8	−33.7	−27.6
		2	11.8	−9.9	0.0	0.0	0.0	12.9	−10.8	0.0	0.0	0.0	14.0	−11.8	0.0	0.0	0.0
	6:12 (26.6°)	1	−16.5	−20.6	−31.7	−28.3	−23.2	−18.0	−22.6	−34.7	−30.9	−25.3	−19.6	−24.6	−37.8	−33.7	−27.6
		2	13.0	−9.9	0.0	0.0	0.0	14.2	−10.8	0.0	0.0	0.0	15.5	−11.8	0.0	0.0	0.0
	9:12 (36.9°)	1	−9.5	−20.6	−31.7	−28.3	−23.2	−10.4	−22.6	−34.7	−30.9	−25.3	−11.4	−24.6	−37.8	−33.7	−27.6
		2	15.6	−9.9	0.0	0.0	0.0	17.0	−10.8	0.0	0.0	0.0	18.5	−11.8	0.0	0.0	0.0
	12:12 (45.0°)	1	−5.4	−20.6	−31.7	−28.3	−23.2	−5.9	−22.6	−34.7	−30.9	−25.3	−6.4	−24.6	−37.8	−33.7	−27.6
		2	15.6	−9.9	0.0	0.0	0.0	17.0	−10.8	0.0	0.0	0.0	18.5	−11.8	0.0	0.0	0.0
50	Flat < 2:12 (9.46°)	1	NA	NA	−30.5	−27.2	−22.3	NA	NA	−33.4	−29.7	−24.4	NA	NA	−36.3	−32.4	−26.6
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−30.0	−20.4	−30.5	−27.2	−22.3	−32.7	−23.6	−33.4	−29.7	−24.4	−35.6	−24.2	−36.3	−32.4	−26.6
		2	4.3	−6.1	0.0	0.0	0.0	4.7	−6.6	0.0	0.0	0.0	5.1	−7.2	0.0	0.0	0.0
	4:12 (18.4°)	1	−24.6	−19.9	−30.5	−27.2	−22.3	−26.9	−21.7	−33.4	−29.7	−24.4	−29.3	−23.6	−36.3	−32.4	−26.6
		2	8.5	−8.7	0.0	0.0	0.0	9.3	−9.5	0.0	0.0	0.0	10.1	−10.4	0.0	0.0	0.0
	5:12 (22.6°)	1	−19.8	−19.9	−30.5	−27.2	−22.3	−21.6	−21.7	−33.4	−29.7	−24.4	−23.5	−23.6	−36.3	−32.4	−26.6
		2	11.3	−9.5	0.0	0.0	0.0	12.4	−10.4	0.0	0.0	0.0	13.5	−11.3	0.0	0.0	0.0
	6:12 (26.6°)	1	−15.9	−19.9	−30.5	−27.2	−22.3	−17.3	−21.7	−33.4	−29.7	−24.4	−18.9	−23.6	−36.3	−32.4	−26.6
		2	12.5	−9.5	0.0	0.0	0.0	13.7	−10.4	0.0	0.0	0.0	14.9	−11.3	0.0	0.0	0.0
	9:12 (36.9°)	1	−9.2	−19.9	−30.5	−27.2	−22.3	−10.0	−21.7	−33.4	−29.7	−24.4	−10.9	−23.6	−36.3	−32.4	−26.6
		2	15.0	−9.5	0.0	0.0	0.0	16.4	−10.4	0.0	0.0	0.0	17.8	−11.3	0.0	0.0	0.0
	12:12 (45.0°)	1	−5.2	−19.9	−30.5	−27.2	−22.3	−5.7	−21.7	−33.4	−29.7	−24.4	−6.2	−23.6	−36.3	−32.4	−26.6
		2	15.0	−9.5	0.0	0.0	0.0	16.4	−10.4	0.0	0.0	0.0	17.8	−11.3	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 50$–70 ft, $V = 130$–150 mi/h

			V (mi/h)														
	V (mi/h)		130					140					150				
h	Roof Slope	Load	Zone					Zone					Zone				
(ft)		Case	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
70	Flat < 2:12 (9.46°)	1	NA	NA	–45.8	–40.8	–33.5	NA	NA	–53.1	–47.3	–38.8	NA	NA	–60.9	–54.3	–44.5
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–44.9	–30.5	–45.8	–40.8	–33.5	–52.1	–35.4	–53.1	–47.3	–38.8	–59.8	–40.6	–60.9	–54.3	–44.5
		2	6.5	–9.1	0.0	0.0	0.0	7.5	–10.6	0.0	0.0	0.0	8.6	–12.1	0.0	0.0	0.0
	4:12 (18.4°)	1	–36.9	–29.8	–45.8	–40.8	–33.5	–42.8	–34.6	–53.1	–47.3	–38.8	–49.1	–39.7	–60.9	–54.3	–44.5
		2	12.8	–13.1	0.0	0.0	0.0	14.8	–15.2	0.0	0.0	0.0	17.0	–17.4	0.0	0.0	0.0
	5:12 (22.6°)	1	–29.6	–29.8	–45.8	–40.8	–33.5	–34.4	–34.6	–53.1	–47.3	–38.8	–39.4	–39.7	–60.9	–54.3	–44.5
		2	17.0	–14.2	0.0	0.0	0.0	19.7	–16.5	0.0	0.0	0.0	22.6	–19.0	0.0	0.0	0.0
	6:12 (26.6°)	1	–23.8	–29.8	–45.8	–40.8	–33.5	–27.6	–34.6	–53.1	–47.3	–38.8	–31.7	–39.7	–60.9	–54.3	–44.5
		2	18.8	–14.2	0.0	0.0	0.0	21.8	–16.5	0.0	0.0	0.0	25.0	–19.0	0.0	0.0	0.0
	9:12 (36.9°)	1	–13.8	–29.8	–45.8	–40.8	–33.5	–16.0	–34.6	–53.1	–47.3	–38.8	–18.3	–39.7	–60.9	–54.3	–44.5
		2	22.5	–14.2	0.0	0.0	0.0	26.0	–16.5	0.0	0.0	0.0	9.6	–19.0	0.0	0.0	0.0
	12:12 (45.0°)	1	–7.8	–29.8	–45.8	–40.8	–33.5	–9.0	–34.6	–53.1	–47.3	–38.8	–10.3	–39.7	–60.9	–54.3	–44.5
		2	22.5	–14.2	0.0	0.0	0.0	26.0	–16.5	0.0	0.0	0.0	29.9	–19.0	0.0	0.0	0.0
60	Flat < 2:12 (9.46°)	1	NA	NA	–44.3	–39.5	–32.4	NA	NA	–51.4	–45.8	–37.6	NA	NA	–59.0	–52.6	–43.1
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–43.5	–29.6	–44.3	–39.5	–32.4	–50.4	–34.3	–51.4	–45.8	–37.6	–57.9	–39.3	–59.0	–52.6	–43.1
		2	6.3	–8.8	0.0	0.0	0.0	7.3	–10.2	0.0	0.0	0.0	8.3	–11.7	0.0	0.0	0.0
	4:12 (18.4°)	1	–35.7	–28.8	–44.3	–39.5	–32.4	–41.4	–33.4	–51.4	–45.8	–37.6	–47.6	–38.4	–59.0	–52.6	–43.1
		2	12.4	–12.7	0.0	0.0	0.0	14.3	–14.7	0.0	0.0	0.0	16.5	–16.9	0.0	0.0	0.0
	5:12 (22.6°)	1	–28.7	–28.8	–44.3	–39.5	–32.4	–33.3	–33.4	–51.4	–45.8	–37.6	–38.2	–38.4	–59.0	–52.6	–43.1
		2	16.5	–13.8	0.0	0.0	0.0	19.1	–16.0	0.0	0.0	0.0	21.9	–18.4	0.0	0.0	0.0
	6:12 (26.6°)	1	–23.0	–28.8	–44.3	–39.5	–32.4	–26.7	–33.4	–51.4	–45.8	–37.6	–30.7	–38.4	–59.0	–52.6	–43.1
		2	18.2	–13.8	0.0	0.0	0.0	21.1	–16.0	0.0	0.0	0.0	24.2	–18.4	0.0	0.0	0.0
	9:12 (36.9°)	1	–13.3	–28.8	–44.3	–39.5	–32.4	–15.5	–33.4	–51.4	–45.8	–37.6	–17.8	–38.4	–59.0	–52.6	–43.1
		2	21.7	–13.8	0.0	0.0	0.0	25.2	–16.0	0.0	0.0	0.0	9.3	–18.4	0.0	0.0	0.0
	12:12 (45.0°)	1	–7.5	–28.8	–44.3	–39.5	–32.4	–8.7	–33.4	–51.4	–45.8	–37.6	–10.0	–38.4	–59.0	–52.6	–43.1
		2	21.7	–13.8	0.0	0.0	0.0	25.2	–16.0	0.0	0.0	0.0	28.9	–18.4	0.0	0.0	0.0
50	Flat < 2:12 (9.46°)	1	NA	NA	–42.6	–38.0	–31.2	NA	NA	–49.4	–44.1	–36.2	NA	NA	–56.8	–50.6	–41.5
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–41.8	–28.4	–42.6	–38.0	–31.2	–48.5	–33.0	–49.4	–44.1	–36.2	–55.7	–37.9	–56.8	–50.6	–41.5
		2	6.0	–8.5	0.0	0.0	0.0	7.0	–9.8	0.0	0.0	0.0	8.0	–11.3	0.0	0.0	0.0
	4:12 (18.4°)	1	–34.4	–27.8	–42.6	–38.0	–31.2	–39.9	–32.2	–49.4	–44.1	–36.2	–45.8	–37.0	–56.8	–50.6	–41.5
		2	11.9	–12.2	0.0	0.0	0.0	13.8	–14.1	0.0	0.0	0.0	15.9	–16.2	0.0	0.0	0.0
	5:12 (22.6°)	1	–27.6	–27.8	–42.6	–38.0	–31.2	–32.0	–32.2	–49.4	–44.1	–36.2	–36.7	–37.0	–56.8	–50.6	–41.5
		2	15.8	–13.3	0.0	0.0	0.0	18.4	–15.4	0.0	0.0	0.0	21.1	–17.7	0.0	0.0	0.0
	6:12 (26.6°)	1	–22.2	–27.8	–42.6	–38.0	–31.2	–25.7	–32.2	–49.4	–44.1	–36.2	–29.5	–37.0	–56.8	–50.6	–41.5
		2	17.5	–13.3	0.0	0.0	0.0	20.3	–15.4	0.0	0.0	0.0	23.3	–17.7	0.0	0.0	0.0
	9:12 (36.9°)	1	–12.8	–27.8	–42.6	–38.0	–31.2	–14.9	–32.2	–49.4	–44.1	–36.2	–17.1	–37.0	–56.8	–50.6	–41.5
		2	20.9	–13.3	0.0	0.0	0.0	24.3	–15.4	0.0	0.0	0.0	8.9	–17.7	0.0	0.0	0.0
	12:12 (45.0°)	1	–7.2	–27.8	–42.6	–38.0	–31.2	–8.4	–32.2	–49.4	–44.1	–36.2	–9.6	–37.0	–56.8	–50.6	–41.5
		2	20.9	–13.3	0.0	0.0	0.0	24.3	–15.4	0.0	0.0	0.0	27.8	–17.7	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \le 160$ ft ($h \le 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 50–70$ ft, $V = 160–200$ mi/h

			V (mi/h)														
			160					180					200				
	Roof Slope	Load Case	Zone					Zone					Zone				
h (ft)			1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
70	Flat < 2:12 (9.46°)	1	NA	NA	−69.3	−61.8	−50.7	NA	NA	−87.7	−78.2	−64.2	NA	NA	−108.3	−96.6	−79.2
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−68.0	−49.0	−69.3	−61.8	−50.7	−86.1	−58.5	−87.7	−78.2	−64.2	−106.3	−72.2	−108.3	−96.6	−79.2
		2	9.8	−13.8	0.0	0.0	0.0	12.4	−17.5	0.0	0.0	0.0	15.3	−21.6	0.0	0.0	0.0
	4:12 (18.4°)	1	−55.9	−45.1	−69.3	−61.8	−50.7	−70.8	−57.1	−87.7	−78.2	−64.2	−87.4	−70.5	−108.3	−96.6	−79.2
		2	19.4	−19.8	0.0	0.0	0.0	24.5	−25.1	0.0	0.0	0.0	30.2	−31.0	0.0	0.0	0.0
	5:12 (22.6°)	1	−44.9	−45.1	−69.3	−61.8	−50.7	−56.8	−57.1	−87.7	−78.2	−64.2	−70.1	−70.5	−108.3	−96.6	−79.2
		2	25.8	−21.6	0.0	0.0	0.0	32.6	−27.3	0.0	0.0	0.0	40.3	−33.7	0.0	0.0	0.0
	6:12 (26.6°)	1	−36.0	−45.1	−69.3	−61.8	−50.7	−45.6	−57.1	−87.7	−78.2	−64.2	−56.3	−70.5	−108.3	−96.6	−79.2
		2	28.4	−21.6	0.0	0.0	0.0	36.0	−27.3	0.0	0.0	0.0	44.5	−33.7	0.0	0.0	0.0
	9:12 (36.9°)	1	−20.9	−45.1	−69.3	−61.8	−50.7	−26.4	−57.1	−87.7	−78.2	−64.2	−32.6	−70.5	−108.3	−96.6	−79.2
		2	34.0	−21.6	0.0	0.0	0.0	43.0	−27.3	0.0	0.0	0.0	53.1	−33.7	0.0	0.0	0.0
	12:12 (45.0°)	1	−11.8	−45.1	−69.3	−61.8	−50.7	−14.9	−57.1	−87.7	−78.2	−64.2	−18.4	−70.5	−108.3	−96.6	−79.2
		2	34.0	−21.6	0.0	0.0	0.0	43.0	−27.3	0.0	0.0	0.0	53.1	−33.7	0.0	0.0	0.0
60	Flat < 2:12 (9.46°)	1	NA	NA	−67.1	−59.8	−49.1	NA	NA	−84.9	−75.7	−62.1	NA	NA	−104.9	−93.5	−76.7
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−65.8	−47.4	−67.1	−59.8	−49.1	−83.3	−56.7	−84.9	−75.7	−62.1	−102.9	−69.9	−104.9	−93.5	−76.7
		2	9.5	−13.4	0.0	0.0	0.0	12.0	−16.9	0.0	0.0	0.0	14.8	−20.9	0.0	0.0	0.0
	4:12 (18.4°)	1	−54.1	−43.7	−67.1	−59.8	−49.1	−68.5	−55.3	−84.9	−75.7	−62.1	−84.6	−68.3	−104.9	−93.5	−76.7
		2	18.7	−19.2	0.0	0.0	0.0	23.7	−24.3	0.0	0.0	0.0	29.3	−30.0	0.0	0.0	0.0
	5:12 (22.6°)	1	−43.4	−43.7	−67.1	−59.8	−49.1	−55.0	−55.3	−84.9	−75.7	−62.1	−67.9	−68.3	−104.9	−93.5	−76.7
		2	24.9	−20.9	0.0	0.0	0.0	31.6	−26.4	0.0	0.0	0.0	39.0	−32.6	0.0	0.0	0.0
	6:12 (26.6°)	1	−34.9	−43.7	−67.1	−59.8	−49.1	−44.2	−55.3	−84.9	−75.7	−62.1	−54.5	−68.3	−104.9	−93.5	−76.7
		2	27.5	−20.9	0.0	0.0	0.0	34.9	−26.4	0.0	0.0	0.0	43.0	−32.6	0.0	0.0	0.0
	9:12 (36.9°)	1	−20.2	−43.7	−67.1	−59.8	−49.1	−25.6	−55.3	−84.9	−75.7	−62.1	−31.6	−68.3	−104.9	−93.5	−76.7
		2	32.9	−20.9	0.0	0.0	0.0	41.7	−26.4	0.0	0.0	0.0	51.4	−32.6	0.0	0.0	0.0
	12:12 (45.0°)	1	−11.4	−43.7	−67.1	−59.8	−49.1	−14.4	−55.3	−84.9	−75.7	−62.1	−17.8	−68.3	−104.9	−93.5	−76.7
		2	32.9	−20.9	0.0	0.0	0.0	41.7	−26.4	0.0	0.0	0.0	51.4	−32.6	0.0	0.0	0.0
50	Flat < 2:12 (9.46°)	1	NA	NA	−64.6	−57.6	−47.2	NA	NA	−81.7	−72.9	−59.8	NA	NA	−100.9	−90.0	−73.8
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−63.4	−45.6	−64.6	−57.6	−47.2	−80.2	−54.5	−81.7	−72.9	−59.8	−99.0	−67.3	−100.9	−90.0	−73.8
		2	9.1	−12.9	0.0	0.0	0.0	11.6	−16.3	0.0	0.0	0.0	14.3	−20.1	0.0	0.0	0.0
	4:12 (18.4°)	1	−52.1	−42.0	−64.6	−57.6	−47.2	−65.9	−53.2	−81.7	−72.9	−59.8	−81.4	−65.7	−100.9	−90.0	−73.8
		2	18.0	−18.5	0.0	0.0	0.0	22.8	−23.4	0.0	0.0	0.0	28.2	−28.8	0.0	0.0	0.0
	5:12 (22.6°)	1	−41.8	−42.0	−64.6	−57.6	−47.2	−52.9	−53.2	−81.7	−72.9	−59.8	−65.3	−65.7	−100.9	−90.0	−73.8
		2	24.0	−20.1	0.0	0.0	0.0	30.4	−25.4	0.0	0.0	0.0	37.5	−31.4	0.0	0.0	0.0
	6:12 (26.6°)	1	−33.6	−42.0	−64.6	−57.6	−47.2	−42.5	−53.2	−81.7	−72.9	−59.8	−52.5	−65.7	−100.9	−90.0	−73.8
		2	26.5	−20.1	0.0	0.0	0.0	33.5	−25.4	0.0	0.0	0.0	41.4	−31.4	0.0	0.0	0.0
	9:12 (36.9°)	1	−19.4	−42.0	−64.6	−57.6	−47.2	−24.6	−53.2	−81.7	−72.9	−59.8	−30.4	−65.7	−100.9	−90.0	−73.8
		2	31.7	−20.1	0.0	0.0	0.0	40.1	−25.4	0.0	0.0	0.0	49.5	−31.4	0.0	0.0	0.0
	12:12 (45.0°)	1	−11.0	−42.0	−64.6	−57.6	−47.2	−13.9	−53.2	−81.7	−72.9	−59.8	−17.1	−65.7	−100.9	−90.0	−73.8
		2	31.7	−20.1	0.0	0.0	0.0	40.1	−25.4	0.0	0.0	0.0	49.5	−31.4	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \le 160$ ft ($h \le 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 20$–40 ft, $V = 110$–120 mi/h

			V (mi/h)														
			110					115					120				
	Roof Slope	Load Case	Zone					Zone					Zone				
h (ft)			1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
40	Flat < 2:12 (9.46°)	1	NA	NA	–29.1	–26.0	–21.3	NA	NA	–31.8	–28.4	–23.3	NA	NA	–34.7	–30.9	–25.3
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–28.6	–19.4	–29.1	–26.0	–21.3	–31.2	–22.5	–31.8	–28.4	–23.3	–34.0	–23.1	–34.7	–30.9	–25.3
		2	4.1	–5.8	0.0	0.0	0.0	4.5	–6.3	0.0	0.0	0.0	4.9	–6.9	0.0	0.0	0.0
	4:12 (18.4°)	1	–23.5	–19.0	–29.1	–26.0	–21.3	–25.7	–20.7	–31.8	–28.4	–23.3	–28.0	–22.6	–34.7	–30.9	–25.3
		2	8.1	–8.3	0.0	0.0	0.0	8.9	–9.1	0.0	0.0	0.0	9.7	–9.9	0.0	0.0	0.0
	5:12 (22.6°)	1	–18.8	–19.0	–29.1	–26.0	–21.3	–20.6	–20.7	–31.8	–28.4	–23.3	–22.4	–22.6	–34.7	–30.9	–25.3
		2	10.8	–9.1	0.0	0.0	0.0	11.8	–9.9	0.0	0.0	0.0	12.9	–10.8	0.0	0.0	0.0
	6:12 (26.6°)	1	–15.1	–19.0	–29.1	–26.0	–21.3	–16.5	–20.7	–31.8	–28.4	–23.3	–18.0	–22.6	–34.7	–30.9	–25.3
		2	12.0	–9.1	0.0	0.0	0.0	13.1	–9.9	0.0	0.0	0.0	14.2	–10.8	0.0	0.0	0.0
	9:12 (36.9°)	1	–8.8	–19.0	–29.1	–26.0	–21.3	–9.6	–20.7	–31.8	–28.4	–23.3	–10.4	–22.6	–34.7	–30.9	–25.3
		2	14.3	–9.1	0.0	0.0	0.0	15.6	–9.9	0.0	0.0	0.0	17.0	–10.8	0.0	0.0	0.0
	12:12 (45.0°)	1	–4.9	–19.0	–29.1	–26.0	–21.3	–5.4	–20.7	–31.8	–28.4	–23.3	–5.9	–22.6	–34.7	–30.9	–25.3
		2	14.3	–9.1	0.0	0.0	0.0	15.6	–9.9	0.0	0.0	0.0	17.0	–10.8	0.0	0.0	0.0
30	Flat < 2:12 (9.46°)	1	NA	NA	–27.4	–24.4	–20.0	NA	NA	–30.0	–26.7	–21.9	NA	NA	–32.6	–29.1	–23.9
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–26.9	–18.3	–27.4	–24.4	–20.0	–29.4	–21.2	–30.0	–26.7	–21.9	–32.0	–21.8	–32.6	–29.1	–23.9
		2	3.9	–5.5	0.0	0.0	0.0	4.2	–6.0	0.0	0.0	0.0	4.6	–6.5	0.0	0.0	0.0
	4:12 (18.4°)	1	–22.1	–17.8	–27.4	–24.4	–20.0	–24.2	–19.5	–30.0	–26.7	–21.9	–26.3	–21.2	–32.6	–29.1	–23.9
		2	7.7	–7.8	0.0	0.0	0.0	8.4	–8.6	0.0	0.0	0.0	9.1	–9.3	0.0	0.0	0.0
	5:12 (22.6°)	1	–17.7	–17.8	–27.4	–24.4	–20.0	–19.4	–19.5	–30.0	–26.7	–21.9	–21.1	–21.2	–32.6	–29.1	–23.9
		2	10.2	–8.5	0.0	0.0	0.0	11.1	–9.3	0.0	0.0	0.0	12.1	–10.2	0.0	0.0	0.0
	6:12 (26.6°)	1	–14.3	–17.8	–27.4	–24.4	–20.0	–15.6	–19.5	–30.0	–26.7	–21.9	–17.0	–21.2	–32.6	–29.1	–23.9
		2	11.3	–8.5	0.0	0.0	0.0	12.3	–9.3	0.0	0.0	0.0	13.4	–10.2	0.0	0.0	0.0
	9:12 (36.9°)	1	–8.3	–17.8	–27.4	–24.4	–20.0	–9.0	–19.5	–30.0	–26.7	–21.9	–9.8	–21.2	–32.6	–29.1	–23.9
		2	13.4	–8.5	0.0	0.0	0.0	14.7	–9.3	0.0	0.0	0.0	16.0	–10.2	0.0	0.0	0.0
	12:12 (45.0°)	1	–4.7	–17.8	–27.4	–24.4	–20.0	–5.1	–19.5	–30.0	–26.7	–21.9	–5.5	–21.2	–32.6	–29.1	–23.9
		2	13.4	–8.5	0.0	0.0	0.0	14.7	–9.3	0.0	0.0	0.0	16.0	–10.2	0.0	0.0	0.0
20	Flat < 2:12 (9.46°)	1	NA	NA	–25.2	–22.4	–18.4	NA	NA	–27.5	–24.5	–20.1	NA	NA	–30.0	–26.7	–21.9
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–24.7	–16.8	–25.2	–22.4	–18.4	–27.0	–19.4	–27.5	–24.5	–20.1	–29.4	–20.0	–30.0	–26.7	–21.9
		2	3.6	–5.0	0.0	0.0	0.0	3.9	–5.5	0.0	0.0	0.0	4.2	–6.0	0.0	0.0	0.0
	4:12 (18.4°)	1	–20.3	–16.4	–25.2	–22.4	–18.4	–22.2	–17.9	–27.5	–24.5	–20.1	–24.2	–19.5	–30.0	–26.7	–21.9
		2	7.0	–7.2	0.0	0.0	0.0	7.7	–7.9	0.0	0.0	0.0	8.4	–8.6	0.0	0.0	0.0
	5:12 (22.6°)	1	–16.3	–16.4	–25.2	–22.4	–18.4	–17.8	–17.9	–27.5	–24.5	–20.1	–19.4	–19.5	–30.0	–26.7	–21.9
		2	9.4	–7.8	0.0	0.0	0.0	10.2	–8.6	0.0	0.0	0.0	11.1	–9.3	0.0	0.0	0.0
	6:12 (26.6°)	1	–13.1	–16.4	–25.2	–22.4	–18.4	–14.3	–17.9	–27.5	–24.5	–20.1	–15.6	–19.5	–30.0	–26.7	–21.9
		2	10.3	–7.8	0.0	0.0	0.0	11.3	–8.6	0.0	0.0	0.0	12.3	–9.3	0.0	0.0	0.0
	9:12 (36.9°)	1	–7.6	–16.4	–25.2	–22.4	–18.4	–8.3	–17.9	–27.5	–24.5	–20.1	–9.0	–19.5	–30.0	–26.7	–21.9
		2	12.3	–7.8	0.0	0.0	0.0	13.5	–8.6	0.0	0.0	0.0	14.7	–9.3	0.0	0.0	0.0
	12:12 (45.0°)	1	–4.3	–16.4	–25.2	–22.4	–18.4	–4.7	–17.9	–27.5	–24.5	–20.1	–5.1	–19.5	–30.0	–26.7	–21.9
		2	12.3	–7.8	0.0	0.0	0.0	13.5	–8.6	0.0	0.0	0.0	14.7	–9.3	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 20$–40 ft, $V = 130$–150 mi/h

			V (mi/h)														
			130					140					150				
			Zone					Zone					Zone				
h (ft)	Roof Slope	Load Case	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
40	Flat < 2:12 (9.46°)	1	NA	NA	–40.7	–36.3	–29.7	NA	NA	–47.2	–42.1	–34.5	NA	NA	–54.2	–48.3	–39.6
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–39.9	–27.1	–40.7	–36.3	–29.7	–46.3	–31.5	–47.2	–42.1	–34.5	–53.1	–36.1	–54.2	–48.3	–39.6
		2	5.8	–8.1	0.0	0.0	0.0	6.7	–9.4	0.0	0.0	0.0	7.7	–10.8	0.0	0.0	0.0
	4:12 (18.4°)	1	–32.8	–26.5	–40.7	–36.3	–29.7	–38.1	–30.7	–47.2	–42.1	–34.5	–43.7	–35.3	–54.2	–48.3	–39.6
		2	11.4	–11.6	0.0	0.0	0.0	13.2	–13.5	0.0	0.0	0.0	15.1	–15.5	0.0	0.0	0.0
	5:12 (22.6°)	1	–26.3	–26.5	–40.7	–36.3	–29.7	–30.5	–30.7	–47.2	–42.1	–34.5	–35.1	–35.3	–54.2	–48.3	–39.6
		2	15.1	–12.7	0.0	0.0	0.0	17.5	–14.7	0.0	0.0	0.0	20.1	–16.9	0.0	0.0	0.0
	6:12 (26.6°)	1	–21.1	–26.5	–40.7	–36.3	–29.7	–24.5	–30.7	–47.2	–42.1	–34.5	–28.2	–35.3	–54.2	–48.3	–39.6
		2	16.7	–12.7	0.0	0.0	0.0	19.4	–14.7	0.0	0.0	0.0	22.2	–16.9	0.0	0.0	0.0
	9:12 (36.9°)	1	–12.2	–26.5	–40.7	–36.3	–29.7	–14.2	–30.7	–47.2	–42.1	–34.5	–16.3	–35.3	–54.2	–48.3	–39.6
		2	20.0	–12.7	0.0	0.0	0.0	23.1	–14.7	0.0	0.0	0.0	8.5	–16.9	0.0	0.0	0.0
	12:12 (45.0°)	1	–6.9	–26.5	–40.7	–36.3	–29.7	–8.0	–30.7	–47.2	–42.1	–34.5	–9.2	–35.3	–54.2	–48.3	–39.6
		2	20.0	–12.7	0.0	0.0	0.0	23.1	–14.7	0.0	0.0	0.0	26.6	–16.9	0.0	0.0	0.0
30	Flat < 2:12 (9.46°)	1	NA	NA	–38.3	–34.1	–28.0	NA	NA	–44.4	–39.6	–32.5	NA	NA	–51.0	–45.4	–37.3
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–37.6	–25.5	–38.3	–34.1	–28.0	–43.6	–29.6	–44.4	–39.6	–32.5	–50.0	–34.0	–51.0	–45.4	–37.3
		2	5.4	–7.6	0.0	0.0	0.0	6.3	–8.8	0.0	0.0	0.0	7.2	–10.1	0.0	0.0	0.0
	4:12 (18.4°)	1	–30.9	–24.9	–38.3	–34.1	–28.0	–35.8	–28.9	–44.4	–39.6	–32.5	–41.1	–33.2	–51.0	–45.4	–37.3
		2	10.7	–10.9	0.0	0.0	0.0	12.4	–12.7	0.0	0.0	0.0	14.2	–14.6	0.0	0.0	0.0
	5:12 (22.6°)	1	–24.8	–24.9	–38.3	–34.1	–28.0	–28.7	–28.9	–44.4	–39.6	–32.5	–33.0	–33.2	–51.0	–45.4	–37.3
		2	14.2	–11.9	0.0	0.0	0.0	16.5	–13.8	0.0	0.0	0.0	18.9	–15.9	0.0	0.0	0.0
	6:12 (26.6°)	1	–19.9	–24.9	–38.3	–34.1	–28.0	–23.1	–28.9	–44.4	–39.6	–32.5	–26.5	–33.2	–51.0	–45.4	–37.3
		2	15.7	–11.9	0.0	0.0	0.0	18.2	–13.8	0.0	0.0	0.0	20.9	–15.9	0.0	0.0	0.0
	9:12 (36.9°)	1	–11.5	–24.9	–38.3	–34.1	–28.0	–13.4	–28.9	–44.4	–39.6	–32.5	–15.3	–33.2	–51.0	–45.4	–37.3
		2	18.8	–11.9	0.0	0.0	0.0	21.8	–13.8	0.0	0.0	0.0	8.0	–15.9	0.0	0.0	0.0
	12:12 (45.0°)	1	–6.5	–24.9	–38.3	–34.1	–28.0	–7.5	–28.9	–44.4	–39.6	–32.5	–8.7	–33.2	–51.0	–45.4	–37.3
		2	18.8	–11.9	0.0	0.0	0.0	21.8	–13.8	0.0	0.0	0.0	25.0	–15.9	0.0	0.0	0.0
20	Flat < 2:12 (9.46°)	1	NA	NA	–35.2	–31.3	–25.7	NA	NA	–40.8	–36.3	–29.8	NA	NA	–46.8	–41.7	–34.2
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–34.5	–23.4	–35.2	–31.3	–25.7	–40.0	–27.2	–40.8	–36.3	–29.8	–45.9	–31.2	–46.8	–41.7	–34.2
		2	5.0	–7.0	0.0	0.0	0.0	5.8	–8.1	0.0	0.0	0.0	6.6	–9.3	0.0	0.0	0.0
	4:12 (18.4°)	1	–28.4	–22.9	–35.2	–31.3	–25.7	–32.9	–26.5	–40.8	–36.3	–29.8	–37.8	–30.5	–46.8	–41.7	–34.2
		2	9.8	–10.0	0.0	0.0	0.0	11.4	–11.7	0.0	0.0	0.0	13.1	–13.4	0.0	0.0	0.0
	5:12 (22.6°)	1	–22.8	–22.9	–35.2	–31.3	–25.7	–26.4	–26.5	–40.8	–36.3	–29.8	–30.3	–30.5	–46.8	–41.7	–34.2
		2	13.1	–10.9	0.0	0.0	0.0	15.2	–12.7	0.0	0.0	0.0	17.4	–14.6	0.0	0.0	0.0
	6:12 (26.6°)	1	–18.3	–22.9	–35.2	–31.3	–25.7	–21.2	–26.5	–40.8	–36.3	–29.8	–24.3	–30.5	–46.8	–41.7	–34.2
		2	14.4	–10.9	0.0	0.0	0.0	16.7	–12.7	0.0	0.0	0.0	19.2	–14.6	0.0	0.0	0.0
	9:12 (36.9°)	1	–10.6	–22.9	–35.2	–31.3	–25.7	–12.3	–26.5	–40.8	–36.3	–29.8	–14.1	–30.5	–46.8	–41.7	–34.2
		2	17.2	–10.9	0.0	0.0	0.0	20.0	–12.7	0.0	0.0	0.0	7.4	–14.6	0.0	0.0	0.0
	12:12 (45.0°)	1	–6.0	–22.9	–35.2	–31.3	–25.7	–6.9	–26.5	–40.8	–36.3	–29.8	–7.9	–30.5	–46.8	–41.7	–34.2
		2	17.2	–10.9	0.0	0.0	0.0	20.0	–12.7	0.0	0.0	0.0	23.0	–14.6	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \le 160$ ft ($h \le 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 20–40$ ft, $V = 160–200$ mi/h

			V (mi/h)														
			160					180					200				
h (ft)	Roof Slope	Load Case	Zone					Zone					Zone				
			1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
40	Flat < 2:12 (9.46°)	1	NA	NA	−61.6	−54.9	−45.1	NA	NA	−78.0	−69.5	−57.0	NA	NA	−96.3	−85.8	−70.4
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−60.5	−43.5	−61.6	−54.9	−45.1	−76.5	−52.0	−78.0	−69.5	−57.0	−94.5	−64.2	−96.3	−85.8	−70.4
		2	8.7	−12.3	0.0	0.0	0.0	11.0	−15.5	0.0	0.0	0.0	13.6	−19.2	0.0	0.0	0.0
	4:12 (18.4°)	1	−49.7	−40.1	−61.6	−54.9	−45.1	−62.9	−50.8	−78.0	−69.5	−57.0	−77.7	−62.7	−96.3	−85.8	−70.4
		2	17.2	−17.6	0.0	0.0	0.0	21.8	−22.3	0.0	0.0	0.0	26.9	−27.5	0.0	0.0	0.0
	5:12 (22.6°)	1	−39.9	−40.1	−61.6	−54.9	−45.1	−50.5	−50.8	−78.0	−69.5	−57.0	−62.3	−62.7	−96.3	−85.8	−70.4
		2	22.9	−19.2	0.0	0.0	0.0	29.0	−24.3	0.0	0.0	0.0	35.8	−30.0	0.0	0.0	0.0
	6:12 (26.6°)	1	−32.0	−40.1	−61.6	−54.9	−45.1	−40.5	−50.8	−78.0	−69.5	−57.0	−50.0	−62.7	−96.3	−85.8	−70.4
		2	25.3	−19.2	0.0	0.0	0.0	32.0	−24.3	0.0	0.0	0.0	39.5	−30.0	0.0	0.0	0.0
	9:12 (36.9°)	1	−18.5	−40.1	−61.6	−54.9	−45.1	−23.5	−50.8	−78.0	−69.5	−57.0	−29.0	−62.7	−96.3	−85.8	−70.4
		2	30.2	−19.2	0.0	0.0	0.0	38.3	−24.3	0.0	0.0	0.0	47.2	−30.0	0.0	0.0	0.0
	12:12 (45.0°)	1	−10.5	−40.1	−61.6	−54.9	−45.1	−13.2	−50.8	−78.0	−69.5	−57.0	−16.3	−62.7	−96.3	−85.8	−70.4
		2	30.2	−19.2	0.0	0.0	0.0	38.3	−24.3	0.0	0.0	0.0	47.2	−30.0	0.0	0.0	0.0
30	Flat < 2:12 (9.46°)	1	NA	NA	−58.0	−51.7	−42.4	NA	NA	−73.4	−65.4	−53.7	NA	NA	−90.6	−80.8	−66.3
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−56.9	−41.0	−58.0	−51.7	−42.4	−72.0	−49.0	−73.4	−65.4	−53.7	−88.9	−60.4	−90.6	−80.8	−66.3
		2	8.2	−11.5	0.0	0.0	0.0	10.4	−14.6	0.0	0.0	0.0	12.8	−18.0	0.0	0.0	0.0
	4:12 (18.4°)	1	−46.8	−37.8	−58.0	−51.7	−42.4	−59.2	−47.8	−73.4	−65.4	−53.7	−73.1	−59.0	−90.6	−80.8	−66.3
		2	16.2	−16.6	0.0	0.0	0.0	20.5	−21.0	0.0	0.0	0.0	25.3	−25.9	0.0	0.0	0.0
	5:12 (22.6°)	1	−37.5	−37.8	−58.0	−51.7	−42.4	−47.5	−47.8	−73.4	−65.4	−53.7	−58.6	−59.0	−90.6	−80.8	−66.3
		2	21.6	−18.1	0.0	0.0	0.0	27.3	−22.9	0.0	0.0	0.0	33.7	−28.2	0.0	0.0	0.0
	6:12 (26.6°)	1	−30.1	−37.8	−58.0	−51.7	−42.4	−38.2	−47.8	−73.4	−65.4	−53.7	−47.1	−59.0	−90.6	−80.8	−66.3
		2	23.8	−18.1	0.0	0.0	0.0	30.1	−22.9	0.0	0.0	0.0	37.2	−28.2	0.0	0.0	0.0
	9:12 (36.9°)	1	−17.5	−37.8	−58.0	−51.7	−42.4	−22.1	−47.8	−73.4	−65.4	−53.7	−27.3	−59.0	−90.6	−80.8	−66.3
		2	28.5	−18.1	0.0	0.0	0.0	36.0	−22.9	0.0	0.0	0.0	44.5	−28.2	0.0	0.0	0.0
	12:12 (45.0°)	1	−9.8	−37.8	−58.0	−51.7	−42.4	−12.5	−47.8	−73.4	−65.4	−53.7	−15.4	−59.0	−90.6	−80.8	−66.3
		2	28.5	−18.1	0.0	0.0	0.0	36.0	−22.9	0.0	0.0	0.0	44.5	−28.2	0.0	0.0	0.0
20	Flat < 2:12 (9.46°)	1	NA	NA	−53.3	−47.5	−38.9	NA	NA	−67.4	−60.1	−49.3	NA	NA	−83.2	−74.2	−60.8
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−52.2	−37.6	−53.3	−47.5	−38.9	−66.1	−45.0	−67.4	−60.1	−49.3	−81.6	−55.5	−83.2	−74.2	−60.8
		2	7.5	−10.6	0.0	0.0	0.0	9.5	−13.4	0.0	0.0	0.0	11.8	−16.6	0.0	0.0	0.0
	4:12 (18.4°)	1	−43.0	−34.7	−53.3	−47.5	−38.9	−54.4	−43.9	−67.4	−60.1	−49.3	−67.1	−54.2	−83.2	−74.2	−60.8
		2	14.9	−15.2	0.0	0.0	0.0	18.8	−19.3	0.0	0.0	0.0	23.2	−23.8	0.0	0.0	0.0
	5:12 (22.6°)	1	−34.5	−34.7	−53.3	−47.5	−38.9	−43.6	−43.9	−67.4	−60.1	−49.3	−53.9	−54.2	−83.2	−74.2	−60.8
		2	19.8	−16.6	0.0	0.0	0.0	25.1	−21.0	0.0	0.0	0.0	30.9	−25.9	0.0	0.0	0.0
	6:12 (26.6°)	1	−27.7	−34.7	−53.3	−47.5	−38.9	−35.0	−43.9	−67.4	−60.1	−49.3	−43.3	−54.2	−83.2	−74.2	−60.8
		2	21.9	−16.6	0.0	0.0	0.0	27.7	−21.0	0.0	0.0	0.0	34.1	−25.9	0.0	0.0	0.0
	9:12 (36.9°)	1	−16.0	−34.7	−53.3	−47.5	−38.9	−20.3	−43.9	−67.4	−60.1	−49.3	−25.0	−54.2	−83.2	−74.2	−60.8
		2	26.1	−16.6	0.0	0.0	0.0	33.1	−21.0	0.0	0.0	0.0	40.8	−25.9	0.0	0.0	0.0
	12:12 (45.0°)	1	−9.0	−34.7	−53.3	−47.5	−38.9	−11.4	−43.9	−67.4	−60.1	−49.3	−14.1	−54.2	−83.2	−74.2	−60.8
		2	26.1	−16.6	0.0	0.0	0.0	33.1	−21.0	0.0	0.0	0.0	40.8	−25.9	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \leq 160$ ft ($h \leq 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 15$ ft, $V = 110–120$ mi/h

			V (mi/h)														
			110					115					120				
			Zone					Zone					Zone				
h (ft)	Roof Slope	Load Case	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
15	Flat < 2:12 (9.46°)	1	NA	NA	−23.7	−21.1	−17.3	NA	NA	−25.9	−23.1	−18.9	NA	NA	−28.2	−25.1	−20.6
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	−23.2	−15.8	−23.7	−21.1	−17.3	−25.4	−18.3	−25.9	−23.1	−18.9	−27.7	−18.8	−28.2	−25.1	−20.6
		2	3.4	−4.7	0.0	0.0	0.0	3.7	−5.2	0.0	0.0	0.0	4.0	−5.6	0.0	0.0	0.0
	4:12 (18.4°)	1	−19.1	−15.4	−23.7	−21.1	−17.3	−20.9	−16.9	−25.9	−23.1	−18.9	−22.7	−18.4	−28.2	−25.1	−20.6
		2	6.6	−6.8	0.0	0.0	0.0	7.2	−7.4	0.0	0.0	0.0	7.9	−8.1	0.0	0.0	0.0
	5:12 (22.6°)	1	−15.3	−15.4	−23.7	−21.1	−17.3	−16.8	−16.9	−25.9	−23.1	−18.9	−18.2	−18.4	−28.2	−25.1	−20.6
		2	8.8	−7.4	0.0	0.0	0.0	9.6	−8.1	0.0	0.0	0.0	10.5	−8.8	0.0	0.0	0.0
	6:12 (26.6°)	1	−12.3	−15.4	−23.7	−21.1	−17.3	−13.5	−16.9	−25.9	−23.1	−18.9	−14.7	−18.4	−28.2	−25.1	−20.6
		2	9.7	−7.4	0.0	0.0	0.0	10.6	−8.1	0.0	0.0	0.0	11.6	−8.8	0.0	0.0	0.0
	9:12 (36.9°)	1	−7.1	−15.4	−23.7	−21.1	−17.3	−7.8	−16.9	−25.9	−23.1	−18.9	−8.5	−18.4	−28.2	−25.1	−20.6
		2	11.6	−7.4	0.0	0.0	0.0	12.7	−8.1	0.0	0.0	0.0	13.8	−8.8	0.0	0.0	0.0
	12:12 (45.0°)	1	−4.0	−15.4	−23.7	−21.1	−17.3	−4.4	−16.9	−25.9	−23.1	−18.9	−4.8	−18.4	−28.2	−25.1	−20.6
		2	11.6	−7.4	0.0	0.0	0.0	12.7	−8.1	0.0	0.0	0.0	13.8	−8.8	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \le 160$ ft ($h \le 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs

Exposure C: $h = 15$ ft, $V = 130–150$ mi/h

			V (mi/h)														
			130					140					150				
h (ft)	Roof Slope	Load Case	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
15	Flat < 2:12 (9.46°)	1	NA	NA	–33.1	–29.5	–24.2	NA	NA	–38.4	–34.2	–28.1	NA	NA	–44.1	–39.3	–32.2
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–32.5	–22.1	–33.1	–29.5	–24.2	–37.7	–25.6	–38.4	–34.2	–28.1	–43.2	–29.4	–44.1	–39.3	–32.2
		2	4.7	–6.6	0.0	0.0	0.0	5.4	–7.6	0.0	0.0	0.0	6.2	–8.8	0.0	0.0	0.0
	4:12 (18.4°)	1	–26.7	–21.5	–33.1	–29.5	–24.2	–31.0	–25.0	–38.4	–34.2	–28.1	–35.5	–28.7	–44.1	–39.3	–32.2
		2	9.2	–9.5	0.0	0.0	0.0	10.7	–11.0	0.0	0.0	0.0	12.3	–12.6	0.0	0.0	0.0
	5:12 (22.6°)	1	–21.4	–21.5	–33.1	–29.5	–24.2	–24.8	–25.0	–38.4	–34.2	–28.1	–28.5	–28.7	–44.1	–39.3	–32.2
		2	12.3	–10.3	0.0	0.0	0.0	14.3	–11.9	0.0	0.0	0.0	16.4	–13.7	0.0	0.0	0.0
	6:12 (26.6°)	1	–17.2	–21.5	–33.1	–29.5	–24.2	–19.9	–25.0	–38.4	–34.2	–28.1	–22.9	–28.7	–44.1	–39.3	–32.2
		2	13.6	–10.3	0.0	0.0	0.0	15.7	–11.9	0.0	0.0	0.0	18.1	–13.7	0.0	0.0	0.0
	9:12 (36.9°)	1	–10.0	–21.5	–33.1	–29.5	–24.2	–11.5	–25.0	–38.4	–34.2	–28.1	–13.3	–28.7	–44.1	–39.3	–32.2
		2	16.2	–10.3	0.0	0.0	0.0	18.8	–11.9	0.0	0.0	0.0	6.9	–13.7	0.0	0.0	0.0
	12:12 (45.0°)	1	–5.6	–21.5	–33.1	–29.5	–24.2	–6.5	–25.0	–38.4	–34.2	–28.1	–7.5	–28.7	–44.1	–39.3	–32.2
		2	16.2	–10.3	0.0	0.0	0.0	18.8	–11.9	0.0	0.0	0.0	21.6	–13.7	0.0	0.0	0.0

continues

Table 27.5-2 (*Continued*). Main Wind Force Resisting System, Part 2 [$h \le 160$ ft ($h \le 48.8$ m)]: Enclosed Simple Diaphragm Buildings—Wind Pressures—Roofs
Exposure C: $h = 15$ ft, $V = 160$–200 mi/h

			V (mi/h)														
			160					180					200				
h (ft)	Roof Slope	Load Case	Zone					Zone					Zone				
			1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
15	Flat < 2:12 (9.46°)	1	NA	NA	–50.1	–44.7	–36.6	NA	NA	–63.4	–56.6	–46.4	NA	NA	–78.3	–69.8	–57.3
		2	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0	NA	NA	0.0	0.0	0.0
	3:12 (14.0°)	1	–49.2	–35.4	–50.1	–44.7	–36.6	–62.2	–42.3	–63.4	–56.6	–46.4	–76.8	–52.2	–78.3	–69.8	–57.3
		2	7.1	–10.0	0.0	0.0	0.0	9.0	–12.6	0.0	0.0	0.0	11.1	–15.6	0.0	0.0	0.0
	4:12 (18.4°)	1	–40.4	–32.6	–50.1	–44.7	–36.6	–51.2	–41.3	–63.4	–56.6	–46.4	–63.2	–51.0	–78.3	–69.8	–57.3
		2	14.0	–14.3	0.0	0.0	0.0	17.7	–18.1	0.0	0.0	0.0	21.9	–22.4	0.0	0.0	0.0
	5:12 (22.6°)	1	–32.4	–32.6	–50.1	–44.7	–36.6	–41.1	–41.3	–63.4	–56.6	–46.4	–50.7	–51.0	–78.3	–69.8	–57.3
		2	18.6	–15.6	0.0	0.0	0.0	23.6	–19.7	0.0	0.0	0.0	29.1	–24.4	0.0	0.0	0.0
	6:12 (26.6°)	1	–26.1	–32.6	–50.1	–44.7	–36.6	–33.0	–41.3	–63.4	–56.6	–46.4	–40.7	–51.0	–78.3	–69.8	–57.3
		2	20.6	–15.6	0.0	0.0	0.0	26.0	–19.7	0.0	0.0	0.0	32.1	–24.4	0.0	0.0	0.0
	9:12 (36.9°)	1	–15.1	–32.6	–50.1	–44.7	–36.6	–19.1	–41.3	–63.4	–56.6	–46.4	–23.6	–51.0	–78.3	–69.8	–57.3
		2	24.6	–15.6	0.0	0.0	0.0	31.1	–19.7	0.0	0.0	0.0	38.4	–24.4	0.0	0.0	0.0
	12:12 (45.0°)	1	–8.5	–32.6	–50.1	–44.7	–36.6	–10.8	–41.3	–63.4	–56.6	–46.4	–13.3	–51.0	–78.3	–69.8	–57.3
		2	24.6	–15.6	0.0	0.0	0.0	31.1	–19.7	0.0	0.0	0.0	38.4	–24.4	0.0	0.0	0.0